

PATHCAST: A Process-Aware Probabilistic Forecasting Method for Software Development Processes

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Espaço dedicado para a ficha catalográfica. Não alterar.

Espaço dedicado para a declaração de conclusão emitida pelo programa. Não
alterar.

*À Solange,
a Letícia, Mateus, Daniel, Rebeca e Elias,
e à minha família,
que me ampararam com generosidade
ao longo deste caminho.*

*Cor hominis disponit viam suam,
sed Domini est dirigere gressus eius.
— Proverbia 16, 9 (Vulgata)*

Abstract

Software development organizations generate large volumes of process data through *issue* management systems, version control repositories, continuous integration *pipelines*, and code review platforms. Despite this extensive instrumentation, the prediction of software development process behavior still relies predominantly on aggregated metrics and heuristic estimation methods that treat development workflows as black boxes. As a result, existing approaches provide a limited understanding of the structural dynamics and path-dependent variability that characterize real software development processes. This thesis investigates whether models derived from process mining can provide a more informative basis for the probabilistic forecasting of software development processes. In particular, the research tests the hypothesis that process-derived features and explicit stochastic models can improve predictive performance in comparison with approaches based exclusively on traditional software repository metrics. To address this problem, the thesis proposes PATHCAST (PATH-aware foreCASTing), a computational method that integrates process mining, Markov chain modeling, Monte Carlo simulation, and machine learning into a unified analytical *pipeline*. The approach transforms heterogeneous data from software repositories into structured event logs, from which process models are discovered to represent actual development flows. These models are then used to derive stochastic state-transition models that enable the probabilistic simulation of process execution trajectories and their outcome distributions. The research follows the *Design Science Research* methodology, in which the proposed method is implemented and empirically evaluated based on historical data from software projects. The evaluation compares the predictive performance of PATHCAST with baseline models based on repository metrics and conventional forecasting techniques. The expected results are that the explicit incorporation of process structure into forecasting models will improve both predictive accuracy and interpretability, thereby contributing to the advancement of process intelligence applied to software engineering.

Key-words: Process Mining, Probabilistic Forecasting, Markov Chains, Monte Carlo Simulation, Machine Learning, Design Science Research

Resumo

Organizações de desenvolvimento de software geram grandes volumes de dados de processo por meio de sistemas de gerenciamento de *issues*, repositórios de controle de versão, *pipelines* de integração contínua e plataformas de revisão de código. Apesar dessa extensa instrumentação, a previsão do comportamento de processos de desenvolvimento de software ainda se baseia predominantemente em métricas agregadas e métodos heurísticos de estimativa que tratam os fluxos de trabalho de desenvolvimento como caixas-pretas. Como consequência, as abordagens existentes fornecem uma compreensão limitada da dinâmica estrutural e da variabilidade dependente do caminho que caracterizam os processos reais de desenvolvimento de software. Esta tese investiga se modelos derivados de mineração de processos podem fornecer uma base mais informativa para a previsão probabilística de processos de desenvolvimento de software. Em particular, a pesquisa testa a hipótese de que características derivadas do processo e modelos estocásticos explícitos podem melhorar o desempenho preditivo em comparação com abordagens baseadas exclusivamente em métricas tradicionais em repositórios de software. Para abordar esse problema, a tese propõe o PATHCAST (PATH-aware foreCASTing), um método computacional que integra mineração de processos, modelagem por cadeias de Markov, simulação de Monte Carlo e aprendizado de máquina em um *pipeline* analítico unificado. A abordagem transforma dados heterogêneos provenientes de repositórios de software em logs de eventos estruturados, a partir dos quais são descobertos modelos de processo que representam os fluxos reais de desenvolvimento. Esses modelos são então utilizados para derivar modelos estocásticos de transição de estados que permitem a simulação probabilística de trajetórias de execução do processo e de suas distribuições de resultados. A pesquisa segue a metodologia *Design Science Research*, na qual o método proposto é implementado e avaliado empiricamente com base em dados históricos de projetos de software. A avaliação compara o desempenho preditivo do PATHCAST com modelos de referência baseados em métricas de repositórios e em técnicas convencionais de previsão. Espera-se que os resultados demonstrem que a incorporação explícita da estrutura do processo em modelos de previsão melhore tanto a precisão preditiva quanto a interpretabilidade, contribuindo, assim, para o avanço da inteligência de processos aplicada à engenharia de software.

Palavras-chave: Mineração de Processos, Previsão Probabilística, Cadeias de Markov, Simulação de Monte Carlo, Aprendizado de Máquina, Design Science Research

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Contents

List of Figures	xiii
List of Tables	xvii
List of Charts	xviii
List of Algorithms	xix
List of Acronyms	xx
1 Introduction	1
1.1 Research Problem	2
1.2 Objectives	3
1.2.1 General Objective	3
1.2.2 Specific Objectives	4
1.3 Research Questions	4
1.4 Justification and Research Gap	5
1.5 Expected Contributions	6
1.6 Research Methodology	7
1.7 Structure of the Document	9
2 Theoretical Foundation	10
2.1 Process Mining	10
2.2 Mining Software Repositories	11
2.3 Stochastic Process Modeling with Markov Chains	12
2.4 Monte Carlo Simulation and Probabilistic Forecasting Metrics	14
2.5 Machine Learning for Software Engineering	15
2.6 Variant Entropy	16
2.7 Chapter Synthesis	16

3	Systematic Literature Review	18
3.1	Introduction	18
3.2	Review Protocol	19
3.2.1	Research Questions	19
3.2.2	Search Strategy	19
3.2.3	Inclusion and Exclusion Criteria	22
3.2.4	Quality Assessment	23
3.2.5	Data Extraction	26
3.3	Search Results and Study Selection	26
3.3.1	Database Retrieval and Deduplication	26
3.3.2	Operational Working Set Construction	27
3.3.3	Title/Abstract Screening	28
3.3.4	Eligibility Screening (Full-Text / Enriched-Abstract)	29
3.4	Results and Synthesis	30
3.4.1	Overview of Included Studies	30
3.4.2	Inclusion Criterion Coverage	33
3.4.3	Prevalent Exclusion Patterns	34
3.4.4	Full-Text Extraction and PDF-Based Triangulation	35
3.4.5	RQ1: Application Landscape	36
3.4.6	RQ2: Techniques and Prediction	38
3.4.7	RQ3: Integration and Gaps	42
3.5	Discussion	45
3.5.1	Research Gap Identification	45
3.5.2	Positioning of PATHCAST	48
3.5.3	Proposed Taxonomy: SPMF	50
3.5.4	Research Agenda	50
3.5.5	Implications for Practitioners	51
3.6	Threats to Validity of the SLR	51
3.6.1	Construct Validity	51
3.6.2	Internal Validity	51
3.6.3	External Validity	54
3.6.4	Conclusion Validity	57
3.6.5	Replication Package	59
3.7	Cross-Domain Synthesis and Gap Map	59
3.7.1	Process Mining Applied to SDLC	59
3.7.2	Stochastic Forecasting and ML for SE	60

3.7.3	Gap Map	60
3.8	Chapter Summary	61
4	The Proposed Method: PATHCAST	64
4.1	Architectural Overview	64
4.1.1	Pipeline Definition	64
4.1.2	Pipeline Architecture	65
4.1.3	Design Principles	66
4.1.4	Component View	67
4.1.5	Stage Contracts	68
4.2	Stage 1: Event Log Construction	69
4.2.1	Event Extraction	69
4.2.2	Case Correlation	70
4.2.3	Activity Mapping	71
4.2.4	Event Log Formatting and Filtering	72
4.2.5	Log Quality Assessment	73
4.3	Stage 2: Process Mining	73
4.3.1	Variant Entropy	73
4.3.2	Process Discovery	74
4.3.3	Conformance Checking	74
4.3.4	Performance Enhancement	75
4.3.5	State Space Extraction	75
4.4	Stage 3: Stochastic Modeling	76
4.4.1	Transition Counting	76
4.4.2	Maximum Likelihood Estimation	77
4.4.3	Confidence Assessment	77
4.4.4	Markov-Order Validation	78
4.4.5	Absorbing Chain Decomposition	79
4.4.6	Empirical Sojourn Time Distributions	80
4.4.7	Expected Occupancy Distribution and WIP Analysis	82
4.4.8	Rework Analysis	82
4.4.9	Numerical Example	82
4.5	Stage 4: Monte Carlo Simulation	83
4.5.1	Simulation Algorithm	83
4.5.2	Forecast Extraction	84
4.5.3	Convergence and Simulation Budget	85

4.5.4	Reproducibility	86
4.5.5	Comparison with Throughput-Based Monte Carlo	86
4.6	Machine Learning Enhancement Component	87
4.6.1	Feature Engineering	87
4.6.2	Label Construction	88
4.6.3	Training Protocol	88
4.6.4	Integration with Monte Carlo Forecasts	89
4.7	Pipeline Integration	89
4.8	Formal Properties and Theoretical Guarantees	90
4.8.1	Statistical Consistency	90
4.8.2	Calibration Property	90
4.8.3	Information-Theoretic Advantage of Process Awareness	91
4.8.4	Complexity Analysis	91
4.8.5	Assumptions and Boundary Conditions	92
4.9	Chapter Summary	92
5	Research Methodology	96
5.1	Design Science Research Paradigm	96
5.1.1	DSR Artifact Characterization	97
5.1.2	DSRM Process Instantiation	97
5.1.3	DSR Knowledge Contribution	97
5.2	Dataset Description	98
5.2.1	Dataset Rationale	98
5.2.2	Repository Selection Criteria	98
5.2.3	Selection Process	99
5.2.4	Data Collection	99
5.2.5	Dataset Descriptive Statistics	101
5.2.6	Ethical Considerations	102
5.3	Experimental Design	103
5.3.1	E1: Pipeline Validation	103
5.3.2	E2: Forecasting Accuracy	105
5.4	Evaluation Metrics	106
5.4.1	Metrics for RQ1: Event Log and Process Model Quality	106
5.4.2	Metrics for RQ2: Stochastic Forecasting Accuracy	107
5.4.3	Metrics for RQ3: ML Feature Comparison	109
5.5	Baseline Methods	110

5.6	Statistical Analysis Procedures	111
5.6.1	Statistical Hypotheses	111
5.6.2	Within-Repository Comparison	112
5.6.3	Cross-Repository Aggregation	112
5.6.4	Statistical Power Analysis	112
5.6.5	Reporting Standards	113
5.7	Threats to Validity	113
5.7.1	Internal Validity	114
5.7.2	External Validity	114
5.7.3	Construct Validity	115
5.7.4	Conclusion Validity	116
5.8	Chapter Summary	116
6	Implementation	118
6.1	Software Architecture	118
6.2	Stage 1: Event Log Construction	119
6.2.1	Preliminary harvest statistics	119
6.3	Interim Data Artifacts	120
6.4	Preliminary E1 Feasibility Pilot	121
6.4.1	E1 batch on ten representative repositories	121
6.5	Confidential Organisational Data	122
6.6	Implementation Roadmap	123
	References	125
A	Markov Chain Derivations	134
A.1	The Fundamental Matrix Identity	134
A.2	Expected Steps to Absorption	135
A.3	Variance of Absorption Time	135
A.4	Absorption Probabilities	136
A.5	Bayesian Estimation and Laplace Smoothing	136
A.6	Semi-Markov Connection	137
A.7	Numerical Example: Six-State SDLC Chain	138
B	Probabilistic Metric Derivations	141
B.1	CRPS: Definition and Equivalent Forms	141
B.2	Propriety of CRPS	142

B.3	CRPS–MAE Relationship	142
B.4	Calibration and the Aggregate Calibration Error	143
B.5	Worked Numerical Examples	143
C	Process Mining Algorithmic Details	144
C.1	Inductive Miner Internals	144
C.2	Alignment-Based Conformance Checking	145
C.3	Fitness, Precision, and Generalization	145
C.4	Worked Example	146
D	End-to-End Walkthrough Example	147
D.1	Synthetic Event Log	147
D.2	Process Discovery	148
D.3	Markov Estimation	148
D.4	Monte Carlo Simulation	149
D.5	Cross-Validation Against Analytical Results	149
D.6	ML Feature Construction	150
E	Search Protocol: Search Strings	151
E.1	Search String Summary by Database	151
E.2	String S1 — Principal ($P \wedge I$)	152
E.3	String S2 — Complementary ($MSR \rightarrow$ Process Modelling)	152
E.4	String S3 — Frontier ($Stochastic \wedge PM \wedge SE$)	152
E.5	String S4 — Markov in Software Testing (Scopus supplement)	153
F	LLM Screening Prompts	154
F.1	Phase 1: Title/Abstract Screening	154
F.2	Phase 2: Full-Text Screening	156
F.3	Phase 2b: <i>Screen-Blanks</i> Pass	156
F.4	Model Parameters by Phase	157
G	Data Extraction Form	158
G.1	Extraction Prompt	158
G.2	Data Extraction Form Fields	159
G.3	Controlled Vocabulary (Enums) for Structured Output	159

H	Assessment and Classification Instruments	162
H.1	Priority Bands for the Full-Text Queue	162
H.2	Abstract Enrichment Algorithm	162
H.3	Quality Assessment Checklist	164
H.4	Control Paper Set	164
H.5	SPMF Integration-Level Taxonomy	167
I	Reading List and Pipeline Artefacts	168
I.1	Multi-criterion Scoring Rubric	168
I.2	Top-32: Ranked Reading List	169
I.3	Review Pipeline Scripts	169
I.4	Persistent Artefact Registry	169
J	Evaluation Dataset: Repository List	172
J.1	Stratification Summary	172
J.2	Complete Repository List	172

List of Figures

1.1	Overview of the DSRM process applied to this research, with six activities, produced artifacts, and a two-stage evaluation.	8
3.1	PRISMA 2020 flow diagram of the study selection process. Phase labels follow the four-stage structure: Identification, Screening, Eligibility, and Included.	31
3.2	Distribution of confirmed studies by publication year ($n = 169$). .	33
3.3	Inclusion criteria matched by the confirmed set ($n = 169$; non-exclusive; IC labels map to IC4a–IC4d in the protocol).	34
3.4	Operational decision distribution in the 73-paper PDF triangulation subset.	36
3.5	Main technique families identified in the 169-study confirmed set. Counts are non-exclusive: a paper may use multiple technique families.	39
3.6	Operational integration-level distribution (L0–L3) in the 73-paper PDF triangulation subset.	43
3.7	Reading-priority distribution (High/Medium/Low) in the 73-paper PDF triangulation subset.	44
3.8	Evidence-type distribution in the 73-paper PDF triangulation subset.	45
4.1	PATHCAST pipeline architecture. Solid arrows indicate the primary sequential pipeline; dashed arrows indicate ML enhancement data flows.	66
4.2	PATHCAST component overview.	67
5.1	Repository selection and enrichment process.	100
5.2	Two-stage experimental design.	103

H.1	Abstract enrichment cascade applied to the 886-paper FT queue. .	163
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List of Tables

1.1	Positioning relative to existing work.	6
3.1	Research questions and expected outputs.	19
3.2	PICO decomposition for search string construction.	20
3.3	Database-specific search adaptations and raw retrieval counts. . .	22
3.4	Control papers for search string validation.	23
3.5	Inclusion and exclusion criteria.	24
3.6	Quality assessment checklist.	25
3.7	Quality-assessment results for the confirmed included set ($n = 169$; full QA applied). Scores were produced by the LLM-assisted QA protocol (Section 3.2.4).	25
3.8	Data extraction form.	26
3.9	Record retrieval by source.	27
3.10	Working set composition.	28
3.11	Title/abstract screening results.	29
3.12	Full-text screening outcomes (886 papers forwarded from T/A phase).	30
3.13	Provenance of headline denominators in the results synthesis. . .	32
3.14	Item-level quality-assessment profile of the confirmed included set ($n = 169$). Percentages are the share of studies scoring 1 on each binary criterion. Scores were produced by the LLM-assisted QA protocol described in Section 3.2.4 and archived in the replication package (Section 3.6.5).	45
3.15	Integration levels in the 169-study confirmed set and PATHCAST positioning.	49

3.16	Inter-rater agreement (Cohen’s κ) between primary screener (claude-haiku-4-5) and verifier (claude-sonnet-4-6) on a stratified random 20% sample.	52
3.17	Auxiliary-tier inter-rater agreement (cross-model verification: Haiku 4.5 primary vs. Sonnet 4.6 verifier on a stratified 20% sample). . .	53
3.18	Auxiliary-corpus sensitivity check. A random sample of n papers from the deferred auxiliary corpus was re-screened with the same T/A LLM protocol used for the working set.	56
3.19	Auxiliary-corpus FT-screening outcome (papers that passed T/A on the 3,807-paper deferred corpus).	56
3.20	Re-FT screening of aux-pending papers that gained an abstract through the post-hoc enrichment cascade.	56
3.21	EC5 recovery attempt against four open-access discovery APIs. .	57
3.22	Quality assessment combined across working-set and auxiliary tiers, after cross-tier de-duplication ($n = 319$). The auxiliary tier is reported as unique studies (150) after removing 60 cross-tier and 2 internal duplicates from the 212 raw first-pass inclusions. .	58
3.23	Process mining applied to software development.	60
3.24	Technique integration ($\checkmark$) across studies. To the best of the author’s knowledge, no reviewed study occupies the full intersection targeted by PATHCAST.	61
4.1	Event extraction mapping per data source.	70
4.2	Default activity mapping from source event types to SDLC taxonomy.	72
4.3	Throughput-based vs. process-aware Monte Carlo.	86
4.4	Feature definitions for the ML component.	87
4.5	Inter-stage data contracts.	90
4.6	Computational complexity per pipeline stage.	92
4.7	Assumptions, implications, and mitigations.	93
4.8	Method components vs. research questions.	95
5.1	Mapping of DSRM activities to thesis components.	97
5.2	Aggregate dataset statistics (Stage-1 harvest of 46/48 GitHub repositories with event logs; 47/48 evaluation repositories confirmed; see <code>repo_manifest.json</code>).	102
5.3	Dataset diversity profile.	102

5.4	Evaluation metrics for RQ1.	107
5.5	Evaluation metrics for RQ2.	108
5.6	Evaluation metrics for RQ3.	109
5.7	Baseline methods.	110
5.8	A priori power estimates ($n = 48$, four baselines).	113
5.9	Summary: research questions, evaluation stages, metrics, baselines.	116
6.1	PATHCAST implementation modules (v0.1.0).	119
6.2	Stage-1 GitHub harvest snapshot (June 2026, $n = 46$ repositories).	120
6.3	Interim artefacts at v0.1.0.	120
6.4	Preliminary E1 pilot on <code>fastapi</code> / <code>fastapi</code> (800 most recent cases; Inductive Miner, $\theta_{\text{noise}} = 0.2$). Not the full 48-repository E1.	121
6.5	E1 preliminary batch on ten representative repositories (IMf, $\theta_{\text{noise}} = 0.2$, up to 800 cases each).	122
6.6	Anonymised aggregate statistics for four confidential organisational projects (denoted P_1 – P_4). Jira transition volumes are available for P_1 only; E1 preview metrics use the PATHCAST Bitbucket activity alphabet (PR/commit lifecycle, no author or title fields).	124
E.1	Search strings by database and field scope.	151
F.1	LLM parameters in each screening phase.	157
G.1	Data extraction form: all fields in <code>extraction_template.csv</code>	160
G.2	Controlled vocabulary for LLM structured output.	161
H.1	Priority-band definitions for the full-text review queue.	163
H.2	Abstract enrichment results by priority band.	164
H.3	Quality assessment checklist with scoring guide.	165
H.4	Complete control-paper set for string validation.	166
H.5	SPMF taxonomy: three dimensions with level definitions.	167
I.1	Multi-criterion scoring rubric for the top-32 reading list.	168
I.2	Top-32 reading list with score, year, IC cluster, and summary of the main finding. Papers marked with † are forced inclusions due to seminal status.	170
I.3	Review pipeline scripts and their functions.	171
I.4	Persistent review artefacts.	171

J.1	Stage-1 harvest coverage by evaluation repository (47/48 with usable logs; snapshot June 2026).	173
J.2	Repository count by primary language stratum ($n = 48$).	174
J.3	Complete list of 48 evaluation repositories with stratification meta-data. Abbreviations: Scale — S = Small (< 5K commits), M = Medium (5–20K), L = Large (> 20K); Gov. — Fdn = Foundation, Corp = Corporate, Comm = Community; WF — B = Basic, St = Standard, A = Advanced.	175

List of Charts

List of Algorithms

1	Case Correlation Cascade	71
2	Transition Count Extraction	76
3	Process-Aware Monte Carlo Simulation	83
4	ML Training with Temporal Hold-Out	88

List of Acronyms

AHP Analytic Hierarchy Process

BPM Business Process Management

CEP Complex Event Processing

CI/CD Continuous Integration / Continuous Delivery

DSR Design Science Research

DSRM Design Science Research Methodology

EDA Event-Driven Architecture

LLM Large Language Model

LSTM Long Short-Term Memory

MCDM Multi-Criteria Decision Making

MSR Mining Software Repositories

PATHCAST PATH-aware foreCASTing

PI Process Intelligence

PM Process Mining

SDLC Software Development Life Cycle

SLR Systematic Literature Review

XES eXtensible Event Stream



Introduction

Contemporary software development operates in a densely instrumented ecosystem. Every state transition recorded in an issue tracker such as Jira, every commit pushed to a Git repository, every build triggered in a continuous integration platform, every pull request reviewed on GitHub generates time-stamped event data that documents who performed which activity, when, and in what sequence. Collectively, these events constitute a high-fidelity chronicle of the software development process as it actually occurs, rather than as it is documented in process manuals or assumed in retrospective meetings.

The discipline of process mining, formalized by van der Aalst (AALST, 2016), has demonstrated that event data can be systematically transformed into process models, compliance diagnostics, performance metrics, and predictions of future behavior. The field has matured along an analytical spectrum that progresses from descriptive capabilities (what happened?), through diagnostic (why did it happen?) and predictive (what will happen?), to prescriptive (what should be done?). In domains such as healthcare, manufacturing, financial services, and public administration, process mining has delivered measurable organizational value (AALST, 2012).

Software engineering, however, presents a paradox. Despite generating some of the richest and most fine-grained event data among all organizational activities, the application of process mining to the Software Development Lifecycle (SDLC) remains limited to descriptive analysis. Development teams continue to rely on aggregated metrics (velocity, throughput, cycle time) computed from project management dashboards, rarely interrogating the underlying process

flows that produce these metrics. When forecasting is attempted, it typically uses throughput-based heuristics that treat the development process as a black box, ignoring the internal state dynamics, rework loops, and path-dependent variability that characterize real software processes.

This thesis proposes a method that bridges the gap between the rich event data generated by software development tools and the probabilistic forecasting capabilities that this data could support. The method, named PATHCAST integrates process mining, Markov chain estimation, Monte Carlo simulation, and machine learning into a unified pipeline that transforms heterogeneous repository data into calibrated probabilistic forecasts of software process outcomes.

1.1 RESEARCH PROBLEM

Three interrelated problems motivate this research.

The first problem is the **descriptive ceiling** in software process mining. The existing literature on process mining applied to software development (COOK; WOLF, 1998; PONCIN; SEREBRENIK; BRAND, 2011; RUBIN et al., 2007) has demonstrated that process models can be discovered from repository data. However, these studies stop at the descriptive level: they reveal what happened, but do not advance to predicting what will happen or quantifying the uncertainty associated with future outcomes. No published work proceeds from software process discovery to stochastic modeling and probabilistic forecasting using the discovered process structure as the foundation for prediction.

The second problem is the **structural blindness** of existing forecasting approaches. The most widely adopted forecasting method in agile software development, throughput-based Monte Carlo simulation (VACANTI, 2015), samples daily throughput values from historical data and simulates backlog completion. While practical, this approach treats the development process as a black box: it does not model internal state transitions, rework probabilities, state-specific delays, or path-dependent variability. An item that has been reworked three times is treated identically to one progressing linearly, because the simulation has no awareness of process structure. This limitation produces forecasts that cannot capture the variance explained by process dynamics.

The third problem is **technical fragmentation**. The techniques required for process-aware forecasting, including process mining for model discovery,

Markov chains for transition modeling, Monte Carlo simulation for distributional prediction, and machine learning for case-level classification, exist independently in the literature. Their integration into a coherent pipeline, where the output of each technique serves as input to the next, has not been formalized or empirically validated in the software engineering domain. Researchers and practitioners who wish to combine these techniques must construct *ad hoc* integrations without guidance from established methods.

1.2 OBJECTIVES

The growing volume of data generated throughout the Software Development Life Cycle (SDLC) offers relevant opportunities for analyzing and forecasting the behavior of software development processes. Systems such as version control repositories, *issue* management platforms, and continuous integration *pipelines* record in detail the activities performed during development. However, a large portion of the approaches used in practice to estimate project deadlines and outcomes still relies on aggregated metrics and heuristic methods that treat development processes as opaque systems. As a result, these approaches have limitations in capturing the structural dynamics of processes and the variability associated with different execution paths.

In this context, this research investigates the potential of process mining, combined with stochastic modeling and machine learning techniques, to improve the forecasting of software development processes.

1.2.1 GENERAL OBJECTIVE

To develop a computational method for the probabilistic forecasting of software development processes, based on the integration of process mining, stochastic modeling through Markov chains, Monte Carlo simulation, and machine learning, capable of transforming data from software repositories into analytical models that enable the prediction of the behavior and outcomes of these processes.

1.2.2 SPECIFIC OBJECTIVES

To achieve the general objective and test the proposed hypotheses, the following specific objectives are defined:

1. Build structured event logs from data originating from software repositories, including issue tracking systems, version control repositories, and continuous integration pipelines, in order to formally represent the actual behavior of development processes.
2. Apply process mining techniques to discover models that represent the actual flow of activities in software development, allowing the identification of states, transitions, and execution patterns of the process..
3. Develop a probabilistic prediction model based on Markov chains and Monte Carlo simulation, capable of estimating execution time distributions, process trajectories, and possible outcomes of development processes.
4. Empirically evaluate the proposed method by integrating it with machine learning techniques and comparing it with predictive models based solely on traditional software repository metrics, analyzing the gains in predictive accuracy and interpretability..

1.3 RESEARCH QUESTIONS

The research is guided by the following central question: Can the explicit incorporation of process structure, obtained through process mining and stochastic modeling techniques, improve the accuracy and interpretability of forecasting software development processes compared to approaches based exclusively on traditional software repository metrics?

H1 (Predictive Superiority of Process-Derived Features): *Predictive models that incorporate features derived from process structure achieve superior predictive performance compared with models based solely on traditional software repository metrics.*

This hypothesis evaluates whether process-aware representations provide additional explanatory and predictive power beyond conventional repository-based indicators.

H2 (Fidelity and Interpretability of Markov-Based Stochastic Models): *Stochastic models based on Markov chains enable a more faithful representation of the dynamics of software development processes, allowing the generation of interpretable probabilistic forecasts.*

This hypothesis examines whether state-transition models can adequately capture process dynamics and support probabilistic predictions that are both analytically grounded and interpretable.

H3 (Performance Gain from Multi-Technique Integration): *The integration of process mining, Monte Carlo simulation, and machine learning improves forecast quality when compared with the isolated use of each of these techniques.*

This hypothesis assesses whether combining complementary analytical techniques produces better forecasting performance than applying each method independently.

1.4 JUSTIFICATION AND RESEARCH GAP

Several systematic reviews and surveys have been published in areas adjacent to the scope of this research, each covering a distinct facet of the landscape without addressing their convergence in the software engineering domain.

In the process mining literature, Augusto et al. (AUGUSTO et al., 2019) conducted a comprehensive review of automated process discovery algorithms. Burattin (BURATTIN, 2022) formalized the emerging subfield of streaming process mining. Neither work addresses the specific challenges of applying process mining to software development event logs, where case correlation across heterogeneous tools constitutes a prerequisite not present in traditional BPM scenarios.

In the Mining Software Repositories (MSR) community, Hassan (HASSAN, 2008) established a foundational research roadmap, and Kagdi, Collard, and Maletic (KAGDI; COLLARD; MALETIC, 2007) produced an extensive taxonomy of MSR approaches. These works focus predominantly on mining individual artifacts (source code, commits, bug reports) rather than mining the process flows that connect these artifacts throughout the development lifecycle.

In the predictive and prescriptive monitoring literature, Márquez-Chamorro et al. (MARQUEZ-CHAMORRO; RESINAS; RUIZ-CORTES, 2018) mapped predictive monitoring techniques and Kubrak et al. (KUBRAK; MILANI; NOLTE, 2022) mapped the state of prescriptive process monitoring. Both surveys treat business processes generically, without examining the specific context of software development.

In the stochastic forecasting literature for software projects, Vacanti (VA-

CANTI, 2015) popularized throughput-based Monte Carlo simulation for agile delivery forecasting, while Whittaker and Thomason (WHITTAKER; THOMASON, 1994) pioneered Markov chain models for software testing. Neither work connects these stochastic methods to process mining-discovered models.

Table 1.1 synthesizes the positioning of this research relative to existing surveys and methods, demonstrating that the addressed intersection, combining process mining, stochastic modeling, and SDLC-specific application, remains unoccupied.

Table 1.1: Positioning relative to existing work.

Work	PM	SDLC Focus	Markov/ MC Sim	ML	Integrated Pipeline
Cook & Wolf (1998)	Yes	Discovery	No	No	No
Kagdi et al. (2007)	No	Artifacts	No	No	No
Hassan (2008)	No	Artifacts	No	No	No
Poncin et al. (2011)	Yes	Discovery	No	No	No
Whittaker & Thomason (1994)	No	Testing	Markov	No	No
Vacanti (2015)	No	Forecasting	MC	No	No
Augusto et al. (2019)	Yes	No	No	No	No
Márquez-Chamorro et al. (2018)	Yes	No	No	Yes	No
Kubrak et al. (2022)	Yes	No	No	Yes	No
Rozinat et al. (2009)	Yes	No	MC	No	Partial
This thesis (2026)	Yes	SDLC	Both	Yes	Yes

The table reveals two patterns. First, works that apply process mining to SDLC data stop at descriptive analysis (discovery), without advancing to stochastic modeling or prediction. Second, works that employ stochastic methods for software forecasting (Vacanti, Whittaker) operate without process mining, treating the process as either a black box (throughput sampling) or a theoretical construct (usage-model Markov chains). The proposed method occupies the intersection: process mining provides the empirical process model that parameterizes the stochastic simulation, applied to the SDLC domain.

1.5 EXPECTED CONTRIBUTIONS

This thesis aims to make the following contributions:

C1 (Method): A formally specified method (PATHCAST) that integrates process mining, Markov chain estimation, Monte Carlo simulation, and machine learning into a four-stage pipeline for probabilistic forecasting of software de-

velopment processes. The method is defined through inter-stage data contracts, formal properties, and correctness guarantees.

C2 (Event Log Construction): A configurable framework for constructing structured event logs from heterogeneous software repositories, addressing the case correlation problem through a cascade of explicit, structural, and heuristic strategies with formal quality metrics. This framework is reusable independently of the forecasting pipeline.

C3 (Process-Aware Stochastic Forecasting): An empirically validated demonstration that Monte Carlo simulation parameterized by Markov chain models estimated from process mining-discovered models produces more accurate and better-calibrated forecasts than throughput-based Monte Carlo, for software development processes.

C4 (Process-Derived Feature Set): A novel set of features derived from process mining and Markov analysis (conformance scores, rework counts, Markov-expected remaining times) for machine learning-based outcome prediction in software processes, with empirical comparison against traditional MSR features.

C5 (Empirical Evidence): A reproducible empirical evaluation on 48 open-source software repositories, providing baseline results for stochastic process forecasting in the SDLC domain and a reusable dataset for future research.

1.6 RESEARCH METHODOLOGY

This research adopts the Design Science Research (DSR) paradigm, following the framework of Hevner et al. (HEVNER et al., 2004) and the methodology of Peffers et al. (PEFFERS et al., 2007). DSR is the appropriate paradigm because the primary goal is to design, implement, and evaluate an artifact (the PATHCAST method and its supporting tool) that addresses a practical problem identified in the literature.

The DSRM process is instantiated as follows:

Activity 1 (Problem Identification): The research problem is identified through analysis of the literature at the intersection of process mining, MSR, and stochastic forecasting, revealing the descriptive ceiling, structural blindness, and technical fragmentation described in Section 1.1.

Activity 2 (Objectives): The objectives are derived from the identified gaps: a method that advances beyond descriptive analysis, incorporates process struc-

ture into forecasting, and integrates complementary techniques into a formally specified pipeline.

Activity 3 (Design and Development): The PATHCAST method is designed as a four-stage pipeline with formal inter-stage contracts, implemented as a Python-based tool using PM4Py for process mining, NumPy for Markov computation, and scikit-learn/XGBoost for machine learning.

Activity 4 (Demonstration): The method is demonstrated on 48 open-source software repositories, producing event logs, process models, Markov estimates, Monte Carlo forecasts, and ML predictions.

Activity 5 (Evaluation): The evaluation is structured into two complementary stages:

- **E1 (Pipeline Validation):** Assesses the quality of event log construction (coverage, correlation rate) and process discovery (fitness, precision, generalization) across the 48 repositories. Addresses RQ1.
- **E2 (Forecasting Accuracy):** Compares the proposed process-aware Monte Carlo against baselines (throughput-based MC, analytical Markov, historical mean) using temporal hold-out validation. Evaluates ML feature sets through cross-validated comparison. Addresses RQ2 and RQ3.

Activity 6 (Communication): Results are communicated through this thesis and through submissions to venues in empirical software engineering and process mining (e.g., EASE, MSR, ICPM).

Figure 1.1 illustrates the DSRM process applied to this research.

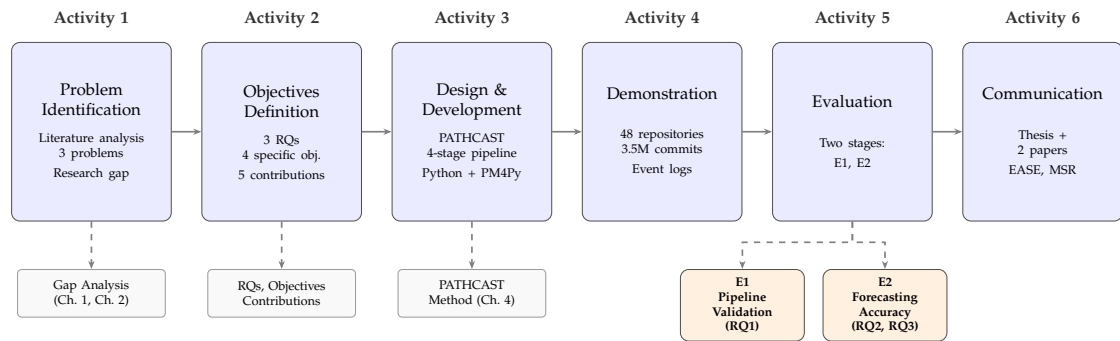


Figure 1.1: Overview of the DSRM process applied to this research, with six activities, produced artifacts, and a two-stage evaluation.

The methodology is detailed in Chapter 5.

1.7 STRUCTURE OF THE DOCUMENT

The remainder of this document is organized as follows.

Chapter 2 presents the theoretical foundations, covering process mining, mining of software repositories, Markov chains for software process modeling, Monte Carlo simulation for process forecasting, and machine learning for software engineering.

Chapter 3 presents a systematic literature review following Kitchenham and Charters' guidelines, mapping the state of the art at the intersection of process mining, stochastic modeling, and machine learning applied to software process forecasting, and identifying the research gap addressed by the proposed method.

Chapter 4 presents the proposed method (PATHCAST), formalizing the four-stage pipeline with input/output contracts, algorithms, formal properties, and theoretical guarantees.

Chapter 5 details the research methodology, including the Design Science Research framework, the experimental design for the two evaluation stages, and threats to validity.

A dedicated implementation chapter (in preparation) will describe the implementation architecture, technology choices, and design decisions.

The empirical-evaluation chapter (in preparation) will present the empirical evaluation, including the results of pipeline validation (E1) and forecasting accuracy assessment (E2), with statistical analysis and comparison against baselines.

The concluding chapter (in preparation) will discuss the results, identify limitations, propose directions for future work, and present the conclusions.

2

Theoretical Foundation

This chapter states the formal vocabulary on which PATHCAST is built. Six concepts are introduced in succession: process mining (Section 2.1), mining of software repositories (Section 2.2), Markov chains (Section 2.3), Monte Carlo simulation and the probabilistic forecasting metrics it requires (Section 2.4), machine learning for software engineering (Section 2.5), and the entropy of trace variants (Section 2.6). Each section is intentionally short: its purpose is to fix notation and quote the central property used elsewhere in the thesis. Derivations, algorithmic internals and worked examples are deferred to Appendices A–D. The literature gap that motivates the method is synthesised in Chapter 3, not here.

2.1 PROCESS MINING

Process mining extracts behavioural knowledge from event logs recorded by information systems (AALST, 2016). Three classes of analysis are distinguished: *discovery* extracts a model M from a log $\mathcal{L}$ alone; *conformance* compares $\mathcal{L}$ against a given M ; *enhancement* enriches M with information derived from $\mathcal{L}$. PATHCAST uses all three.

Definition 2.1 (Event, Trace, Event Log). An event $e = (c, a, t, d)$ is a tuple with case identifier $c \in \mathcal{C}$, activity $a \in \mathcal{A}$, timestamp $t \in \mathbb{R}^+$ and attribute map $d \in \mathcal{D}$. Projections $\kappa(e), \alpha(e), \tau(e)$ return case, activity and timestamp. A trace $\sigma_c = \langle e_1, \dots, e_n \rangle$ is the timestamp-ordered sequence of events of case c . An event log $\mathcal{L} =$

$\{\sigma_{c_1}, \dots, \sigma_{c_N}\}$ is a multiset of traces. Since each case corresponds to exactly one trace, we use the case identifier c and its trace σ_c interchangeably where no ambiguity arises.

Definition 2.2 (Alignment and Alignment-Based Fitness). An alignment γ between trace σ and model M is a sequence of synchronous, log and model moves (ADRIANSYAH; DONGEN; AALST, 2011). With γ^* the cost-optimal alignment,

$$\text{fitness}(\sigma, M) = 1 - \frac{\text{cost}(\gamma^*)}{\text{cost}(\gamma_{\text{worst}})}, \quad (2.1)$$

$$\text{conf}(c) = 1 - \frac{\text{cost}(\gamma_c^*)}{\text{cost}(\gamma_{\text{worst},c})}. \quad (2.2)$$

Thus $\text{conf}(c)$ is the per-case (trace-level) alignment-based fitness: the cost of the optimal alignment γ_c^* of trace σ_c normalized by its worst-case alignment cost, yielding a score in $[0, 1]$. Log-level fitness is the trace-weighted average of $\text{conf}(c)$. The case-level conformance $\text{conf}(c)$ is reused, under the same definition, as an ML feature in Chapter 4 (Eq. 4.24).

PATHCAST adopts the Inductive Miner with infrequent-behaviour filtering (IMf) (LEEMANS; FAHLAND; AALST, 2013) because it is among the few discovery algorithms that guarantee soundness by construction, producing a block-structured process tree compatible with SDLC workflows. Discovery quality is quantified along four dimensions—fitness, precision, generalisation and simplicity (AALST, 2016); precision and generalisation are estimated by the ETC metric and state-visit frequency analysis, respectively. The internals of IMf, the alignment search and the precision/generalisation estimators are presented in Appendix A.

We refer to this sound, block-structured model M discovered by IMf—the one that captures the dominant, infrequent-behaviour-filtered control flow of the SDLC—as the *process backbone*. The term is used throughout the thesis to denote the discovered control-flow skeleton over which the stochastic and forecasting layers are subsequently built.

2.2 MINING SOFTWARE REPOSITORIES

Mining Software Repositories (MSR) extracts evidence from version control, issue trackers, code review platforms and CI/CD pipelines (HASSAN, 2008; KAGDI; COLLARD; MALETIC, 2007). The bridge with process mining was

established by Cook and Wolf (COOK; WOLF, 1998) and formalised by Poncin et al. (PONCIN; SEREBRENIK; BRAND, 2011): software events are reframed as sequences belonging to process instances.

Modern toolchains emit heterogeneous, partly overlapping streams: issue-tracker status transitions (Jira, GitHub Issues), version-control commits (Git, SVN), code-review activities (PRs, Gerrit) and CI/CD outcomes (Jenkins, GitHub Actions). The empirical challenge is integration (KOCHHAR et al., 2019; DIXIT; VERI et al., 2021): a unified event log requires identifier mapping, temporal alignment and semantic harmonisation across tools. Curated datasets and collectors—SEART GitHub Search (DABIC; AGHAJANI; BAVOTA, 2021), GH Archive (GRIGORIK, 2012), SmartSHARK (TRAUTSCH et al., 2020), GrimoireLab (DUEÑAS et al., 2021) and the Public Jira Dataset (MONTGOMERY; LÜDERS; MAALEJ, 2022)—reduce the effort required to identify and collect suitable repositories for analysis, without eliminating known validity threats (open-source bias, bot noise, incompleteness). The construction sub-tasks (extraction, case correlation, activity mapping) and their archaeological caveats (stratigraphy, contextual interpretation, taphonomic bias (ZIMMERMANN et al., 2004)—i.e., distortions caused by the incomplete preservation of historical repository data) are operationalised by PATHCAST Stage 1 (Chapter 4). Software-specific construction patterns are catalogued in Appendix C, Section C.4.

2.3 STOCHASTIC PROCESS MODELING WITH MARKOV CHAINS

Software delivery evolves through discrete states (Backlog, In Progress, Review, Testing, Done) under queuing, rework and variable service times. Discrete-time Markov chains (DTMCs) provide the mathematical formalism for the state-transition layer of PATHCAST (SPEDICATO, 2017; HÄGGSTRÖM, 2002).

Definition 2.3 (Discrete-Time Markov Chain). *A stochastic process $\{X_t\}_{t \in \mathbb{N}}$ on finite state space $S = \{1, \dots, m\}$ is a DTMC if*

$$P(X_{t+1} = j \mid X_t, \dots, X_0) = P(X_{t+1} = j \mid X_t). \quad (2.3)$$

The (time-homogeneous) chain is fully described by a row-stochastic transition matrix $P = [p_{ij}]$, with $p_{ij} = P(X_{t+1} = j \mid X_t = i)$ satisfying $p_{ij} \geq 0$ and $\sum_j p_{ij} = 1$ (Definition 2.4).

Definition 2.4 (Transition Matrix). *P as in Definition 2.3; multi-step probabilities follow the Chapman–Kolmogorov relation $P^{(n)} = P^n$.*

Definition 2.5 (Absorbing Chain and Fundamental Matrix). *For an absorbing chain with t transient and r absorbing states the canonical form is*

$$P = \begin{pmatrix} Q & R \\ \mathbf{0} & I_r \end{pmatrix}, \quad (2.4)$$

$$N = (I_t - Q)^{-1} = \sum_{k=0}^{\infty} Q^k, \quad (2.5)$$

where N_{ij} is the expected number of visits to transient state j starting from transient state i before absorption (Definition 2.6).

Definition 2.6 (Fundamental Matrix). *The matrix N in Equation 2.5 is well-defined because $\rho(Q) < 1$.*

Theorem 2.1 (Properties of Absorbing Chains). *With $N = (I - Q)^{-1}$: (i) $E[T_i] = (N\mathbf{1})_i$; (ii) $\text{Var}(T_i) = (N(2N_{dg} - I)\mathbf{1})_i - E[T_i]^2$; (iii) $B = NR$ is the absorption probability matrix. Proofs in Appendix A.*

Counts c_{ij} of observed $i \rightarrow j$ transitions yield the maximum likelihood estimator

$$\hat{p}_{ij} = \frac{c_{ij}}{\sum_k c_{ik}}; \quad (2.6)$$

Bayesian smoothing under a symmetric Dirichlet prior is derived in Appendix A. Augmenting a DTMC with state-dependent sojourn distributions $\{H_{ij}\}$ yields a semi-Markov process (ROGGE-SOLTI; WESKE, 2015):

Definition 2.7 (Semi-Markov Process). *The pair (DTMC $\{X_n\}$, sojourn distributions $\{H_{ij}\}$) under which the process remains in X_n for a random time drawn from $H_{X_n, X_{n+1}}$ before transitioning.*

The first-order assumption hides history dependence and time-varying transitions; PATHCAST partially mitigates both through optional state-space expansion and ML-based residual correction (Section 2.5), neither of which fully removes violations of the Markov property.

2.4 MONTE CARLO SIMULATION AND PROBABILISTIC FORECASTING METRICS

Monte Carlo simulation estimates statistical properties of intractable systems through repeated random sampling (METROPOLIS; ULAM, 1949; ROBERT; CASELLA, 2004). In PATHCAST, MC generates synthetic process trajectories by alternating sojourn draws from $\hat{F}_s$ and next-state draws from $P[s, \cdot]$ until absorption (Chapter 4, Algorithm 3).

Convergence rests on two classical results. The strong law of large numbers gives, for i.i.d. $X_1, X_2, \dots$ with $E[X] = \mu$,

$$\bar{X}_M = \frac{1}{M} \sum_{m=1}^M X_m \xrightarrow{a.s.} \mu. \quad (2.7)$$

The central limit theorem yields, with $\text{Var}(X) = \sigma^2 < \infty$,

$$\frac{\bar{X}_M - \mu}{\sigma/\sqrt{M}} \xrightarrow{d} \mathcal{N}(0, 1), \quad (2.8)$$

so for any trajectory functional g the MC estimator and confidence interval are

$$\hat{\mu}_g = \frac{1}{M} \sum_{m=1}^M g(\omega^{(m)}), \quad (2.9)$$

$$\text{CI} = \hat{\mu}_g \pm z_{\alpha/2} \cdot \frac{\hat{\sigma}_g}{\sqrt{M}}. \quad (2.10)$$

The Monte Carlo convergence rate remains $O(1/\sqrt{M})$ regardless of dimensionality (although the estimator variance σ^2 may itself grow with dimension), and seed control via the Mersenne Twister makes the simulation reproducible under fixed seeds and implementation settings.

Forecasting metrics. A distributional forecast must be evaluated jointly for sharpness and calibration. The continuous ranked probability score (GNEITING; RAFTERY, 2007) is the operational choice in this thesis:

Definition 2.8 (CRPS). For predicted distribution $\hat{F}$ and observed value y ,

$$\text{CRPS}(\hat{F}, y) = \int_{-\infty}^{\infty} [\hat{F}(t) - \mathbf{1}(t \geq y)]^2 dt. \quad (2.11)$$

CRPS is a strictly proper scoring rule (GNEITING; RAFTERY, 2007) and reduces to mean absolute error when $\hat{F}$ degenerates to a point mass, so distributional forecasters and point baselines remain comparable on a common scale. Calibration is summarised by the average calibration error (ACE), the quantile-coverage calibration measure of Kuleshov et al. (KULESHOV; FENNER; ERMON, 2018),

$$\text{ACE} = \frac{1}{|\mathcal{A}|} \sum_{\alpha \in \mathcal{A}} |\hat{p}_{\alpha} - \alpha|, \quad (2.12)$$

averaged across the quantile checkpoints $\mathcal{A}$. ACE is the unweighted mean of the absolute deviations between nominal and empirical coverage at a fixed set of probability levels; unlike the expected calibration error (ECE) (NAEINI; COOPER; HAUSKRECHT, 2015), which weights bins by their population, ACE assigns equal weight to each checkpoint and is therefore more sensitive to miscalibration in sparsely populated tail quantiles—the regime most relevant to the percentile-based forecasts produced by PATHCAST. Worked CRPS-versus-MAE derivations and numerical examples are in Appendix B.

2.5 MACHINE LEARNING FOR SOFTWARE ENGINEERING

Machine learning is widely applied to defect prediction, effort estimation and code quality assessment (WAHONO, 2015; PACHOULY et al., 2022); process-oriented prediction (throughput, workflow optimisation) remains comparatively underexplored (KOTTI; GALANOPOULOU; SPINELLIS, 2023). PATHCAST formulates outcome prediction as supervised learning: a binary classifier $\hat{f}: \mathbb{R}^d \rightarrow \{0, 1\}$ for on-time versus delayed cases, and a regressor $\hat{f}_T: \mathbb{R}^d \rightarrow \mathbb{R}^+$ for lead time, both trained on a feature vector $\mathbf{x}_c \in \mathbb{R}^d$ derived per case. The on-time/delayed label presupposes a reference target for each case (a due date, an SLA, or a sprint deadline); the concrete delay-threshold strategies used to derive this label when no explicit target is recorded are specified in Chapter 5. The candidate algorithms (Random Forest (BREIMAN, 2001), XGBoost (CHEN; GUESTRIN, 2016), logistic/linear regression baselines, and an LSTM (TAX et al., 2017) as a consolidated predictive-process-monitoring baseline—included

to quantify the gain of process-oriented modelling rather than to compete with more recent Transformer-based architectures—for the sequence comparator) and the training protocol (temporal hold-out, hyperparameter search, SHAP interpretation) are specified in Chapter 5. The working hypothesis is that process-derived features—conformance, rework count, expected remaining transitions, completion probability, path entropy—add predictive value beyond traditional MSR features; the feature catalogue and the PM-only/MSR-only/Combined comparison protocol are in Chapter 4, Table 4.4.

2.6 VARIANT ENTROPY

The trace-variant distribution $V = \{v_1, \dots, v_k\}$ with empirical frequencies $p(v_i)$ summarises the diversity of execution paths in an event log. The Shannon entropy

$$H(V) = - \sum_{v \in V} p(v) \log_2 p(v) \quad (2.13)$$

quantifies path uncertainty in bits; its normalised form $H_{\text{norm}}(V) = H(V)/\log_2 |V| \in [0, 1]$ is a scale-independent process-complexity proxy. Higher H_{norm} signals greater behavioural heterogeneity and is expected to be associated with lower forecast confidence; PATHCAST uses it as both a diagnostic indicator at Stage 2 (Section 4.3.1) and a feature of the ML component.

2.7 CHAPTER SYNTHESIS

The six concepts compose the PATHCAST pipeline as follows: process mining yields the discovered *process backbone*—the sound, block-structured process model (Section 2.1) that captures the dominant control-flow of the SDLC—and the per-case conformance feature; mining of software repositories provides the heterogeneous event sources from which that log is built; the Markov chain layer turns the backbone into a probabilistic state-transition model with closed-form expressions for expected completion time and absorption probabilities; Monte Carlo simulation, paired with CRPS and ACE, produces and evaluates the distributional forecasts; machine learning consumes process-derived features alongside MSR features to refine case-level outcome predictions; and the entropy of variants provides a proxy for process complexity and expected fore-

casting difficulty. The empirical evidence that this integration has not, to the best of our knowledge, been previously proposed for software development is the subject of Chapter 3.

3

Systematic Literature Review

3.1 INTRODUCTION

This chapter presents a systematic literature review (SLR) conducted to map the state of the art at the intersection of process mining, stochastic modeling, and machine learning applied to software process forecasting. The review follows the guidelines established by Kitchenham and Charters (KITCHENHAM; CHARTERS, 2007) for evidence-based software engineering, complemented by the snowballing procedures described by Wohlin (WOHLIN, 2014) (EASE 2014) and the mapping study guidelines of Petersen et al. (PETERSEN; VAKKALANKA; KUZNIARZ, 2015). The primary objective is to identify, classify, and synthesize existing approaches that combine process-aware techniques with predictive modeling for software development outcomes, establishing the research gap that motivates PATHCAST proposed in Chapter 4.

The review addresses three overarching research questions, each decomposed into two sub-questions covering the application landscape of process mining in SDLC contexts, the techniques and prediction methods employed, and the degree of integration between process mining and stochastic forecasting methods.

3.2 REVIEW PROTOCOL

Title/abstract screening was conducted with the assistance of a large language model as the primary filter at the title-and-abstract stage, operating on contexts enriched with baseline-document exemplars. All inclusion decisions were verified manually by the author; the cross-model inter-rater check and the prompts used are reported as replication artefacts in Appendix F.¹

3.2.1 RESEARCH QUESTIONS

The SLR is guided by the following research questions, structured to progressively map the landscape from application scope to technique integration:

Table 3.1: Research questions and expected outputs.

RQ	Sub-RQ	Question	Expected Output
RQ1	RQ1.1	Which SDLC phases have been analyzed with process mining?	Frequency distribution by phase
RQ1	RQ1.2	What event log sources are used and how are they constructed?	Taxonomy of sources and construction methods
RQ2	RQ2.1	Which process discovery algorithms have been applied to SDLC?	Comparative matrix
RQ2	RQ2.2	Which predictive and stochastic techniques have been used?	Technique classification
RQ3	RQ3.1	What is the level of integration between PM and stochastic methods?	Integration maturity assessment
RQ3	RQ3.2	What gaps exist for process-aware forecasting in the SDLC?	Gap analysis matrix

3.2.2 SEARCH STRATEGY

The search strategy combines automated database search with backward and forward snowballing. The automated search targets five digital libraries: IEEE Xplore, ACM Digital Library, Scopus, Web of Science, and SpringerLink.

¹The use of LLMs as primary screeners in systematic literature reviews has been studied by Syriani et al. (SYRIANI; DAVID; KUMAR, 2023) and Khraisha et al. (KHRAISHA et al., 2024).

PICO DECOMPOSITION

The search string construction follows the PICO framework adapted for software engineering, as shown in Table 3.2.

Table 3.2: PICO decomposition for search string construction.

Facet	Role	Concepts
Population	Application domain	Software engineering, SDLC, software process, DevOps, CI/CD
Intervention	Analytical technique	Process mining, Markov chain, Monte Carlo, predictive PM, ML
Comparison	N/A	(omitted)
Outcome	Result	Forecasting, prediction, lead time, cycle time, process model

SEARCH STRINGS

Three complementary search strings are executed to maximize coverage while maintaining precision.

Principal String ($P \wedge I$). Block P captures the software engineering context:

("software engineering" OR "software development" OR "software process" OR "software lifecycle" OR "software life cycle" OR "SDLC" OR "development lifecycle" OR "development life cycle" OR "DevOps" OR "CI/CD" OR "continuous integration" OR "continuous delivery" OR "agile development" OR "agile process" OR "issue tracking" OR "bug tracking" OR "version control" OR "code review" OR "pull request" OR "software project" OR "software repository" OR "software maintenance" OR "software evolution")

Block I captures the process mining and stochastic modeling techniques:

("process mining" OR "process discovery" OR "conformance checking" OR "workflow mining" OR "predictive process monitoring" OR "process prediction" OR "remaining time prediction" OR "process forecasting" OR "event log" OR "process simulation" OR "Monte Carlo simulation" OR "Markov chain" OR "Markov model" OR "stochastic process model" OR "transition probability" OR "absorbing chain" OR "lead time prediction" OR "cycle time prediction")

Machine-learning terms (e.g., "machine learning", "deep learning", "neural network", "LSTM", "transformer") are deliberately *not* added to Block I. The review targets the intersection of process mining and stochastic modeling with software-process forecasting; ML is treated as an *enhancement* dimension rather than a defining technique. Including generic ML terms in the intervention block would flood the retrieval with the large body of ML-for-SE work that is unrelated to process-aware forecasting, degrading precision without improving coverage of the target intersection. Instead, the use of ML is captured *post hoc* during extraction via the *ML Technique* field (Table 3.8), so that ML-enhanced process-forecasting studies are recorded when they surface through the process-mining and forecasting terms already present in Blocks I and O. Predictive-monitoring terms such as "predictive process monitoring" and "process prediction" further ensure that the dominant ML-based PPM literature is retrieved.

Complementary String (MSR $\rightarrow$ Process Modeling). Captures studies using MSR terminology without canonical process mining terms:

*("mining software repositories" OR "software repository mining"
OR "GitHub mining" OR "Jira mining") AND ("process model" OR "Markov"
OR "Monte Carlo" OR "stochastic" OR "lead time" OR "cycle time"
OR "throughput" OR "transition matrix")*

Frontier String (Stochastic $\wedge$ PM $\wedge$ SE). Targets the intersection of stochastic methods with process mining in software contexts:

*("Monte Carlo" OR "Markov chain" OR "stochastic simulation" OR "probabilistic
forecast" OR "transition matrix") AND ("process mining" OR "event
log" OR "workflow mining" OR "process model" OR "business process")
AND ("software development" OR "software engineering" OR "software
process" OR "DevOps" OR "SDLC" OR "issue tracker" OR "commit log"
OR "build pipeline")*

The strings are adapted to each database's syntax and field conventions as summarized in Table 3.3. The eligibility window for publications spans January 1994 to December 2025 (full calendar years), and the searches were executed on 12 April 2026; publications dated 2026 that were already indexed at execution time are therefore retained.

Table 3.3: Database-specific search adaptations and raw retrieval counts.

Database	Search Field	Strings	Filters	Records
Scopus	TITLE-ABS-KEY	3	ar, cp; EN; 1994–2026	2,377
IEEE Xplore	All Metadata	4 (split)	Conf. + Journals; 1994–2026	1,386
Web of Science	TS (Topic)	3	Article, Proc.; EN; 1994–2026	997
SpringerLink	Simplified	8 queries	Chapter, Article; EN	2,884
ACM DL	Abstract	3	Date: 1994–2026	96
Snowballing	–	–	Backward + forward	595
Control set	–	–	Manual	12
Total				8,347

SNOWBALLING PROCEDURE

Backward snowballing examines the reference lists of all included primary studies. Forward snowballing uses Google Scholar citation tracking to identify subsequent works that cite the included studies. Both rounds apply the same inclusion and exclusion criteria defined in Section 3.2.3.

SEARCH STRING VALIDATION

The search strings are validated against a control set of 10 papers that are known to be relevant to the review scope (Table 3.4). The acceptance threshold is $\geq 8/10$ papers found; the validation returned 10/10 (100%, PASS).

3.2.3 INCLUSION AND EXCLUSION CRITERIA

The inclusion criteria are organized in two tiers. Formal criteria (IC1–IC3) filter on publication attributes and are applied during the database search configuration. Content criteria (IC4a–IC4d) require that the study’s research contribution addresses at least one of four specific technique-domain intersections; they are applied during title/abstract and full-text screening. Exclusion criteria (EC1–EC7) are applied cumulatively across both screening phases. Criteria EC1–EC4 are evaluated by the LLM-assisted screener; the access/quality criteria EC5–EC7 are applied programmatically. EC5 (full text inaccessible) is applied after the inter-library-loan recovery pass for Band D/E papers fails to retrieve a copy; EC6 (insufficient metadata) closes papers for which no usable abstract could be recovered after the multi-source enrichment cascade; and EC7 (unre-

Table 3.4: Control papers for search string validation.

#	Paper	Justification	Captured via
V1	Cook & Wolf (1998)	Seminal PM + SE	ACM
V2	Rubin et al. (2007)	PM framework for SW processes	Scopus
V3	Poncin et al. (2011)	PM + MSR intersection	IEEE
V4	Caldeira & Cardoso (2019)	PM + SW process assessment	IEEE
V5	Lemos et al. (2011)	PM for SW process management	IEEE
V6	Whittaker & Thomason (1994)	Markov + SW testing (seminal)	Scopus
V7	Tax et al. (2017)	PPM seminal (BPM Control baseline)	Control
V8	Rogge-Solti & Weske (2015)	Stochastic + PM forecasting	Scopus
V9	Rozinat et al. (2009)	PM → Simulation bridge	Control
V10	Gupta & Sureka (2017)	PM multi-repository in SE	ACM

solvable screening disagreement) closes, conservatively, the cases in which the primary and verifier raters produced an irreconcilable decision that no accessible full text could adjudicate—rather than escalating them to open-ended human arbitration. This three-way split replaces the single umbrella “EC5-extended” code used in earlier drafts and improves the transparency of the PRISMA exclusion accounting.

At least one of IC4a–IC4d must be satisfied for inclusion. Criterion IC4b ensures that seminal works applying Markov chains to software processes—such as Whittaker and Thomason (WHITTAKER; THOMASON, 1994)—are captured despite predating the term “process mining.”

3.2.4 QUALITY ASSESSMENT

Each included study is assessed against a quality checklist adapted from Dybå and Dingsøyr (DYBÅ; DINGSØYR, 2008), with eight binary criteria:

Studies scoring below 4.0/8 (50%) are excluded from the evidence synthesis (Findings F1–F5) but are retained in the PRISMA included set as context. QA8 reflects the centrality of process-model quality in the PATHCAST pipeline. Be-

Table 3.5: Inclusion and exclusion criteria.

Type	ID	Criterion
Inclusion	IC1	Publication period: January 1994 to the search-execution date (searches run on 12 April 2026; full calendar years 1994–2025 plus 2026 records available at execution), from Whittaker & Thomason onward.
	IC2	Language: English.
	IC3	Peer-reviewed journal article, conference paper, or workshop paper (≥ 4 pages).
	IC4a	Applies process mining techniques (discovery, conformance checking, predictive process monitoring) to software development artifacts (commits, issues, pull requests, CI/CD logs, bug trackers, VCS).
	IC4b	Uses stochastic models (Markov chains, Monte Carlo, stochastic Petri nets, absorbing chains) to analyze software development processes or workflows.
	IC4c	Proposes or evaluates forecasting of software process metrics (lead time, cycle time, remaining time, throughput, defect rates) from event logs or repository data.
	IC4d	Mines software repositories (GitHub, Jira, VCS, CI/CD platforms) to discover, analyze, or improve software process models.
Exclusion	EC1	Application domain is exclusively outside software development (healthcare, manufacturing, supply chain) without demonstrated relevance to SDLC processes.
	EC2	Software appears only as the implementation platform; the process being analyzed is not a software development process.
	EC3	Purely algorithmic or theoretical contribution without empirical evaluation in a software development context.
	EC4	Secondary study (SLR, mapping study, survey) not specifically addressing PM or stochastic methods in software engineering.
	EC5	Full text inaccessible despite all retrieval attempts (open access, institutional, ILL, author-provided preprint). Applied programmatically after Band D/E recovery.
	EC6	Insufficient metadata: no usable abstract could be recovered after the multi-source enrichment cascade (Semantic Scholar, OpenAlex, CORE, Crossref), so the study cannot be assessed against the content criteria.
	EC7	Unresolvable screening disagreement: the primary screener and the verifier produced an irreconcilable include/exclude decision and no accessible full text was available to adjudicate it; closed conservatively (biased toward exclusion).

Table 3.6: Quality assessment checklist.

#	Criterion	Score
QA1	Are the research objectives clearly stated?	0/1
QA2	Is the software engineering context described in detail?	0/1
QA3	Is the data source (event log) described reproducibly?	0/1
QA4	Is the PM or stochastic technique formally defined?	0/1
QA5	Are results validated empirically?	0/1
QA6	Are threats to validity discussed?	0/1
QA7	Is the study reproducible (data and/or code available)?	0/1
QA8	Are process model quality metrics reported (fitness, precision)?	0/1

cause QA8 rewards the reporting of conformance metrics (fitness, precision) that are native to process-mining studies but are not expected from purely Markov-, Monte-Carlo-, or forecasting-oriented work, it is scored as “not applicable” (no penalty) for studies whose methodological family does not produce a discovered process model, so that the criterion does not systematically disadvantage non-process-mining contributions.

QA scoring is performed by the same LLM-assisted protocol used for screening, with a strict per-criterion rubric returning a binary score and a brief justification. The full prompt and the per-study scores are part of the replication package (Section 3.6.5). Table 3.7 reports the quality-assessment outcome for the complete confirmed set ($n = 169$). All 169 studies were scored.

Table 3.7: Quality-assessment results for the confirmed included set ($n = 169$; full QA applied). Scores were produced by the LLM-assisted QA protocol (Section 3.2.4).

QA Outcome	Count	Notes
Studies assessed	169	Absolute count and 100.0% of included set
Studies with score $\geq 4/8$	136	Retained for synthesis
Studies with score $< 4/8$	33	Excluded after QA
Mean QA score	4.88	Mean and standard deviation (1.40)
Median QA score	5.00	Median and interquartile range (4.00–6.00)

The 33 studies below the QA threshold are predominantly short workshop papers, position papers, or extended abstracts that fail QA3 (reproducible data source) and QA4 (formally defined technique) simultaneously. They are not

removed from the PRISMA flow; they are retained in the included set as context but are not used in the F1–F5 evidence base reported in Section 3.5.1.

3.2.5 DATA EXTRACTION

For each included study, the attributes in Table 3.8 are extracted.

Table 3.8: Data extraction form.

Field	Description
ID	Unique identifier
Authors, Year, Venue	Bibliographic metadata
SDLC Phase(s)	Lifecycle phases covered
Event Log Source	Jira, GitHub, GitLab, Jenkins, etc.
Event Log Construction	Manual, semi-automatic, or automatic
PM Technique Category	Discovery / Conformance / Enhancement / Prediction
Specific Algorithm(s)	Alpha, Heuristics, Inductive Miner, etc.
Stochastic Method	Markov / Monte Carlo / Stochastic Petri Net / None
ML Technique	RF, XGBoost, LSTM, etc. / None
Prediction Target	Remaining time, outcome, lead time, delay, etc.
Integration Level	L0 None / L1 Sequential / L2 Pipeline / L3 Unified
Validation Type	Case study / Experiment / Industrial / Simulation
Tool/Platform	PM4Py, ProM, Disco, custom, etc.
Dataset Size	Number of events and cases analyzed
Process Model Quality	Fitness, precision, generalization (if reported)

3.3 SEARCH RESULTS AND STUDY SELECTION

3.3.1 DATABASE RETRIEVAL AND DEDUPLICATION

The three search strings executed across five databases returned 7,740 records. Backward and forward snowballing from the control set yielded an additional 595 records. Together with 12 control papers maintained as a validation-only set, the initial corpus totaled 8,347 records. Table 3.9 summarizes the retrieval by source.

Table 3.9: Record retrieval by source.

Source	Raw Records	After Internal Dedup
Scopus (6 queries)	2,377	1,852
SpringerLink (10 exports)	2,884	1,543
IEEE Xplore (manual)	1,386	1,263
Snowballing	595	575
Web of Science (4 queries)	997	514
ACM Digital Library (3 queries)	96	34
Control set (validation only)	12	2
Total	8,347	5,783

Cross-source deduplication removed 2,564 duplicate records (30.7%), yielding 5,781 unique primary-corpus papers plus 2 unique control papers. Deduplication used normalized DOI matching, followed by fuzzy title matching (Levenshtein similarity ≥ 0.9) for records without DOIs.

3.3.2 OPERATIONAL WORKING SET CONSTRUCTION

Screening 5,781 records in a single pass would require substantial resources. A working set of 2,340 papers was therefore constructed from the highest-signal primary queries—those whose query design directly targets the PATHCAST intersection (process mining, stochastic modeling, and software development)—as shown in Table 3.10. All remaining deduplicated records that are not matched to the working set by identifier constitute a *high-recall auxiliary corpus* of 3,807 records, retained as a frozen baseline and used for supplementary robustness analysis rather than as a primary evidence source (Section 3.6.3). Across the two tiers, the protocol therefore subjected $2,340 + 3,807 = 6,147$ deduplicated records to LLM-assisted title/abstract screening, of which the working set is the analytical core. Because the working set was frozen before the high-recall re-deduplication snapshot, approximately 364 working-set papers carry divergent identifiers in that snapshot and therefore also appear, as near-duplicates, in the auxiliary pool; the working set (2,340) and the auxiliary corpus (3,807) consequently overlap by about 364 records. A cross-check confirms that 60 of the 212 first-pass auxiliary inclusions duplicate working-set studies (56 by shared DOI, 4 by title under divergent identifiers); the combined-tier counts reported in this chapter therefore include this overlap and are interpreted as recall upper

bounds rather than fully de-duplicated totals (Section 3.6.3). The substantive findings are anchored on the 169-study working set and are unaffected by this overlap.

Table 3.10: Working set composition.

Database	Queries included	Unique papers
Scopus	scopus_principal, scopus_complementar, scopus_frenteira, scopus_markov_testing, scopus_stochastic_petri, scopus_simulation_bridge	2,197
IEEE Xplore	ieee_manual_01 (PM + SE subgroup)	97
ACM DL	acm_principal, acm_msr	46
Total		2,340

Table 3.10 lists, per database, the query identifiers whose records were promoted into the deduplicated working set, together with the resulting unique-paper counts. These identifiers index the working-set composition and are not a re-count of the strings executed at retrieval time: the number of strings executed against each database (e.g., three for ACM DL) is the one reported in Table 3.3, while the records they returned are consolidated, after cross-string and cross-database deduplication, into the unique counts shown here.

The working set preserves 10/10 control papers when the validation-only control set is included, and 8/10 from primary queries alone—both above the 80% threshold.

3.3.3 TITLE/ABSTRACT SCREENING

All 2,340 working-set papers were screened against IC4a–IC4d and EC1–EC4 using the LLM-assisted protocol introduced in the review-protocol preamble. The screening was executed across five sequential batches of up to 500 papers each via the Anthropic Batches API, producing structured JSON outputs with decision, rationale, and matched criteria for each record.

The include set (111 papers) and the maybe set (775 papers) together form the full-text review queue of 886 papers. The exclude decisions were driven primarily by EC1 (domain outside software development, principally generic business process management) and EC3 (theoretical algorithm proposals without SE evaluation).

Table 3.11: Title/abstract screening results.

Decision	Count	%
Include	111	4.7
Maybe	775	33.1
Exclude	1,454	62.2
Total	2,340	100.0

3.3.4 ELIGIBILITY SCREENING (FULL-TEXT / ENRICHED-ABSTRACT)

This phase is termed *eligibility screening* rather than strictly full-text screening because, although a locally accessible PDF was available for only a subset of papers, every record was assessed against the eligibility criteria using the best available evidence—full text where accessible, and enriched abstract plus meta-data otherwise. The 886 papers forwarded from the T/A phase were subjected to LLM-assisted eligibility screening with a strict binary prompt (decisions limited to include, exclude, or pending; maybe is not permitted at this stage). To maximise the number of papers assessable by the LLM, a multi-source abstract-enrichment cascade was executed prior to screening—combining DOI-based lookups via Semantic Scholar, OpenAlex, CORE, and Crossref, followed by title-based fallback lookups—raising abstract coverage from 16.6% to 72.6% (643 of 886 papers).²

Table 3.12 summarises the outcome.

Among the 717 excluded papers, EC3 (algorithmic or theoretical contribution without SE evaluation, $n = 232$) and EC1 (domain outside software development, $n = 205$) jointly account for the largest share of FT-stage exclusions. EC5 (full text inaccessible despite all retrieval attempts, active post-audit $n = 145$) corresponds to the residual portion of the original 148-paper Band D/E stratum for which no open-access copy, institutional access, or author-provided preprint could be located; these are recorded in a dedicated traceability file and documented in the PRISMA flow as a named exclusion stratum. EC2 (software

²Papers were processed in order of a five-band priority scheme: Band A (91 papers, T/A include ≥ 2 ICs), Band B (20, T/A include 1 IC), Band C (39, T/A maybe + abstract), Band D (152, T/A maybe + IC signal, no abstract), Band E (584, T/A maybe, no IC signal, no abstract). Abstract enrichment and LLM re-screening were applied to Bands A, B, C, and the subset of D/E that gained an abstract after enrichment.

Table 3.12: Full-text screening outcomes (886 papers forwarded from T/A phase).

Outcome	Count	%
Included	169	19.1
Excluded	717	80.9
EC3 (algorithmic, no SE eval.)	(232)	(26.2)
EC1 (non-SE domain)	(205)	(23.1)
EC5 (full text inaccessible, post-audit active stratum)	(145)	(16.4)
EC2 (SW as implementation)	(110)	(12.4)
EC4 (out-of-scope secondary)	(60)	(6.8)
Total	886	100.0

Note: EC counts are non-exclusive — a single paper may trigger multiple ECs when more than one applies (e.g., a non-SE-domain paper whose contribution is purely theoretical). The EC5 count of 145 is the active post-audit stratum after re-screening the original historical EC5 band of 148 papers and reclassifying 3 of them to EC3. The sum of per-EC counts is $232 + 205 + 145 + 110 + 60 = 752$, which exceeds the 717 excluded papers by 35 (corresponding to papers triggering more than one EC). The percentages above are with respect to the 886-paper FT queue.

as implementation platform rather than analyzed process, $n = 110$) and EC4 (out-of-scope secondary studies, $n = 60$) account for the remainder.

Figure 3.1 presents the complete PRISMA 2020 flow diagram (PAGE et al., 2021) of the selection process.

3.4 RESULTS AND SYNTHESIS

This section reports results at multiple corpus tiers. Three headline denominators recur throughout the synthesis; Table 3.13 maps each count to its definition, composition, and analytical role so that percentages quoted below can be traced to a single, explicit evidence base.

3.4.1 OVERVIEW OF INCLUDED STUDIES

The SLR is reported in two complementary tiers: (i) the **working-set tier**, comprising the 2,340-paper operational corpus that was deduplicated and processed through the LLM-assisted T/A and FT screening, yielding **169 confirmed primary studies**; and (ii) the **combined tier**, which adds the 3,807-paper deferred auxiliary corpus after a full T/A and FT pass on the same protocol, yielding 212 first-pass auxiliary inclusions—of which **150 are unique studies** not

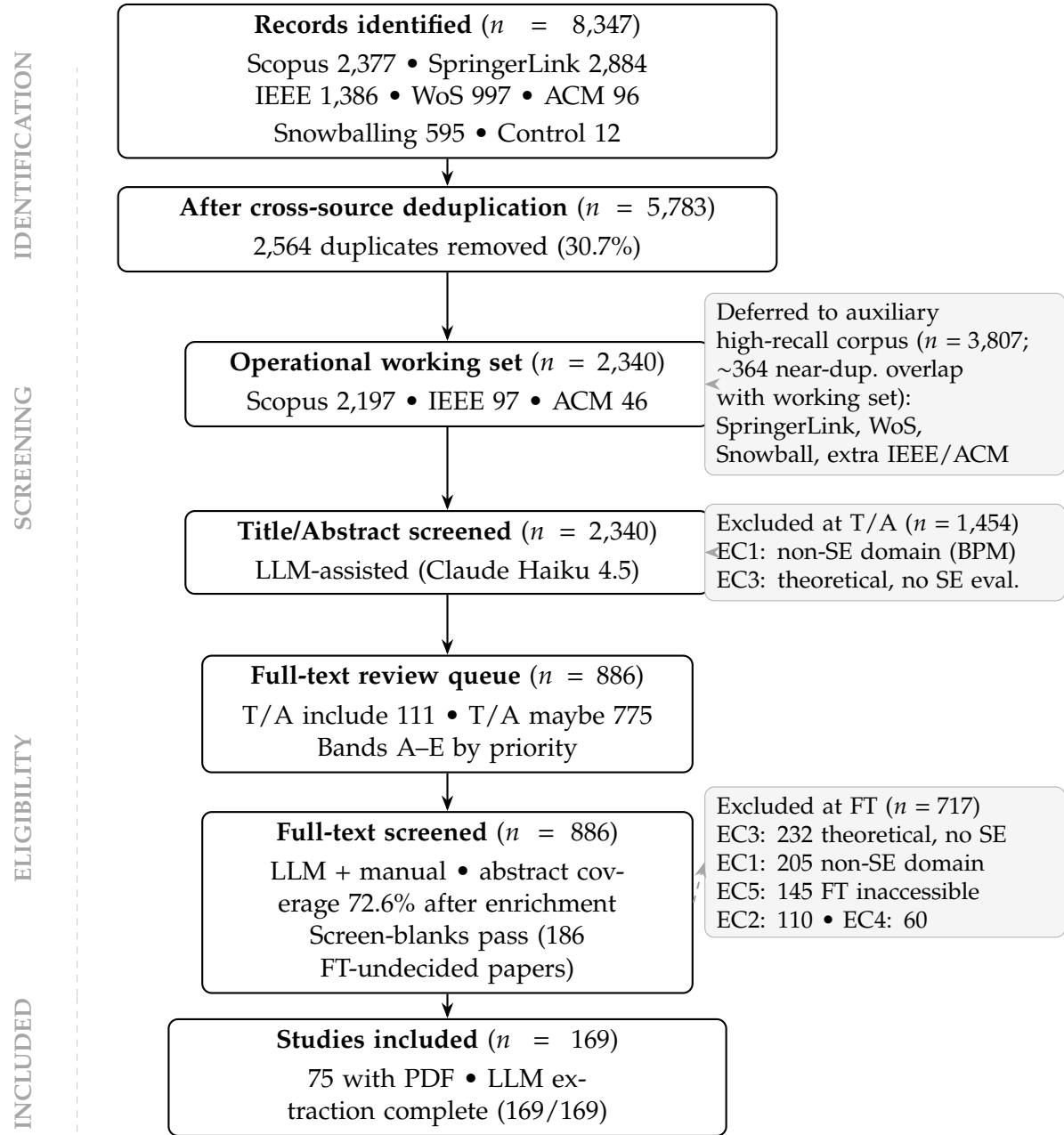


Figure 3.1: PRISMA 2020 flow diagram of the study selection process. Phase labels follow the four-stage structure: Identification, Screening, Eligibility, and Included.

Table 3.13: Provenance of headline denominators in the results synthesis.

<i>n</i>	Tier name	Composition	Primary analyses in this section	Why it coexists with the others
169	Working-set confirmed set	FT-included papers from the 2,340-paper operational corpus after LLM-assisted T/A and FT screening	Narrative spine: year and document-type distributions, inclusion-criterion coverage, structured extrac-tion, technique fam-ilies, evidence-type breakdown, and the working-set branch of Findings F1–F5	Highest-signal query tier; full structured ex-traction (PDF or abstract/metadata); unaffected by cross-tier overlap with the auxiliary corpus
319	De-duplicated combined analytical subset	$169_{ws} + 150_{aux1}^{unique}$: working set plus first-pass auxiliary in-clusions after removing 60 cross-tier near-duplicates and 2 internal duplicates; frozen before the second auxiliary re-FT pass	+ Combined-tier F5 distributions, combination event-log source technique tables, and QA-filtered combined synthesis (260/319 above the $\geq 4/8$ threshold)	F1– Recall-bounded expansion of the same protocol over the 3,807-paper auxiliary corpus; cross-tier duplicates reconciled so com-bined percentages are not inflated by papers counted twice
381	Raw combined tier (pre de-duplication)	$169_{ws} + 212_{aux1}$: working set plus all first-pass auxiliary FT-includes, before cross-tier reconcilia-tion	Not used as a synthe-sis denominator; cited in Section 3.6.3 and the conclusion-validity discussion to contrast the auxiliary screening yield (315/381 would pass QA) with the de-duplicated analytical subset reported in Ta-ble 3.22 (260/319)	Arithmetic sum before cross-tier reconciliation; the 62 papers removed to obtain 319 (60 cross-tier + 2 inter-nal duplicates) are named explicitly so that combined-tier percentages are not silently inflated

Note: A fourth headline count, **341** ($169 + 150 + 22$), denotes the final de-duplicated combined evidence base after a second auxiliary re-FT pass over enriched abstracts; it extends the recall bound but is not the basis of the detailed combined-tier tables below, which refer to the 319-study first-pass subset unless stated otherwise.

already present in the working set, after removing 60 cross-tier near-duplicates and 2 internal duplicates—plus 22 further unique confirmations from a second auxiliary pass over enriched abstracts, for a de-duplicated combined total of **341 confirmed primary studies**. The working-set tier is the narrative spine of this chapter; the combined-tier numbers are reported in parallel for the findings F1–F5 and provide the recall-bounded view of the SLR. Unless stated otherwise, detailed combined-tier distributions refer to the **319-study de-duplicated analytical subset** (working set plus 150 unique first-pass auxiliary studies), frozen before the second auxiliary re-FT pass. The 169-study set spans three decades (1995–2026), with a clear acceleration from 2014 onward (Figure 3.2). Seventy-one percent of confirmed papers (working-set tier) were published between 2015 and 2026, reflecting the maturation of open-source process mining toolkits (PM4Py, ProM) and the broader adoption of deep learning frameworks in software engineering research.

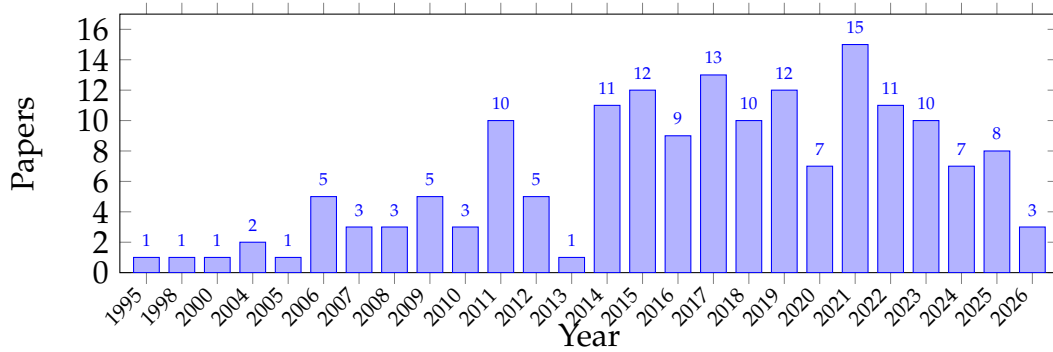


Figure 3.2: Distribution of confirmed studies by publication year ($n = 169$).

By document type, the majority are conference papers and journal articles. By source database, Scopus is the dominant contributor, followed by IEEE Xplore and ACM DL. Data extraction was completed for all 169 studies using a structured LLM-assisted pipeline (*claude-haiku-4-5-20251001*) operating on locally available PDFs (75/169, of which 73 were operationally coded in the triangulation subset) or abstract/title metadata (94/169).

3.4.2 INCLUSION CRITERION COVERAGE

Figure 3.3 shows the distribution of inclusion criteria among the 169 confirmed papers (criteria are non-exclusive; a paper may satisfy multiple).

IC1/IC4a (process mining applied to SE artifacts) is the dominant criterion,

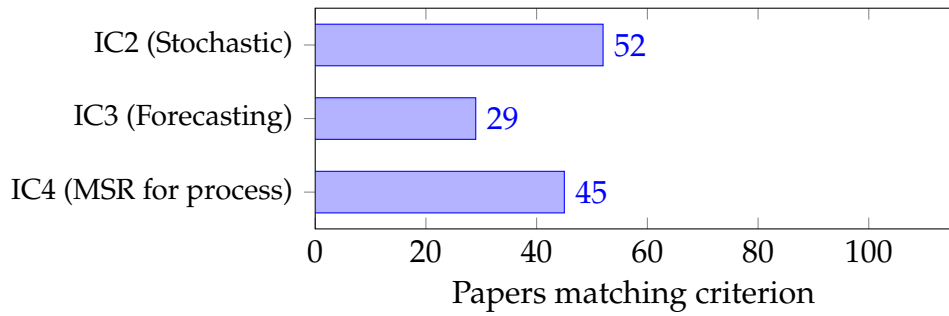


Figure 3.3: Inclusion criteria matched by the confirmed set ($n = 169$; non-exclusive; IC labels map to IC4a–IC4d in the protocol).

matched by 121 papers (71.6%). IC4/IC4d (repository mining for process modeling) is second at 45 papers (26.6%), reflecting the MSR community’s overlapping contributions. IC2/IC4b (stochastic modeling) appears in 52 papers (30.8%) and IC3/IC4c (forecasting of process metrics) in 29 papers (17.2%). Notably, 68 papers (40.2%) match IC1 only, without stochastic or forecasting components, providing direct empirical evidence for the “Descriptive Ceiling” identified in Finding F1 below.

3.4.3 PREVALENT EXCLUSION PATTERNS

Among the 717 FT-excluded papers (Table 3.12), the largest exclusion reason is EC3 (algorithmic or theoretical contribution without software-engineering evaluation, $n = 232$), indicating a substantial body of process mining and stochastic modeling research that has not been empirically applied to software engineering contexts. EC1 (domain exclusively outside software development, $n = 205$) is the second-largest reason and consists primarily of generic business process management (BPM) studies that use the same terminology as the search strings but apply techniques to organizational processes unrelated to SDLC. EC5 (full text inaccessible, active post-audit $n = 145$; original historical stratum $n = 148$) closes the group of large strata, followed by EC2 (software as implementation platform rather than analyzed process, $n = 110$) and EC4 (out-of-scope secondary study, $n = 60$). EC counts are non-exclusive: the sum of per-EC counts ($232 + 205 + 145 + 110 + 60 = 752$) exceeds the 717 excluded papers by 35, corresponding to papers that triggered more than one EC simultaneously. The dominance of EC3 and EC1 directly supports Findings F3 and F4 below: process mining and stochastic methods are widely studied in the

abstract or in non-SE domains, but their application to SDLC processes remains comparatively narrow.

3.4.4 FULL-TEXT EXTRACTION AND PDF-BASED TRIANGULATION

Structured data extraction was performed on the full 169-study set using the extraction template defined in Table 3.8. For each study, the LLM extracted the research question, study type, research contribution, PM technique, stochastic technique, software artifact, software process, data source, dataset availability, tool used, main finding, limitations, and replication-package status. Title-and-abstract metadata was used as the primary input for studies without locally available PDFs (94/169); for the remaining 75/169 studies a locally accessible PDF was available and used to triangulate against the LLM-extracted fields. Of these 75 PDFs, 73 were fully coded in the operational triangulation spreadsheet used for the figures and positioning tables below; the remaining 2 PDFs were excluded from operational coding because they were corrupted or duplicate exports with insufficient extractable content, and their studies are retained in the 169-study set on the basis of their abstract/metadata extraction.

The extraction was run with the LLM configured for deterministic output (`temperature = 0`, fixed model `claude-haiku-4-5-20251001`, JSON-schema-constrained responses), and each extracted field carried an explicit `not_reported` sentinel. Fields that require full-text inspection and that were unavailable in title/abstract metadata—most notably dataset size, process-model quality scores, validation type, and tool/platform for the 94 studies without a locally available PDF—were coded as missing rather than imputed, and are excluded from any analysis that depends on them; the corresponding tables report the effective n for each variable. For the 75-study PDF subset, the LLM-extracted fields were triangulated against the source PDF, and the observed field-level disagreement rate is reported in the replication package (Section 3.6.5).

The 73-paper PDF triangulation subset is therefore not a separate selection stage and does not override the PRISMA counts. It is used in this chapter as a verification layer for the LLM-extracted fields used in Findings F1–F5.

Figure 3.4 summarizes the operational triage profile of the 73 coded PDFs (54 confirm-include, 13 confirm-include with caveats, 6 confirm-exclude after full-text inspection); the formal selection decisions reported in the PRISMA flow (Figure 3.1) are unchanged.

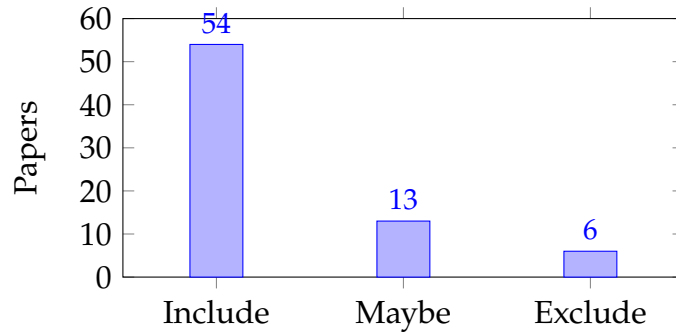


Figure 3.4: Operational decision distribution in the 73-paper PDF triangulation subset.

3.4.5 RQ1: APPLICATION LANDSCAPE

RQ1.1: SDLC PHASES COVERED

The structured extraction over the full 169-study set confirms a strongly uneven coverage of the SDLC. **Development activities** (coding, commits, pull requests, repository evolution) dominate at 133 papers (78.7%); **quality-related activities** (testing, defect handling, code review, process evaluation) appear in 74 papers (43.8%); **maintenance** appears in 38 papers (22.5%); **requirements engineering** in only 22 papers (13.0%); and **delivery/DevOps contexts** (CI/CD, build and release pipelines) in just 18 papers (10.7%). **Design and architecture** were not identified as the primary focus of any of the confirmed studies, and only 3 papers (1.8%) center the analysis on project management.

Two conclusions follow. First, the literature is concentrated on phases where digital traces are abundant and already timestamped, such as commits, issues, test events, and pipeline executions. Second, the early SDLC phases remain structurally underrepresented, which weakens the ability of current approaches to support end-to-end forecasting from requirements to delivery. For PATH-CAST, this supports a pragmatic focus on operational phases with reliable event data while leaving room for future extensions toward requirements and design logs.

RQ1.2: EVENT LOG SOURCES AND CONSTRUCTION

Across the 169-study set, event logs are constructed primarily from four source families: **version-control repositories** (Git/GitHub/SVN commits) appear in 84 papers (49.7%); **issue tracking systems** (especially Jira and GitHub

Issues) in 37 papers (21.9%); **IDE interaction logs** in 22 papers (13.0%); and **CI/CD or build pipelines** in 15 papers (8.9%). The common pattern is not a native event log but rather a *derived* event log assembled from repository metadata, commit histories, issue transitions, and (where available) pipeline records.

The dominant construction problem is case correlation across tools. Studies frequently analyze one source in isolation or perform ad hoc joins between issues and commits, which limits reproducibility. This confirms that event-log construction is not a preparatory detail but a central methodological layer in software process mining. The finding directly justifies Stage 1 of PATHCAST as an explicit transformation pipeline rather than an implicit data-cleaning step.

A non-trivial fraction of studies report data sources only in coarse terms (*repository data, VCS history, open-source projects*) without naming specific projects, time windows, or extraction queries. Across the 169-study set, only 25 papers (14.8%) explicitly state that their dataset is publicly available and only 4 papers (2.4%) declare a replication package. This itself is evidence of incomplete reporting in the literature and provides direct empirical grounding for Finding F2 below.

A complementary body of work constructs Markov models directly from GitHub repository traces, bypassing explicit process discovery in favour of manual state-space definition. Jo et al. (JO; KWON; KWON, 2023) modelled developer activities in GitHub repositories as Discrete-Time Markov Chains (DTMCs), finding that commits account for 62.73% of long-term activity, and employed probabilistic temporal logic to extract collaboration patterns relevant to project management. This approach was extended in (JO; KWON; KWON, 2024), where Probabilistic Model Checking with Computation Tree Logic was applied across five repositories to characterise temporal dynamics, state-transition probabilities, and key behavioural drivers, yielding transparent and explainable project forecasts. Ortu et al. (ORTU et al., 2023) complemented these structural analyses by combining Markov chains with social network analysis to identify fault-insertion and fault-fixing behavioural patterns across Apache Software Foundation projects, demonstrating that developer expertise significantly moderates defect introduction likelihood. Collectively, these studies confirm that SDLC event logs extracted from public repositories are sufficiently rich to support state-transition modelling, satisfying a foundational prerequisite for PATHCAST Stage 1. However, the event-to-state mapping adopted across these works remains unstandardised and ad hoc, representing a methodological gap that

PATHCAST’s process mining discovery stage aims to address systematically.

3.4.6 RQ2: TECHNIQUES AND PREDICTION

RQ2.1: DISCOVERY ALGORITHMS APPLIED TO SDLC

The canonical origin of process mining applied to software development contexts can be traced directly to the foundational contributions of Cook and Wolf in the mid-1990s. In their seminal work, (COOK; WOLF, 1995) introduced three automated process discovery methods—algorithmic grammar inference, Markov models, and neural networks—demonstrating that finite-state machine representations of software processes could be extracted automatically from event traces, substantially lowering the barrier to formal process modeling. This line of inquiry was consolidated in (COOK; WOLF, 1998), which refined the event-based discovery methodology and established the conceptual vocabulary of inferring workflow structures from historical execution data, a paradigm that remains central to the field today. Notably, both contributions employed Markov-chain representations as a *modeling and inference* mechanism for discovered process structure—not as a probabilistic forecasting layer in the modern sense—thereby anticipating, rather than implementing, the predictive stochastic extensions that would later characterise mature process mining frameworks. The translation of these principles into an explicit, named framework for software engineering contexts was accomplished by (RUBIN et al., 2007), who applied the ProM toolkit to source code management logs and demonstrated multi-perspective analysis spanning control-flow, informational, and organisational dimensions. Together, these three works constitute the seminal arc of IC1, establishing automated discovery from SDLC event logs as both technically feasible and analytically productive, and providing PATHCAST with its foundational methodological lineage.

The complete extracted set confirms that **process mining remains the dominant analytical family**: 121 of 169 studies (71.6%) match IC1 in the form of discovery, conformance checking, or process-oriented repository analysis. At the technique level, the most frequently named PM algorithms across the 169 studies are the **inductive miner** (51 papers), **alpha miner** (45 papers), **conformance checking** (30 papers), and **heuristic miner** (22 papers); 11 papers use social network analysis on developer interactions. The literature is markedly

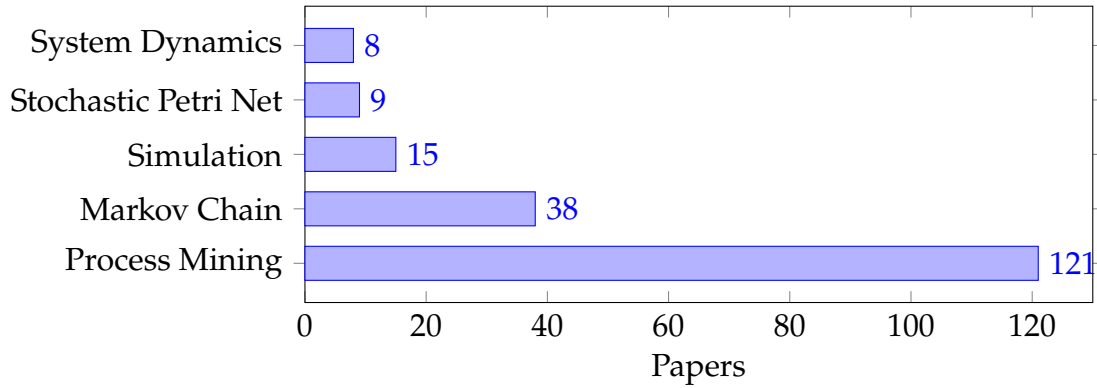


Figure 3.5: Main technique families identified in the 169-study confirmed set. Counts are non-exclusive: a paper may use multiple technique families.

richer in *process representation* than in probabilistic forecasting: only 29 papers (17.2%) match IC3 (forecasting of process metrics), and the dominant contribution type is discovery or conformance checking rather than predictive analytics. This is the transition that PATHCAST targets: moving from discovered process structures to state-based probabilistic forecasting.

The stochastic-technique distribution at the same 169-study level shows the same fragmentation. Only 52 of 169 studies (30.8%) name an explicit stochastic technique: 38 papers (22.5%) use Markov chains, 15 (8.9%) use Monte Carlo simulation, 9 (5.3%) use stochastic Petri nets, and 8 (4.7%) use system dynamics. The remaining studies either do not adopt a stochastic component (110 papers) or place stochastic methods in an auxiliary role (*other*, 29 papers). The forecasting literature thus remains fragmented across adjacent technique families rather than consolidated around a single process-aware predictive paradigm, as shown in Figure 3.5.

RQ2.2: PREDICTIVE AND STOCHASTIC TECHNIQUES

Three distinct sub-families of stochastic techniques emerge from the confirmed literature. The first, **Markov-chain-based prediction**, encompasses models that represent software processes as discrete state transitions, including reinforcement learning agents grounded in Markov Decision Processes for pull-request outcome prediction (JOSHI; KAHANI, 2024), probabilistic risk quantification from repository data (JEON; KIM; IN, 2015), defect removal effectiveness under code churn (TIAN; NOORE, 2005), agile state-space planning and tracking (TYAGI et al., 2021), release-time forecasting via generalised re-

liability models (WASHIZAKI; HONDA; FUKAZAWA, 2015), and Stochastic Automata Networks for Waterfall team estimation (MASSITELA et al., 2018). The second sub-family, **stochastic Petri nets**, models concurrency and asynchrony in DevOps and CI/CD pipelines through non-Markovian specifications (MESMIA; ESCHEIKH; BARKAOUI, 2021) and sensitivity analyses that identify bottleneck stages governing delivery probability (BHADRA, 2022; BHADRA et al., 2023). The third sub-family, **Monte Carlo simulation**, drives project-level forecasting in hybrid System Dynamics models (LI et al., 2020), agile risk assessment frameworks (LUNESU et al., 2021), cycle-time economic analysis (MAGENNIS, 2015), and the PRIMAD digital-twin architecture for SDLC (GUINEA-CABRERA; HOLGADO-TERRIZA; PICO-VALENCIA, 2025).

Despite their methodological diversity, all three sub-families share a common parametrisation strategy that constitutes a structural limitation. Transition probabilities, firing rates, and distributional parameters are invariably derived from aggregate project statistics—historical defect counts, phase-level duration averages, or summary velocity metrics—rather than from transition frequencies extracted through systematic process mining of fine-grained event logs (NAFREEN et al., 2020; BHADRA, 2022; LI et al., 2020). Consequently, model states remain anchored to coarse lifecycle phases and cannot reflect the richer, empirically grounded activity sequences that event logs encode. This observation constitutes **Finding F3**: *stochastic forecasting models for software development are parametrised from aggregate statistics rather than from mined process structures, confining their state spaces to coarse lifecycle granularity and decoupling model topology from observed execution behaviour.*

Three studies exemplify the productive convergence of process mining and stochastic modelling at the component level. Incerto (INCERTO; VANDIN; AHRABI, 2025) demonstrated that variable-length Markov chains outperform eighteen existing stochastic conformance-checking techniques across ten of eleven benchmark datasets, establishing that higher-order sequential dependencies in event logs can be captured without sacrificing computational tractability. López-Pintado (LÓPEZ-PINTADO; DUMAS, 2023) extended process-simulation fidelity by replacing deterministic resource-availability calendars with probabilistic counterparts discovered directly from event logs, showing that stochastic resource representations more accurately replicate empirical cycle-time distributions. Jalote (JALOTE; KAMMA, 2021) took a complementary perspective, modelling programmer task sequences as first-order Markov chains derived

from IDE interaction logs and demonstrating that transferring the process patterns of high-productivity developers to average-productivity peers yields measurable productivity gains. Together, these contributions confirm that stochastic extensions of process models improve both explanatory and predictive validity; however, each study confines its stochastic layer to a single analytical concern without integrating multiple stages into a unified forecasting pipeline. These three results also map directly onto design commitments of the method proposed in Chapter 4: the variable-length-Markov evidence of Incerto (INCERTO; VANDIN; AHRABI, 2025) motivates an explicit *empirical test of Markov order* (rather than an assumed first-order chain); the log-discovered resource calendars of López-Pintado (LÓPEZ-PINTADO; DUMAS, 2023) indicate a concrete path to relax the case-independence assumption; and the non-Poisson, per-activity sojourn distributions reported for PRIMAD (GUINEA-CABRERA; HOLGADO-TERRIZA; PICO-VALENCIA, 2025) (with empirical inter-event times far from exponential) justify a fully nonparametric sojourn estimator with explicit heavy-tail handling rather than a fitted parametric family.

The closest published work to a fully integrated multi-stage architecture is PRIMAD, proposed by Guinea-Cabrera (GUINEA-CABRERA; HOLGADO-TERRIZA; PICO-VALENCIA, 2025), which orchestrates pm4py-based process discovery, Akka actor-system simulation, and LightGBM supervised learning within an evolutionary Digital Twin framework for the SDLC. Nevertheless, PRIMAD remains at what may be characterised as Level 2 integration: validation is reported per component rather than across the assembled pipeline, no formal inter-stage contracts govern the handoffs between discovery, simulation, and learning modules, and the architecture incorporates no dedicated Monte Carlo forecasting layer capable of propagating uncertainty estimates end-to-end. The present review therefore concludes that the Level 3 integration niche—a formally contracted, end-to-end pipeline combining process discovery, absorbing Markov chain construction, Monte Carlo simulation, and residual correction—remains unoccupied in the extant literature.

Studies that combine process mining with forecasting objectives ($IC1 \cap IC3$) tend to approach prediction through *process enhancement* rather than through explicit stochastic modelling. Gupta (GUPTA, 2017) demonstrated that combining inductive miner discovery with Markov chain-based predictive analytics enables identification of recurring failure patterns in software maintenance workflows. Pourbafrani et al. (POURBAFRANI; JIAO; AALST, 2021) presented

SIMPT, a tool that extracts process trees with time parameters from event logs and enables interactive what-if simulation; its successor work (BULIGA et al., 2025) further integrates generative AI with business process simulation to produce conformance-preserving what-if scenarios. Caldeira et al. (CALDEIRA et al., 2022) mine IDE interaction logs and show that process complexity metrics derived from ProM correlate moderately ($r \approx 0.43$) with software cyclomatic complexity, achieving predictive accuracies exceeding 92%, demonstrating the feasibility of process-derived features for product-quality prediction.

Taken together, this cluster confirms that PM and prediction are being connected in the SE literature, but always at the level of sequential application—PM output feeds a downstream model—rather than through a unified pipeline with formal contracts. Notably, none of these systems can express, as a first-class output, the probability that a given process instance will reach completion within k activity cycles—the precise forecasting primitive that distinguishes absorbing Markov chain formulations from ad hoc simulation runs. This gap motivates PATHCAST’s design, wherein the mined process structure feeds directly into an absorbing Markov chain whose transient-state absorption probabilities constitute a formally defined, queryable forecasting interface.

3.4.7 RQ3: INTEGRATION AND GAPS

RQ3.1: PM AND STOCHASTIC INTEGRATION LEVEL

Mapping the confirmed papers onto the L0–L3 integration scale reveals a clear stratification. The largest cluster, comprising papers that touch IC2 and IC3 without IC1—including stochastic Petri net models of CI/CD pipelines (BHADRA, 2022; BHADRA et al., 2023), Markov-based project estimation (MASSITELA et al., 2018; TYAGI et al., 2021), and Monte Carlo risk simulation (LUNESU et al., 2021; MAGENNIS, 2015)—operates almost entirely at **L1**, applying stochastic modelling to software processes without any process discovery layer. Papers spanning IC1 and IC2, such as the VLMC conformance checker of (INCERTO; VANDIN; AHRABI, 2025) and the probabilistic resource-calendar discoverer of (LÓPEZ-PINTADO; DUMAS, 2023), reach **L2** by coupling discovered process structure with stochastic reasoning, yet neither closes the loop toward distributional schedule forecasting. Papers combining IC1 with IC3, such as (GUPTA, 2017) and (CALDEIRA et al., 2022), use process mining descrip-

tively to assess workflows but lack the stochastic machinery required for Monte Carlo distributional forecasts. The PRIMAD framework (GUINEA-CABRERA; HOLGADO-TERRIZA; PICO-VALENCIA, 2025) is the closest antecedent, assembling elements from all three ICs—event-log discovery, simulation, and ML prediction—but its evaluation remains component-wise, placing it at L2 rather than a fully validated end-to-end pipeline. Notably, no paper in the reviewed corpus instantiates an L3 architecture in which process mining discovery feeds an absorbing Markov chain, whose transient distributions drive Monte Carlo forecasting, with residuals corrected by a learned model, all validated on SDLC event logs. PATHCAST is proposed to fill this gap; to the best of our knowledge, no study identified in this review implements such an L3 system.

Quantitative evidence over the full 169-study set confirms the pattern. Only 1 of 169 papers (0.6%) jointly satisfies IC1, IC2, and IC3—that is, combines process mining, stochastic modeling, and forecasting in the same study—and even that single paper does not specify a formal end-to-end pipeline contract. Pairwise convergence is also low: $IC1 \cap IC2$ (PM and stochastic) appears in 10 papers (5.9%), $IC1 \cap IC3$ (PM and forecasting) in 7 papers (4.1%), and $IC2 \cap IC3$ (stochastic and forecasting) in 18 papers (10.7%). **No paper** in the reviewed corpus implements an L3 architecture. The literature has many partial bridges but no end-to-end process-aware forecasting architecture for SDLC data. The operational coding of the 73-PDF triangulation subset is consistent with the full-set numbers and is summarized in Figures 3.6 and 3.7.

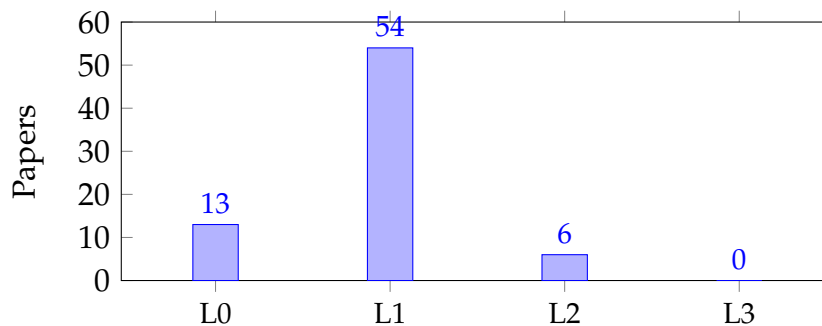


Figure 3.6: Operational integration-level distribution (L0–L3) in the 73-paper PDF triangulation subset.

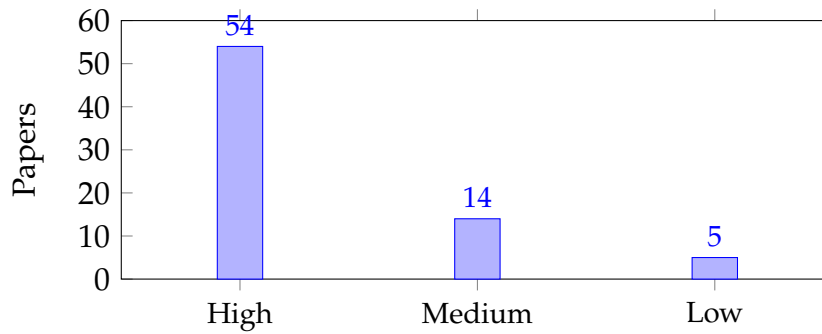


Figure 3.7: Reading-priority distribution (High/Medium/Low) in the 73-paper PDF triangulation subset.

RQ3.2: GAP ANALYSIS

Three structural observations emerge from the 169-study extraction. First, the field is **descriptively strong but forecastively weak**: 121 papers (71.6%) discover or assess process behavior, but only 29 (17.2%) estimate future paths or probabilistic completion distributions. Second, **evidence quality is uneven**: of the 169 confirmed studies, 69 (40.8%) are case studies, 34 (20.1%) are tool papers, 33 (19.5%) are theoretical, 13 (7.7%) are simulation-only, 10 (5.9%) are controlled experiments, 5 (3.0%) are surveys, and 5 (3.0%) fall in residual categories (mixed, framework, exploratory; sum: $69 + 34 + 33 + 13 + 10 + 5 + 5 = 169$). The literature is therefore method-heavy and applied-evidence light. Third, **reproducibility is poor**: only 25 of 169 papers (14.8%) declare their dataset publicly, and only 4 (2.4%) declare a replication package. The item-level QA profile (Table 3.14) confirms and sharpens this pattern: QA7 (data/code availability) is satisfied by only 6.5% of studies (11/169) and is the single weakest criterion across the included set; **validity threats** are even more revealing—QA6 (threats to validity discussed) is satisfied by only 22.5% (38/169), so more than three quarters of the corpus reports no validity discussion at all. Tooling is largely unreported as well: 74% of studies (125/169) name no process-mining tool, and among those that do, ProM dominates; and 39% (66/169) do not even specify whether their dataset is industrial, open-source, academic, or synthetic. These three item-level deficits—reproducibility (QA7), validity reflection (QA6), and tooling transparency—are the concrete face of the maturity gap that motivates PATHCAST’s emphasis on a determinism-and-replication-first design.

Taken together, these patterns explain why PATHCAST requires both a formal pipeline and an evaluation strategy grounded in real SDLC traces. The

Table 3.14: Item-level quality-assessment profile of the confirmed included set ($n = 169$). Percentages are the share of studies scoring 1 on each binary criterion. Scores were produced by the LLM-assisted QA protocol described in Section 3.2.4 and archived in the replication package (Section 3.6.5).

Item	Criterion	Pass	% Pass
QA1	Research objectives clearly stated	168/169	99.4%
QA2	Software-engineering context described	164/169	97.0%
QA3	Data source described reproducibly	108/169	63.9%
QA4	PM/stochastic technique formally defined	139/169	82.2%
QA5	Results validated empirically	138/169	81.7%
QA6	Threats to validity discussed	38/169	22.5%
QA7	Reproducible (data and/or code)	11/169	6.5%
QA8	Process-model quality metrics reported	58/169	34.3%
Mean QA total		4.88 / 8 (SD 1.40)	

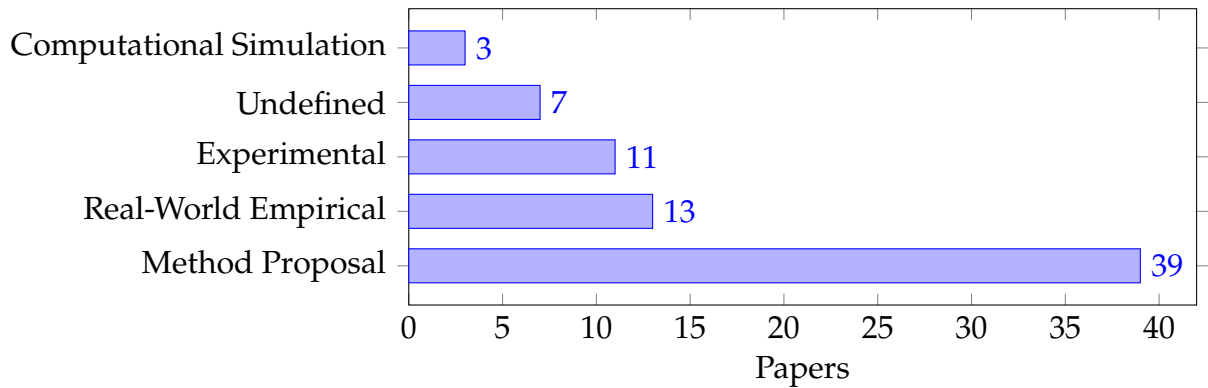


Figure 3.8: Evidence-type distribution in the 73-paper PDF triangulation subset.

gap is not the absence of isolated techniques, but the absence of robust, end-to-end, process-aware forecasting workflows validated on heterogeneous software engineering data. The evidence-type distribution observed in the 73-PDF triangulation subset (Figure 3.8) is consistent with the full-set characterization above and is reported here as an analytical view, not as a separate selection layer.

3.5 DISCUSSION

3.5.1 RESEARCH GAP IDENTIFICATION

The analysis of 2,340 screened records and 169 confirmed primary studies, together with the criterion coverage patterns, supports five key findings:

F1: Descriptive Ceiling. In the **working-set tier** (169 papers), 121 (71.6%) match IC1/IC4a (process mining in SE) but only 29 (17.2%) advance to IC3/IC4c (forecasting of process metrics). In the **de-duplicated combined analytical subset** (319 papers, working-set + 150 unique auxiliary studies), the proportion shifts because the auxiliary corpus pulled in more stochastic-only and forecasting-only studies that did not employ explicit process mining: 162 (50.8%) match IC1, 100 (31.3%) match IC3, and 78 (24.5%) match IC1 alone. In both tiers the dominant contribution type is discovery or conformance checking, not predictive analytics or distributional forecasting; the descriptive ceiling identified in the working-set tier persists in the combined tier.

F2: Event Log Construction Challenge. The reproducibility deficit is uniform across both tiers. Working-set tier: only 25 of 169 papers (14.8%) declare the dataset as publicly available and only 4 (2.4%) declare any replication package. De-duplicated combined analytical subset: 44 of 319 papers (13.8%) declare a public dataset and 9 (2.8%) declare a replication package (3 “yes” + 6 “partial”). The majority of studies that mention data collection procedures rely on ad hoc joins between version-control commits (115 of 319 papers, 36.1%), issue trackers (39, 12.2%), Jira (41, 12.9%), GitHub (42, 13.2%), and IDE interaction logs (32, 10.0%) rather than on a reusable construction protocol with explicit case-correlation semantics. The absence of standardized construction frameworks limits reproducibility and cross-study comparability and is the most common cause of QA3 (data source described reproducibly) failures observed in the QA scoring of the working-set tier.

F3: Stochastic Integration Gap. Working-set tier: 52 of 169 papers (30.8%) apply stochastic models (IC2/IC4b); 38 (22.5%) Markov chains, 15 (8.9%) Monte Carlo, 9 (5.3%) stochastic Petri nets, 8 (4.7%) system dynamics; $IC2 \cap IC3$ covers 18 papers (10.7%); $IC1 \cap IC2$ covers 10 (5.9%). De-duplicated combined analytical subset: the auxiliary corpus contributes a substantial number of stochastic-only papers not previously surfaced by the working-set queries; 158 of 319 (49.5%) match IC2 (with 85 Monte Carlo, 58 Markov, 62 system dynamics, 15 SPN), 81 (25.4%) combine IC2 with IC3 (forecasting), and 13 (4.1%) combine IC2 with IC1 (process mining). Markov chains and Monte Carlo simulation are therefore used extensively in BPM

and software reliability engineering, and the auxiliary expansion confirms a broader stochastic literature than the working-set tier alone suggested; nevertheless, the integration with process mining for SDLC forecasting remains minimal in both tiers (5.9% and 4.1% respectively for $IC1 \cap IC2$). EC3 being the largest exclusion reason at FT in the working-set tier ($n = 232$, 26.2% of the 886-paper FT queue) reinforces the point: the stochastic modeling community has not converged on applying its techniques to software development process data.

F4: Pipeline Fragmentation. Studies apply techniques in isolation. The formal composition of a $PM \rightarrow \text{Markov} \rightarrow MC \rightarrow ML$ pipeline with inter-stage data contracts was not identified in the reviewed literature for SDLC contexts. The IC-combination analysis is robust across both tiers. Working-set tier (169 papers): the joint convergence $IC1 \cap IC2 \cap IC3$ appears in only 1 paper (0.6%), and that single paper does not specify a formal end-to-end pipeline contract; pairwise convergences are $IC1 \cap IC2$: 10 (5.9%), $IC1 \cap IC3$: 7 (4.1%), $IC2 \cap IC3$: 18 (10.7%). De-duplicated combined analytical subset (319 papers): $IC1 \cap IC2 \cap IC3$ remains 1 paper (0.3%), with pairwise convergences $IC1 \cap IC2$: 13 (4.1%), $IC1 \cap IC3$: 10 (3.1%), $IC2 \cap IC3$: 81 (25.4%). The auxiliary expansion increased the $IC2 \cap IC3$ cluster substantially without producing any new $IC1 \cap IC2 \cap IC3$ paper, which is independent empirical confirmation that the gap is not an artefact of the working-set queries but a structural property of the literature. The closest antecedent in the corpus is PRIMAD (GUINEA-CABRERA; HOLGADO-TERRIZA; PICO-VALENCIA, 2025), which assembles discovery, simulation, and learning components but validates each independently and leaves the unified pipeline unevaluated. Even when stochastic components are present, validation usually stops at process analysis or simulation without a downstream learning component.

PRIMAD is the only paper in the corpus that simultaneously satisfies $IC1 \cap IC2 \cap IC3$. Its scope is explicitly SDLC-oriented: it builds a digital twin from issue-tracking flows in JIRA to analyze and improve value streams in software organizations. Methodologically, it combines empirical process discovery (pm4py), a Python-based simulation core, an Akka actor system to model issues and resources, and supervised machine learning for context-aware prediction. The reported results are preliminary and

component-wise rather than end-to-end: the paper shows promising independent results for the process-mining, simulation, and prediction components, but it does not yet validate a formally coupled PM → stochastic simulation → ML pipeline as a unified operational architecture. This is precisely why PRIMAD is adherent to the PATHCAST theme while still remaining an antecedent rather than a completed L3 instance.

F5: Process-Derived Features Unexplored. ML-based prediction studies in the confirmed set rely overwhelmingly on traditional features (code metrics, commit characteristics, defect counts) rather than on features derived from process mining. Working-set tier: 32 papers whose research contribution centers on ML/prediction; only 1 explicitly couples ML with a named process-mining algorithm (alpha, inductive, heuristic, conformance, fuzzy, or Petri-net-based discovery). De-duplicated combined analytical subset: the auxiliary corpus added a large number of ML/prediction papers (111 in the combined 319 set), but the rate of process-mining coupling remains low: only 2 of 111 (1.8%) jointly use a named PM algorithm. Conformance scores, rework counts, Markov-expected remaining time, and path entropy—features that are naturally produced by a process-mining stage—remain essentially absent from the SE prediction literature, in both tiers.

These five findings map directly to the five deficits identified in Section 3.7.3: F1 corresponds to D1 (Descriptive Ceiling), F3 to D3 (Stochastic Integration Absence), F4 to D5 (Pipeline Fragmentation), and F5 to D4 (Feature Engineering Void). F2 provides empirical evidence for a challenge that PATHCAST addresses through Stage 1 (Event Log Construction).

3.5.2 POSITIONING OF PATHCAST

Table 3.15 positions PATHCAST against the integration levels identified in the 169-study confirmed set, using both the IC-combination counts derived from the structured extraction and the operational coding obtained from the 73-PDF triangulation subset.

PATHCAST is positioned at level L3—the first proposal, to the best of the author’s knowledge, to formally compose process mining, absorbing Markov chain analysis, Monte Carlo forecasting, and ML enhancement into a single architecturally specified pipeline applied to SDLC data.

Table 3.15: Integration levels in the 169-study confirmed set and PATHCAST positioning.

Level	Description	Papers (169-set, IC-based)	Papers (73-PDF subset, oper.)	Representative example
L0	Single analytical family (≤ 1 IC matched); no cross-technique chaining	95	13	Contextual or single-technique studies
L1	Two ICs matched; two families present, but no explicit reproducible pipeline contract	70	54	Stochastic models or PM without formal coupling
L2	Three ICs matched; partial bridge linking PM, stochastic, and forecasting, but incomplete pipeline or missing ML closure	4	6	PRIMAD (GUINEA-CABRERA; HOLGADO-TERRIZA; PICO-VALENCIA, 2025); Incerto (INCERTO; VANDIN; AHRABI, 2025); López-Pintado (LÓPEZ-PINTADO; DUMAS, 2023)
L3	Four ICs matched; unified framework: PM $\rightarrow$ Markov $\rightarrow$ MC $\rightarrow$ ML with formal inter-stage contracts	0	0	PATHCAST (proposed)
Total	—	169	73	—

3.5.3 PROPOSED TAXONOMY: SPMF

Based on the systematic analysis, this review proposes the Software Process Mining and Forecasting (SPMF) taxonomy with three dimensions:

Analytical Depth classifies studies along four levels, denoted AD1–AD4 to avoid confusion with the architectural integration levels L0–L3 used elsewhere in this chapter: AD1 (Descriptive: discovery, conformance), AD2 (Diagnostic: bottleneck, rework), AD3 (Predictive: outcome, remaining time), and AD4 (Forecasting: distributional, probabilistic).

SDLC Coverage maps which lifecycle phases are addressed: Requirements and Design, Development (coding, commits, PRs), Quality (testing, reviews, bugs), Delivery (CI/CD, release), and Operations (incidents, maintenance).

Technique Integration classifies methodological combination: Single (PM only, ML only, MC only), Dual (PM + ML, PM + Markov, Markov + MC), and Pipeline (PM → Markov → MC → ML).

3.5.4 RESEARCH AGENDA

The gaps identified motivate four research directions:

RA1: Standardized Event Log Construction for SDLC. Frameworks for transforming heterogeneous repository data into comparable event logs with cross-tool case correlation. Maps to Stage 1 of PATHCAST.

RA2: Process-Aware Stochastic Forecasting. Formal integration of process mining with Markov chains and Monte Carlo simulation for distributional SDLC forecasting. Maps to Stages 2–4 of PATHCAST.

RA3: Process-Derived Features for ML. Systematic exploration of features derived from process mining for outcome prediction. Maps to the ML Enhancement Component of PATHCAST.

RA4: Empirical Benchmarking. Evaluation of forecasting methods on comparable OSS repository datasets with temporal hold-out and standardized baselines. Maps to the empirical evaluation (in preparation).

3.5.5 IMPLICATIONS FOR PRACTITIONERS

Practitioners developing software measurement programs can extract three actionable insights from the 169-study confirmed set. First, the availability of structured event log data in issue tracking systems (Jira, GitHub Issues) is a prerequisite for process mining; organizations should invest in consistent workflow state management and transition logging before attempting process analysis. Second, open-source toolkits (PM4Py, ProM, Disco) suffice for discovery and conformance checking in most SDLC contexts, as suggested by the dominance of the inductive miner (51 papers) and alpha miner (45 papers) in the included studies' tool usage patterns. Third, the stochastic extension (Markov chains for transition probability estimation, Monte Carlo for throughput forecasting) requires less infrastructure than typically assumed: the data already exists in the event logs; the challenge is methodological composition, not data collection.

3.6 THREATS TO VALIDITY OF THE SLR

3.6.1 CONSTRUCT VALIDITY

The search strings may exclude studies that use non-canonical terminology. This threat is mitigated through three mechanisms: the complementary MSR string, the frontier stochastic-PM string, and snowballing. The 10/10 control paper recovery provides empirical evidence of high recall.

3.6.2 INTERNAL VALIDITY

LLM-based screening introduces the risk of systematic decision bias. This threat is addressed by five mechanisms, of which the first four were executed directly in the protocol and the fifth is encoded as a conservative closure rule rather than a mandatory human-arbitration step.

- (i) **Structured outputs.** All screening responses use a strict JSON schema, enabling post-hoc analysis of decision patterns and rationale distributions.
- (ii) **Manual verification of all include decisions.** Every paper retained at the FT stage was inspected by the author against the PRISMA flow before being added to the 169-study confirmed set.
- (iii) **Cross-model inter-rater agreement (Cohen's κ).** A stratified random

20% sample of T/A ($n_{\text{theoretical}} = 468$ from the 2,340-paper working set) and FT ($n_{\text{theoretical}} = 177$ from the 886-paper FT queue) decisions was independently re-rated by a different model family (*claude-sonnet-4-6*) acting as a second rater. Of the 468 T/A papers in the sample, 420 produced a parseable decision from the verifier and entered the agreement computation; the remaining 48 papers returned a non-parseable response (typically because the abstract was absent, leading the verifier to refuse a categorical answer) and were excluded from the κ calculation as missing-at-random. All 177 FT sample papers produced a parseable decision. Both the multi-class κ (over the original include/exclude/maybe or include/exclude/pending categories) and the binary κ (collapsing the rater outputs to include versus not-include, the only distinction that affects the final selection set) are reported in Table 3.16. The binary κ is the operationally relevant measure for SLR selection outcomes. Both binary κ values (0.695 for T/A and 0.694 for FT) exceed the substantial-agreement threshold $\kappa \geq 0.61$ (LANDIS; KOCH, 1977). The multi-class FT κ is lower (0.479, moderate) because the verifier returned pending on 25/177 FT papers where the primary screener committed to a binary decision; this represents a difference in conservativeness, not a difference in the final selection outcome. The cross-model verification design follows the guidance of Khraisha et al. (KHRAISHA et al., 2024) and Syriani et al. (SYRIANI; DAVID; KUMAR, 2023) for LLM-assisted SLRs and is intended as a tractable proxy for a fully independent human second-rater protocol.

Table 3.16: Inter-rater agreement (Cohen’s κ) between primary screener (*claude-haiku-4-5*) and verifier (*claude-sonnet-4-6*) on a stratified random 20% sample.

Stage	N	P_o	P_e	κ_{multi}	κ_{binary}	Interpretation (binary)
TA	420	82.1%	50.2%	0.641	0.695	substantial
FT	177	76.3%	54.5%	0.479	0.694	substantial

The cross-model verification protocol was also extended to the auxiliary tier at the FT stage on a stratified random 20% sample ($n = 276$). On the auxiliary tier the verifier (Sonnet 4.6) returned pending on every paper that the primary screener (Haiku 4.5) had rated as include, yielding an observed agreement of $P_o = 54.3\%$ with a multi-class $\kappa = 0.004$ and a binary $\kappa = 0.000$. The near-zero κ is a direct consequence of the verifier’s blanket pending verdict on every Haiku-include paper: when one rater assigns essentially all items to

a single category, $P_e \rightarrow P_o$ and κ collapses toward zero even when the raw overlap on the exclude side is non-trivial. This is a known degeneracy of the κ statistic under severe class imbalance rather than evidence of random labelling. Nevertheless, a κ of zero *does* bound the evidentiary weight that can be placed on the auxiliary inclusions, and we do not claim otherwise. For this reason the auxiliary tier is positioned throughout as a *robustness and recall-bounding* layer rather than as a primary evidence source: the substantive Findings F1–F5 are established on the 169-study working set (which achieves substantial agreement under the same protocol), and the auxiliary results are reported only to test whether those findings are stable under a high-recall expansion. The verifier’s conservativeness is itself explained by the auxiliary corpus’s weaker abstract coverage (the working-set tier benefited from a multi-source abstract enrichment cascade that the auxiliary tier did not receive in full). Accordingly, the protocol adopts a conservative closure rule: papers that remain abstract-insufficient after the enrichment cascade are closed under EC6 (insufficient metadata), and those for which the verifier can only return *pending* without accessible full text are closed under EC7 (unresolvable screening disagreement). The auxiliary-tier findings are therefore interpreted as supplementary evidence from the subset that remained classifiable under the same retrieval constraints, not as evidence from an unresolved arbitration queue.

Table 3.17: Auxiliary-tier inter-rater agreement (cross-model verification: Haiku 4.5 primary vs. Sonnet 4.6 verifier on a stratified 20% sample).

Stage	N	P_o	κ_{multi}	κ_{binary}
FT	276	54.3%	0.004 (slight)	0.000 (slight)

(iv) **T/A false-positive rate.** The LLM confirmatory pass at FT level (Round 1) rejected 33 of 108 T/A include papers (30.6%), primarily for EC3 (algorithm without SE evaluation). This false-positive rate is consistent with the T/A-screening false-positive rates reported by Khraisha et al. (KHRAISHA et al., 2024) for medical SLRs (in the 25–35% range, depending on domain specificity) and by Syriani et al. (SYRIANI; DAVID; KUMAR, 2023) for SE SLRs. False-positives are caught at the FT phase before any synthesis claim is made and therefore do not affect the F1–F5 evidence base. The Round 2 screening reprocessed 495 maybe papers with enriched abstracts and brought the confirmed set to its final value of 169.

(v) **Conservative closure of unresolved disagreement cases.** Papers where the LLM screener (Haiku 4.5) and the verifier (Sonnet 4.6) disagreed on the binary include/exclude outcome were not treated as an open-ended manual-arbitration backlog. Instead, when the disagreement could not be resolved because the abstract remained insufficient and no accessible full text was available, the paper was closed under EC7 (unresolvable screening disagreement). Across both tiers, 52 such cases were flagged (10 at working-set T/A and 42 at auxiliary FT). This rule intentionally biases the protocol toward false negatives rather than unverifiable false positives. The complete disagreement log is archived in the replication package (Section 3.6.5) for optional post-submission audit; human arbitration is not required for the valid evidence set reported here.

3.6.3 EXTERNAL VALIDITY

The review is limited to English-language peer-reviewed publications, potentially excluding relevant work in other languages or grey literature. The broad time span (1994–2025, search executed 12 April 2026) and five-database coverage mitigate geographic and temporal bias. The deferred high-recall auxiliary corpus (3,807 records from SpringerLink, WoS, Snowball, and other low-precision queries) consists of every deduplicated record not matched to the 2,340-paper working set by identifier. Because the working set was frozen before the high-recall re-deduplication snapshot, roughly 364 working-set papers carry divergent identifiers in that snapshot and thus also appear as near-duplicates in the auxiliary pool; the two sets therefore overlap by about 364 records. A cross-check of the first-pass auxiliary inclusions against the working-set studies shows that 60 of the 212 auxiliary includes are duplicates of working-set studies: 56 share a normalized DOI with their working-set counterpart and 4 carry divergent or missing identifiers and are matched only by normalized title. This cross-tier overlap arises because the working-set and auxiliary tiers were screened as separate pipelines, so duplicates were resolved within each tier but only reconciled across tiers by this post-hoc check. After removing these 60 cross-tier duplicates and 2 internal duplicates, the auxiliary tier contributes 150 genuinely new first-pass studies (and 22 from the second pass) rather than 212 and 23; accordingly, the combined-tier distributions reported in this chapter use the de-duplicated counts—319 studies in the first-pass analytical subset (169 + 150) and 341 in the final combined set (169 + 150 + 22)—rather than the raw sums of 381 and 404.

The auxiliary corpus provides a mechanism for robustness testing and for expansion if gaps in the confirmed 169-study set are identified, and is not treated as a primary evidence source.

A four-stage sensitivity check was executed against the auxiliary corpus. First, a random 200-paper subsample was re-screened with the same T/A LLM protocol used for the working set; 12 of 200 (6.0%) returned a T/A include decision and 59 (29.5%) returned maybe, yielding a point-estimated 228 additional candidate includes if the auxiliary corpus were screened in full. Given that this estimate exceeded the working-set include count, the second stage was triggered: a **full-corpus T/A and FT screening pass** over all 3,807 auxiliary papers, executed with the same dual-stage LLM protocol used for the working set. The result is reported in Table 3.19: of the 3,807 auxiliary papers, 232 were T/A-included and 1,151 T/A-marked-maybe, of which 212 survived FT screening as include. Of these 212 first-pass inclusions, 150 are unique studies after removing 60 cross-tier near-duplicates of working-set papers and 2 internal duplicates (Section 3.6.3). Combined with the 169 working-set confirmations, the SLR therefore covers a de-duplicated set of 319 primary studies after the first auxiliary FT pass ($169 + 150$); this first-pass de-duplicated subset is the analytical basis used by the detailed auxiliary extraction, QA, and IC-combination tables.

The protocol then proceeded to a third stage: a multi-source abstract enrichment cascade (Semantic Scholar, OpenAlex, CORE, Crossref) was applied to the 880 auxiliary FT-pending papers, recovering an abstract for 285 papers (32.4%). The fourth stage re-ran the FT screening on these 285 enriched papers with the same eligibility protocol, confirming 23 additional includes, 246 excludes, and 16 still-pending. Under the conservative closure rule, those 16 residual abstract-insufficient papers are closed under EC6 (insufficient metadata) rather than carried forward as a manual-validation queue. The 23 second-pass inclusions reduce to 22 unique studies after cross-tier de-duplication, so the de-duplicated final combined set is **341 primary studies** ($169_{ws} + 150_{aux1}^{unique} + 22_{aux2}^{unique}$; the raw sum before de-duplication was 404). The remaining 595 papers without abstract after enrichment are likewise closed under EC6, rather than queued for manual retrieval, because they remain unclassifiable after the full multi-source enrichment cascade. In total, the auxiliary tier closes 611 papers under EC6 (595 without a recoverable abstract plus the 16 residual re-FT pending).

A second external-validity exposure concerns the original 148-paper EC5 stratum (full text inaccessible at first pass). To bound the residual recall risk, an

Table 3.18: Auxiliary-corpus sensitivity check. A random sample of n papers from the deferred auxiliary corpus was re-screened with the same T/A LLM protocol used for the working set.

Quantity	Value
Sample size	200 (seed=20260428)
Auxiliary corpus total	3807
Working-set confirmed includes	169
Sample T/A include rate	6.0% ($n_{\text{inc}} = 12$)
Sample T/A maybe rate	29.5% ($n_{\text{maybe}} = 59$)
Sample T/A exclude rate	64.5% ($n_{\text{exc}} = 129$)
Expected new includes (point estimate)	228
Expected new includes (upper bound, maybe→include)	1351
Sensitivity ratio vs. confirmed set (point)	135.2%
Sensitivity ratio vs. confirmed set (upper)	799.7%

Table 3.19: Auxiliary-corpus FT-screening outcome (papers that passed T/A on the 3,807-paper deferred corpus).

Outcome	Count	%
Forwarded from T/A	1383	100.0
Included (raw)	212	15.3
Included (unique, after de-dup.)	150	10.8
Excluded	291	21.0
Pending / unparsed	880	63.6
Combined SLR set (de-dup.: working-set + 150 unique)	319	—

Table 3.20: Re-FT screening of aux-pending papers that gained an abstract through the post-hoc enrichment cascade.

Outcome	Count	%
Re-FT include	23	8.1
Re-FT exclude	246	86.3
Re-FT pending	16	5.6
Total re-FT	285	100.0

automated open-access (OA) recovery pass was executed against four public discovery services (Unpaywall, Semantic Scholar Graph API, OpenAlex, CORE). The result is summarised in Table 3.21: 27 of the 148 EC5 papers (18.2%) returned at least one OA URL during the recovery pass; of those, 17 PDFs were successfully downloaded from the recovered OA URLs in the EC5 stratum, corresponding to 11.5% of the 148-paper EC5 stratum recovered into the read-set. A targeted PDF audit was then executed on those 17 downloads (audit log in the replication package, Section 3.6.5): 14 OA links resolved to unrelated papers or generic proceedings and therefore remained under EC5, while 3 papers were substantively re-screened and reclassified to EC3. The active PRISMA EC5 count after this audit is thus 145 rather than 148. The remaining 121 papers (81.8%) remain inaccessible and continue to be reported under EC5 in the PRISMA flow. A parallel OA-download pass over the 212 auxiliary includes recovered 49 PDFs (20.5% of 212; the lower yield reflects the publisher-paywall mix of the auxiliary corpus and is documented in the replication package (Section 3.6.5)).

Table 3.21: EC5 recovery attempt against four open-access discovery APIs.

Source	Recovered	% of EC5
Unpaywall	6	4.1%
Semantic_Scholar	6	4.1%
Openalex	6	4.1%
Core	20	13.5%
Any source	27	18.2%

3.6.4 CONCLUSION VALIDITY

Synthesis findings F1–F5 are derived from the structured data extraction applied to two extraction layers: (i) the full 169-study working-set extraction (Section 3.2.5; Table 3.8), partly triangulated against the 73-PDF subset; and (ii) the auxiliary-tier extraction performed from abstract metadata only, using the same LLM-assisted extraction template (Section 3.2.5), over the 212 first-pass inclusions, of which 150 are unique studies after cross-tier de-duplication. Both extractions feed the IC-combination counts and the technique distributions reported under F1–F5. The QA-based filtering reported in Table 3.7 retains 136 of the 169 working-set studies (80.5%); QA scoring was subsequently extended

to the auxiliary tier with the same eight-criterion rubric (Section 3.2.4), retaining 124 of the 150 unique auxiliary studies (82.7%, mean 5.09 ± 1.43). The de-duplicated combined 319-study analytical subset therefore retains 260 papers (81.5%, mean 5.00 ± 1.42) above the $\geq 4/8$ threshold. Table 3.22 reports the QA breakdown over the *raw* combined tiers (working set 169 plus the 212 first-pass auxiliary inclusions = 381, of which 315 pass), *before* cross-tier de-duplication; the 260/319 figure quoted here is the corresponding count after removing the 60 cross-tier and 2 internal duplicates, and is the value used in all combined-tier analyses.

Table 3.22: Quality assessment combined across working-set and auxiliary tiers, after cross-tier de-duplication ($n = 319$). The auxiliary tier is reported as unique studies (150) after removing 60 cross-tier and 2 internal duplicates from the 212 raw first-pass inclusions.

Tier	N	Mean	SD	$\geq 4/8$ (pass)	$< 4/8$ (fail)
Working-set	169	4.88	1.40	136	33
Auxiliary (unique)	150	5.09	1.43	124	26
Combined (de-dup.)	319	5.00	1.42	260	59

Table 3.22 reports a counterintuitive pattern: the auxiliary tier’s mean QA score (5.09) exceeds that of the working set (4.88), even though auxiliary inclusions were screened with weaker textual evidence. This ordering is a *selection effect*, not evidence that the auxiliary corpus is methodologically superior. The auxiliary inclusion set retains only studies that survived full-text or enriched-abstract screening under the same eight-criterion rubric; papers closed under EC6 (insufficient metadata) or EC7 (unresolvable disagreement) never enter the denominator, biasing the retained pool toward publications with richer abstracts that explicitly document objectives (QA1), data sources (QA3), and formal technique definitions (QA4). The working set, by contrast, includes the full analytical core regardless of abstract richness, and therefore retains a larger share of short workshop papers, position papers, and legacy publications with sparse methodology reporting—precisely the studies that depress the per-item QA profile (Section 3.2.4, Table 3.14). The higher auxiliary mean thus confirms that screening quality co-varies with metadata availability, and reinforces why F1–F5 are anchored on the working set while the auxiliary tier serves recall-bounding only.

A residual conclusion-validity threat is the LLM-based extraction of PM-technique, stochastic-technique, and SDLC-phase fields. To mitigate this, the 73-PDF triangulation subset was used to spot-check the extracted fields against the corresponding full texts; the spot-check did not reveal any systematic deviation that would materially alter the IC-combination distributions or the F1–F5 findings. As a matter of good methodological practice—rather than as a remedy for an observed problem—a complete human re-extraction on a stratified sample is planned before camera-ready submission.

3.6.5 REPLICATION PACKAGE

A complete replication package is deposited at Zenodo (DOI 10.5281/zenodo.20130276) and includes the search strings for all five databases, the deduplication and enrichment scripts (`pipeline/dedup.py`, `pipeline/enrich.py`), the screening and QA prompts and JSON schema (`config/screening_criteria.py`, `pipeline/qa_llm.py`), the cross-model verification computation (`pipeline/kappa.py`), the decision CSVs at each stage, the QA-scored 169-study table, the structured extraction table, the inter-rater agreement reports, and a PRISMA flow snapshot. The package is intended to be self-contained: any reviewer can re-run the LLM-assisted protocol on the same working set and reproduce the 169-study confirmed selection within the stated κ confidence band. AI tools were used as primary screeners under the manual verification protocol stated in the review-protocol preamble; they are not listed as authors and bear no responsibility for editorial judgements.

3.7 CROSS-DOMAIN SYNTHESIS AND GAP MAP

The findings synthesised above are positioned here against the broader related-work landscape, completing the gap map that motivates PATHCAST.

3.7.1 PROCESS MINING APPLIED TO SDLC

The principal contributions are summarised in Table 3.23: from the seminal work of Cook and Wolf (COOK; WOLF, 1998), through the MSR-PM methodology of Poncin et al. (PONCIN; SEREBRENIK; BRAND, 2011), to the industrial applications by Caldeira and Cardoso (CALDEIRA et al., 2019). The consistent

pattern is concentration at the descriptive and diagnostic levels: discovery and conformance, rarely advancing to prediction or forecasting (Finding F1).

Table 3.23: Process mining applied to software development.

Study	PM Type	Data Source	SDLC Phase	Pred.
Cook & Wolf (1998)	Discovery	Build logs	Build/Test	No
Rubin et al. (2007)	Framework	Conceptual	All	No
Poncin et al. (2011)	Discovery	VCS, Issues, ML	Multi-repo	No
Lemos et al. (2011)	Disc., Con- form.	Issue tracker	Dev. lifecy- cle	No
Mannhardt et al. (2016)	Conform.	Multi- system	Healthcare *	No
Caldeira & Cardoso (2019)	Disc., En- hance.	Jira, Git	Issue lifecy- cle	No

* Methodological reference for multi-perspective conformance.

3.7.2 STOCHASTIC FORECASTING AND ML FOR SE

Markov chains have been applied to software testing (Whittaker and Thomason (WHITTAKER; THOMASON, 1994)) and reliability modelling (WHITTAKER; REKAB; THOMASON, 2000), but not parameterised by process-mining-discovered structures. Vacanti (VACANTI, 2015) popularised throughput-based Monte Carlo for agile forecasting, but this approach treats the process as a black box. Rogge-Solti and Weske (ROGGE-SOLTI; WESKE, 2015) integrated PM with stochastic Petri nets for BPM remaining-time prediction, but did not target SDLC. Defect prediction (ZIMMERMANN et al., 2004; WAHONO, 2015; PACHOULY et al., 2022) relies overwhelmingly on artefact-level features; PPM benchmarks (TAX et al., 2017; MARQUEZ-CHAMORRO; RESINAS; RUIZ-CORTES, 2018) operate on BPM event logs, not software development; Camargo et al. (CAMARGO; DUMAS; GONZÁLEZ-ROJAS, 2022) combine deep learning with simulation in BPM domains.

3.7.3 GAP MAP

Table 3.24 maps the intersection of techniques and SDLC focus.

Table 3.24: Technique integration (✓) across studies. To the best of the author’s knowledge, no reviewed study occupies the full intersection targeted by PATHCAST.

Approach	PM	Markov	MC Sim.	ML	SDLC
Cook & Wolf (1998)	✓				✓
Whittaker & Thomson (1994)		✓			✓
Vacanti (2015)			✓		✓
Zimmermann et al. (2005)				✓	✓
Tax et al. (2017)	✓			✓	
Rogge-Solti & Weske (2015)	✓		✓		
Rozinat et al. (2009)	✓		✓		
Camargo et al. (2019)	✓		✓	✓	
Marquez-Ch. et al. (2018)	✓			✓	
Caldeira & Cardoso (2019)	✓				✓
PATHCAST	✓	✓	✓	✓	✓

The gap decomposes into five deficits, each constituting a contribution dimension of PATHCAST: **(D1) Descriptive Ceiling** (PM in SDLC stops at discovery); **(D2) Structural Blindness** (forecasting methods ignore process structure); **(D3) Stochastic Integration Absence** (Markov/MC not parameterised by PM in SE); **(D4) Feature Engineering Void** (ML predictors do not incorporate PM-derived features); and **(D5) Pipeline Fragmentation** (no formal end-to-end PM → Markov → MC → ML pipeline for SDLC). PATHCAST (Chapter 4) addresses all five simultaneously through a four-stage pipeline with formal inter-stage contracts.

3.8 CHAPTER SUMMARY

This chapter presented a systematic literature review of the intersection of process mining, stochastic modeling, and machine learning applied to software process forecasting. The review executed three complementary search strings across five databases, yielding 8,347 records, 5,783 after deduplication, and a 2,340-paper operational working set for systematic screening. Title and ab-

abstract screening on the working set using a LLM-assisted protocol produced 111 candidate include decisions and 775 maybe decisions (886 papers for full-text review).

Full-text screening combined LLM-assisted passes (Rounds 1 and 2, with cascade abstract enrichment raising coverage from 16.6% to 72.6%, i.e., 643/886 papers) with a final *screen-blanks* pass that processed the 186 papers that still lacked an FT decision after the earlier rounds. In the final screening artifact, 238/886 papers still lack an abstract field, but all have a recorded FT decision. Papers whose full text remained inaccessible despite all retrieval attempts were closed under EC5; after the OA recovery audit, 145 papers remain in that stratum. The working-set result is **169 confirmed primary studies** and 717 FT excluded (EC3: 232, EC1: 205, EC5: 145, EC2: 110, EC4: 60), with zero pending decisions. Data extraction for all 169 studies was completed using a structured LLM-assisted pipeline operating on locally available PDFs (75/169, with 73 operationally coded in the triangulation subset) or abstract/title metadata (94/169).

A second tier of the protocol then applied the same T/A and FT screening to the previously deferred 3,807-paper auxiliary corpus (every deduplicated record not matched to the 2,340-paper working set by identifier, including ~364 near-duplicates of working-set papers with divergent identifiers), yielding 232 T/A includes and 1,151 T/A maybes; FT screening of the 1,383 forwarded papers produced 212 first-pass inclusions, 291 excludes, and 880 pending. A subsequent re-screening pass over 285 abstract-enriched auxiliary papers confirmed 23 further inclusions (second auxiliary pass). After cross-tier de-duplication—which removes 60 first-pass and 1 second-pass near-duplicates of working-set studies, plus 2 internal duplicates—these contribute 150 and 22 unique new studies, giving a de-duplicated combined total of **341 primary studies** ($169_{ws} + 150_{aux1}^{unique} + 22_{aux2}^{unique}$; the raw sum was 404). Under the conservative closure policy, the residual auxiliary cases without usable abstract are treated as EC6 (insufficient metadata) exclusions and the unresolved disagreement cases as EC7, rather than as an open manual-validation backlog. The original EC5 recovery pass was also closed: its 17 downloaded PDFs were audited, yielding 3 substantive EC3 exclusions and 14 residual EC5 mismatches. The de-duplicated final total is therefore 341, and the combined-tier analytical tables below refer to the de-duplicated 319-study first-pass subset (169 + 150).

Analysis of the working-set tier (169 studies), together with the 341-study

de-duplicated combined evidence base and the 319-study combined analytical subset, supports five key findings. First, the field exhibits a descriptive ceiling: 71.6% of confirmed working-set papers (50.8% combined) apply process mining to software artifacts (IC1), but only 17.2% advance (working-set; 31.3% combined) to forecasting of process metrics (IC3) (F1). Second, event log construction from heterogeneous repository sources is a pervasive challenge, with ad hoc joins between issues, commits, and pipeline logs the norm (F2). Third, the integration of stochastic models with process mining for SDLC forecasting is minimal: only 10.7% of working-set papers (25.4% combined) combine IC2 and IC3 (F3). Fourth, no confirmed study implements a formally composed PM $\rightarrow$ Markov $\rightarrow$ MC $\rightarrow$ ML pipeline for software processes; PRIMAD (GUINEA-CABRERA; HOLGADO-TERRIZA; PICO-VALENCIA, 2025) is the closest antecedent at L2 (F4). Fifth, ML studies in SE rarely incorporate process-derived features (F5).

Synthesis of the top-ranked papers by IC cluster sharpens the picture. The IC2+IC3 cluster (18 papers) concentrates at analytical depth AD1, parametrising stochastic models from aggregate statistics rather than mined transitions. The IC1+IC2 cluster reaches AD2 through stochastic conformance and simulation, but stops short of distributional forecasting. The IC1+IC3 cluster chains PM and prediction sequentially without a stochastic layer. The IC2+IC4 cluster demonstrates that public repository traces support state-transition modelling, but event-to-state mapping remains unstandardised. No paper occupies L3.

These five gaps directly motivate the PATHCAST method presented in Chapter 4. PATHCAST is positioned at integration level L3 in the SPMF taxonomy; to the best of our knowledge, no study identified in this review composes all four technique families into a unified, architecturally specified framework for SDLC process forecasting.

Methodologically, the LLM-assisted protocol used in this review achieves substantial inter-rater agreement under cross-model verification (binary $\kappa = 0.695$ for T/A and $\kappa = 0.694$ for FT) and full QA coverage of the 169 confirmed studies (mean QA score = 4.88/8, with 136/169 retained above the $\geq 4/8$ threshold). The complete replication package and the inter-rater agreement reports are deposited at Zenodo (DOI 10.5281/zenodo.20130276).

4

The Proposed Method: PATHCAST

This chapter presents the formal specification of the proposed method, PATHCAST. The method integrates process mining, Markov chain estimation, Monte Carlo simulation, and machine learning into a unified pipeline that transforms heterogeneous software repository data into probabilistic process forecasts.

Section 4.1 provides the architectural overview and defines the formal pipeline. Sections 4.2 through 4.5 formalize each pipeline stage with input/output contracts, algorithms, and correctness properties. Section 4.6 specifies the machine learning enhancement component. Section 4.7 formalizes the inter-stage data contracts. Section 4.8 establishes the formal properties and theoretical guarantees. Section 4.9 synthesizes the contributions.

4.1 ARCHITECTURAL OVERVIEW

4.1.1 PIPELINE DEFINITION

PATHCAST is defined as a composition of four sequential transformation stages and one parallel enhancement component:

Definition 4.1 (PATHCAST Pipeline). *The PATHCAST pipeline is a function composition:*

$$\mathcal{F} = \mathcal{S}_4 \circ \mathcal{S}_3 \circ \mathcal{S}_2 \circ \mathcal{S}_1 \quad (4.1)$$

where:

$$\mathcal{S}_1 : \mathcal{R} \rightarrow \mathcal{L} \quad (\text{Event Log Construction}) \quad (4.2)$$

$$\mathcal{S}_2 : \mathcal{L} \rightarrow (M, \Phi) \quad (\text{Process Mining}) \quad (4.3)$$

$$\mathcal{S}_3 : (\mathcal{L}, M) \rightarrow (P, N, B, \hat{F}) \quad (\text{Markov Estimation}) \quad (4.4)$$

$$\mathcal{S}_4 : (P, \hat{F}, s_0, M_{sim}) \rightarrow \hat{\mathcal{D}} \quad (\text{Monte Carlo Simulation}) \quad (4.5)$$

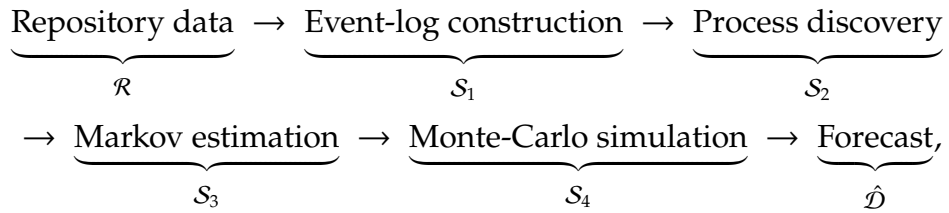
and a parallel component:

$$\mathcal{S}_{ML} : (\mathcal{L}, \Phi, P, N) \rightarrow \hat{f} \quad (\text{ML Enhancement}) \quad (4.6)$$

The notation: $\mathcal{R}$ denotes software repositories; $\mathcal{L}$ the structured event log; M the discovered process model; Φ the process metrics vector; P the Markov transition matrix; N the fundamental matrix; B the absorption probability matrix; $\hat{F}$ the empirical sojourn time distributions; $\hat{\mathcal{D}}$ the empirical forecast distribution; $\hat{f}$ the trained ML predictor.

4.1.2 PIPELINE ARCHITECTURE

Before the algorithmic details of each stage, this subsection presents the conceptual architecture of PATHCAST as an end-to-end pipeline, so that the formal stage specifications that follow can be read against a single mental model. Conceptually, PATHCAST is a five-step transformation of repository evidence into a probabilistic forecast:



with a machine-learning enhancement component that consumes the process-derived artefacts of $\mathcal{S}_2$ and $\mathcal{S}_3$ to refine case-level predictions. Each step is a function with a formally specified input/output contract (Section 4.1.5), and the stages compose into the directed acyclic graph of Figure 4.1; the remainder of this chapter then specifies each function in turn.

Figure 4.1 illustrates the complete pipeline architecture with data flows be-

tween stages.

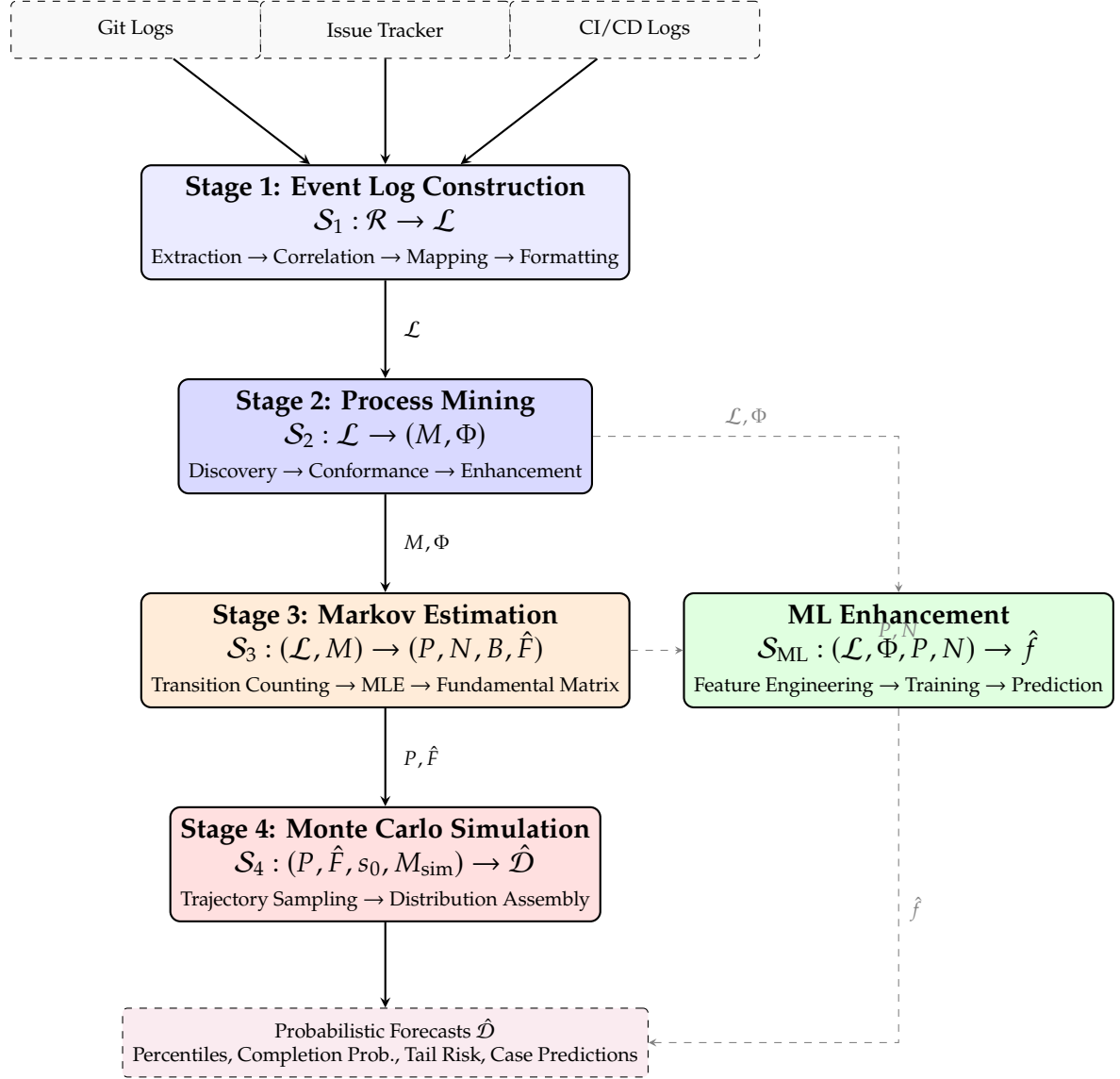


Figure 4.1: PATHCAST pipeline architecture. Solid arrows indicate the primary sequential pipeline; dashed arrows indicate ML enhancement data flows.

4.1.3 DESIGN PRINCIPLES

The pipeline architecture is governed by three design principles:

Principle 1 (Empirical Grounding). Every model parameter is estimated from observed data, never assumed *a priori*. The transition matrix P is estimated from event log counts; sojourn distributions $\hat{F}$ are constructed from observed

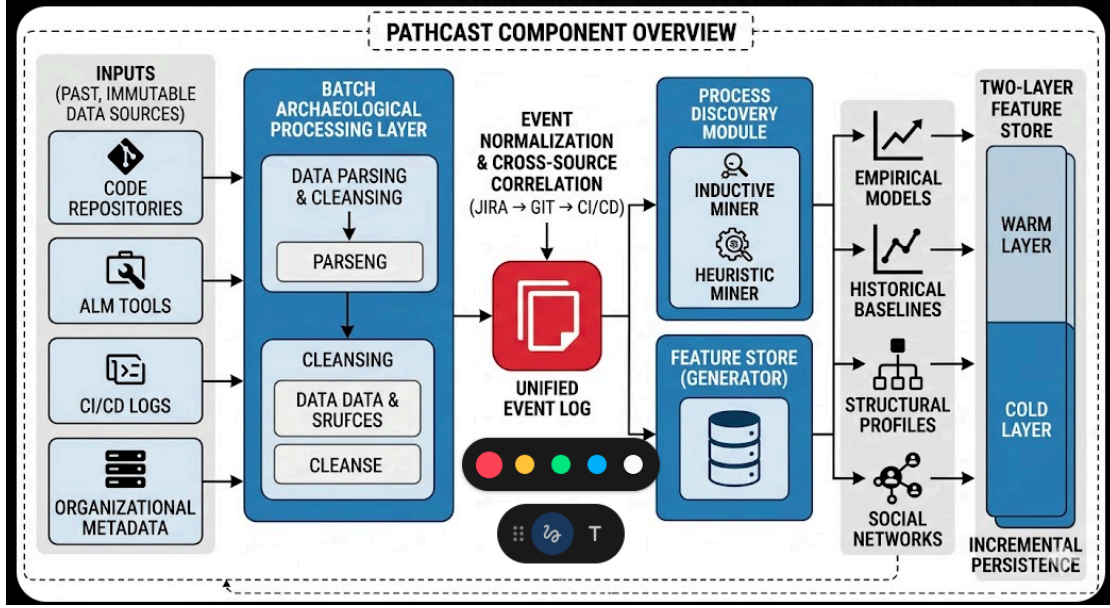


Figure 4.2: PATHCAST component overview.

timestamps; ML features are extracted from empirical traces. No distributional assumptions are imposed unless empirically validated.

Principle 2 (Stage Composability). Each stage has a formally defined input/output contract (Definitions 4.2 through 4.5). Stages can be independently tested, replaced, or extended without affecting other stages, provided the contract is preserved.

Principle 3 (Distributional Outputs). The pipeline produces probability distributions, not point estimates. Every forecast is a collection of M_{sim} simulated outcomes from which percentiles, intervals, and tail probabilities are extracted.

4.1.4 COMPONENT VIEW

Figure 4.2 presents the component view of PATHCAST; this view focuses on the historical-data processing layer that precedes forecasting (the past-facing portion of the pipeline that builds the models later used to project the future). The framework ingests historical, immutable data sources (code repositories, ALM tools, CI/CD logs, organizational metadata) through a batch archaeological processing layer. The pipeline produces a unified event log through event normalization and cross-source correlation (Jira $\rightarrow$ Git $\rightarrow$ CI/CD), feeding both the process discovery module (Inductive Miner, Heuristic Miner) and the feature store. Outputs of the PATHCAST component—empirical models,

historical baselines, structural profiles, and (optionally) organizational social-network features—are persisted incrementally in a two-layer feature store (cold and warm) that serves as input for the forecasting stages described in Sections 4.3 through 4.5. The social-network features are an optional input to the machine-learning enhancement component (Section 4.6) and are not required by the core process-aware forecasting pipeline; they are included here for architectural completeness and may be omitted without affecting Stages 1–4.

Figure 4.2 is the *architectural realization* of the method: it shows implementation concerns such as the cold / warm feature store and the persistence layer. The *functional abstraction* of the method—the mathematical mapping from repositories to forecast distributions—is given by the stage contracts of Section 4.1.5, in which the feature store is the persistence mechanism for the formally specified stage outputs $(\mathcal{L}, M, \Phi, S, P, N, B, \hat{F})$ rather than an additional formal object. The fundamental matrix N and absorption matrix B produced by Stage 3, in particular, are consumed by the machine-learning component as predictive features (see the $S3 \rightarrow ML$ contract in Table 4.5), not merely computed for descriptive analysis.

4.1.5 STAGE CONTRACTS

Each stage of the pipeline is specified by an input/output contract with pre- and postconditions; the contracts are gathered here once and referenced—rather than restated—in the stage sections that follow. Stage composability (Principle 2) depends on these contracts: stages may be independently replaced or extended as long as their contracts are preserved.

Definition 4.2 (Stage 1 Contract). ***Input:** repositories $\mathcal{R} = \{r_1, \dots, r_n\}$, each r_i a set of time-ordered raw-event streams $\{D_i^{(j)}\}$. **Output:** structured event log $\mathcal{L} = \{\sigma_c \mid c \in C\}$ satisfying Definition 2.1. **Postconditions:** every event has a valid case identifier, activity name and timestamp; events are timestamp-ordered within each trace; activity names belong to the SDLC taxonomy $\mathcal{A}$. $S_1 = \text{FORMAT} \circ \text{MAP} \circ \text{CORRELATE} \circ \text{EXTRACT}$ (Equation 4.7).*

Definition 4.3 (Stage 2 Contract). ***Input:** an event log $\mathcal{L}$ satisfying Definition 4.2. **Output:** a triple (M, Φ, S) with discovered model M , metrics vector $\Phi = (\phi_{\text{fit}}, \phi_{\text{prec}}, \phi_{\text{gen}}, \phi_{\text{perf}})$, and state space $S = (S_T, S_A)$. **Postconditions:** M is sound; $\phi_{\text{fit}}, \phi_{\text{prec}} \in [0, 1]$; ϕ_{perf} contains per-activity sojourn statistics; $|S_T| \geq 1$ and $|S_A| \geq 1$; every $a \in \mathcal{A}$ observed in $\mathcal{L}$ maps to exactly one state.*

Definition 4.4 (Stage 3 Contract). **Input:** $\mathcal{L}$ and $S = S_T \cup S_A$. **Output:** $(P, N, B, \hat{F}, \tilde{\pi})$ with transition matrix $P \in \mathbb{R}^{|S| \times |S|}$, fundamental matrix $N = (I - Q)^{-1} \in \mathbb{R}^{|S_T| \times |S_T|}$, absorption matrix $B = NR \in \mathbb{R}^{|S_T| \times |S_A|}$, sojourn distributions $\hat{F} = \{\hat{F}_s\}_{s \in S_T}$, and expected transient-occupancy distribution $\tilde{\pi}$ (Equation 4.45). We avoid the term stationary distribution here: an absorbing chain admits only the degenerate stationary law supported on S_A , so the informative quantity for forecasting is the expected occupancy of the transient states before absorption, not a stationary vector. **Preconditions:** $|S_T| \geq 1$, $|S_A| \geq 1$, and every transient state can reach some absorbing state in M (so that $(I - Q)$ is invertible; a mere outgoing transition is insufficient, as a closed transient cycle would violate invertibility). **Postconditions:** P row-stochastic; $(I - Q)$ invertible; $N_{ij} \geq 0$; $\sum_j B_{ij} = 1$; each $\hat{F}_s$ has at least $n_{\min}$ observations. $S_3 = \text{OCCUPANCY} \circ \text{SOJOURN} \circ \text{ABSORB} \circ \text{ESTIMATE} \circ \text{COUNT}$ (Equation 4.30).

Definition 4.5 (Stage 4 Contract). **Input:** $P, \hat{F}$, initial state s_0 (or initial distribution π_0), and the simulation budget $M_{\text{sim}} \in \mathbb{N}$. **Output:** the empirical forecast distribution $\hat{\mathcal{D}} = \{(T^{(m)}, o^{(m)}, \psi^{(m)})\}_{m=1}^{M_{\text{sim}}}$ of lead times, terminal states and trajectories (Equation 4.48). **Postconditions:** $|\hat{\mathcal{D}}| = M_{\text{sim}}$, every trajectory absorbs, $T^{(m)} > 0$.

Definition 4.6 (ML Component Contract). **Input:** $\mathcal{L}, \Phi, P, N$. **Output:** trained predictor $\hat{f}$ and feature importance vector $\mathbf{w}$. **Postconditions:** $\hat{f}$ is evaluated on a temporally disjoint test set; AUC-ROC, F1, Brier score, MAE and MAPE are reported; $\mathbf{w}$ ranks features.

4.2 STAGE 1: EVENT LOG CONSTRUCTION

Stage 1 transforms heterogeneous software repository data into structured event logs suitable for process mining.

Stage 1 satisfies the contract of Definition 4.2; the decomposition is

$$S_1 = \text{FORMAT} \circ \text{MAP} \circ \text{CORRELATE} \circ \text{EXTRACT}, \quad (4.7)$$

detailed in the four subsections that follow.

4.2.1 EVENT EXTRACTION

Definition 4.7 (Raw Event). A raw event is a tuple $e^{\text{raw}} = (ts, \text{src}, \text{type}, \text{payload})$ with timestamp, source, type, and source-specific attribute map.

Definition 4.8 (Source Extractors).

$$EXTRACT_{git} : D^{(git)} \rightarrow \{e^{raw}\} \quad (4.8)$$

$$EXTRACT_{jira} : D^{(jira)} \rightarrow \{e^{raw}\} \quad (4.9)$$

$$EXTRACT_{ci} : D^{(ci)} \rightarrow \{e^{raw}\} \quad (4.10)$$

$$EXTRACT_{review} : D^{(review)} \rightarrow \{e^{raw}\} \quad (4.11)$$

The raw event pool is $E^{raw} = \bigcup_{i,j} EXTRACT_{src(D_i^{(j)})}(D_i^{(j)})$.

Table 4.1 specifies the extraction mapping per source type.

Table 4.1: Event extraction mapping per data source.

Source	Raw Event Type	Timestamp Source	Key Payload
Git	commit	Author date	hash, author, files
	merge	Committer date	branches
Jira	status_change	Changelog timestamp	from/to status
	created	Creation timestamp	reporter, type
	assigned	Changelog timestamp	assignee
CI/CD	build_start	Build start	pipeline, branch
	build_end	Build end	result, duration
	deploy	Deploy timestamp	environment
Review	pr_opened	PR creation	author, base
	review_submitted	Review timestamp	reviewer, decision
	pr_merged	Merge timestamp	merge method
	pr_closed	Close timestamp	reason

4.2.2 CASE CORRELATION

Definition 4.9 (Case Correlation Function). $\kappa : E^{raw} \rightarrow C \cup \{\perp\}$, where $\perp$ indicates uncorrelated events.

The correlation function is implemented as a cascade of strategies, applied in order of decreasing specificity (Algorithm 1).

Strategy κ_1 (Explicit Reference): extracts issue keys from structured fields (Jira issue key, Git commit message regex, GitHub PR links). Formally:

$$\kappa_1(e) = \begin{cases} \text{issue_key}(e.\text{payload}) & \text{if extractable} \\ \perp & \text{otherwise} \end{cases} \quad (4.12)$$

Algorithm 1 Case Correlation Cascade**Require:** Raw event pool E^{raw} , correlation configuration $\mathcal{K}$ **Ensure:** Case-annotated event set $E^{\text{corr}} \subseteq E^{\text{raw}} \times \mathcal{C}$

```

1: Initialize  $E^{\text{corr}} \leftarrow \emptyset$ ,  $E^{\text{pending}} \leftarrow E^{\text{raw}}$ 
2: for each strategy  $\kappa_k \in \mathcal{K}$  in priority order do
3:   for each event  $e \in E^{\text{pending}}$  do
4:      $c \leftarrow \kappa_k(e)$ 
5:     if  $c \neq \perp$  then
6:        $E^{\text{corr}} \leftarrow E^{\text{corr}} \cup \{(e, c)\}$ 
7:        $E^{\text{pending}} \leftarrow E^{\text{pending}} \setminus \{e\}$ 
8:     end if
9:   end for
10: end for
11: Report  $|E^{\text{pending}}|$  uncorrelated events return  $E^{\text{corr}}$ 

```

Strategy κ_2 (Structural Link): platform-provided relationships (PR→issue, build→commit, sub-task→parent).

Strategy κ_3 (Heuristic): temporal proximity, author identity, and file overlap, applied to events not resolved by κ_1 or κ_2 .

Definition 4.10 (Correlation Quality Metrics).

$$\text{Coverage} = |E^{\text{corr}}|/|E^{\text{raw}}| \quad (4.13)$$

$$\text{Ambiguity Rate} = |\{e : |\kappa(e)| > 1\}|/|E^{\text{raw}}| \quad (4.14)$$

4.2.3 ACTIVITY MAPPING

Definition 4.11 (SDLC Activity Taxonomy).

$$\mathcal{A} = \{\text{created, backlog, in_progress, review, testing, build, deploy, done, cancelled}\} \quad (4.15)$$

The taxonomy in Equation 4.15 is the *canonical minimal* SDLC taxonomy used by PATHCAST; it is intentionally compact to keep the discovered state space interpretable. Project-specific extensions (e.g., blocked, changes_requested, reopened, failed_build, hotfix) are permitted and are accommodated by extending $\mathcal{A}$ and the mapping function μ without altering the formalism; finer-grained statuses that are not modelled explicitly are collapsed onto the nearest canonical activity.

Definition 4.12 (Activity Mapping Function). $\mu : (src, type, payload) \rightarrow \mathcal{A} \cup \{\perp\}$.

The mapping is implemented as a configurable lookup table. The default mapping for common source events is in Table 4.2.

Table 4.2: Default activity mapping from source event types to SDLC taxonomy.

Source	Raw Event / Condition	Mapped Activity
Jira	status→"To Do"	backlog
Jira	status→"In Progress"	in_progress
Jira	status→"In Review"	review
Jira	status→"Done"	done
Git	first commit in case	in_progress
Review	pr_opened	review
Review	review_submitted (approved)	review
CI/CD	build_end	build
CI/CD	deploy	deploy

4.2.4 EVENT LOG FORMATTING AND FILTERING

Definition 4.13 (Structured Event). A structured event is $e = (\kappa(e), \mu(e), \tau(e), \rho(e), \omega(e))$ with case ID, activity, timestamp, resource, and attribute map.

Two filters are applied: (i) **bot filtering**, removing events from known automated accounts and from accounts exceeding a commit-frequency threshold; (ii) **deduplication**, retaining only the first event for each $(case, activity, timestamp)$ triple.

The output event log is

$$\mathcal{L} = \{ \sigma_c = \langle e_1, \dots, e_{|\sigma_c|} \rangle \mid c \in C, \tau(e_k) \leq \tau(e_{k+1}) \forall k \}. \quad (4.16)$$

4.2.5 LOG QUALITY ASSESSMENT

Definition 4.14 (Log Quality Metrics).

$$\text{Closed-Case Ratio} = |\{c : \sigma_c \text{ reaches absorbing}\}|/|C| \quad (4.17)$$

$$\text{Trace Length} = \frac{1}{|C|} \sum_c |\sigma_c| \quad (4.18)$$

$$\text{Activity Coverage} = |\{\mu(e) : e \in \mathcal{L}\}|/|\mathcal{A}| \quad (4.19)$$

$$\text{Variant Count} = |\{\text{unique activity sequences}\}| \quad (4.20)$$

We deliberately name Equation 4.17 the *closed-case ratio* rather than “completeness”: in a live repository a substantial fraction of cases (open issues, in-progress pull requests, pending builds) are legitimately non-terminal, so a low value reflects ongoing work rather than a defective log. Logs with a closed-case ratio below θ_{comp} (default 0.5) or activity coverage below θ_{cov} (default 0.6) trigger a quality warning, which flags that a large share of cases is still in flight (and thus right-censored for forecasting) rather than indicating data corruption.

4.3 STAGE 2: PROCESS MINING

4.3.1 VARIANT ENTROPY

Definition 4.15 (Variant Entropy). Let $V = \{v_1, \dots, v_k\}$ be the distinct trace variants in $\mathcal{L}$, with $p(v_i) = |\{c : \sigma_c = v_i\}|/|\mathcal{L}|$. The variant entropy is

$$H(V) = - \sum_{v \in V} p(v) \log_2 p(v). \quad (4.21)$$

The normalized form $H_{\text{norm}}(V) = H(V)/\log_2 |V|$ provides a scale-independent measure in $[0, 1]$. PATHCAST uses H_{norm} as a complexity proxy and as a candidate ML feature. Using the diversity of trace variants as an indicator of behavioural complexity is consistent with established process-mining practice: behavioural variability is a recognised driver of process-model complexity and of the difficulty of conformance checking and prediction, as discussed in trace-alignment diagnostics (BOSE; AALST, 2012) and in large-scale discovery benchmarks (AUGUSTO et al., 2019).

Stage 2 satisfies the contract of Definition 4.3; the remainder of this sec-

tion specifies the discovery, conformance and performance-enhancement sub-operations.

4.3.2 PROCESS DISCOVERY

$$M = \text{INDUCTIVEMINER-INFREQUENT}(\mathcal{L}, \theta_{\text{noise}}) \quad (4.22)$$

with $\theta_{\text{noise}} \in [0, 1]$ the infrequent-behaviour filtering threshold, which controls the fitness–precision (and simplicity) trade-off: larger θ_{noise} prunes low-frequency edges, raising precision and simplicity at the cost of fitness.

Property 4.1 (Soundness Guarantee). *The Inductive Miner (IM, and its infrequent variant IMf) produces a sound process tree by construction (LEEMANS; FAHLAND; AALST, 2013).*

Property 4.2 (Replay Guarantee under Noise-Free Filtering). *With $\theta_{\text{noise}} = 0$, IMf reduces to the base Inductive Miner (IM), which returns a process tree that can replay all behaviour represented in the (filtered) log; under the assumptions of a complete and noise-free log, the measured fitness($\mathcal{L}, M$) $\rightarrow 1.0$. In practice, residual noise, duplicated events, or incompleteness in the underlying log may yield a measured fitness slightly below 1.0 when evaluated post-hoc by alignment.*

4.3.3 CONFORMANCE CHECKING

$$\phi_{\text{fit}} = \text{fitness}(\mathcal{L}, M), \quad \phi_{\text{prec}} = \text{precision}(\mathcal{L}, M), \quad \phi_{\text{gen}} = \text{generalization}(\mathcal{L}, M) \quad (4.23)$$

computed via alignment-based methods (Definition 2.2). Per-case conformance reuses, under the same definition, the per-case alignment-based fitness $\text{conf}(c) \in [0, 1]$ introduced in Chapter 2 (Eq. 2.2); since each case c corresponds to exactly one trace σ_c , we keep the symbol $\text{conf}(c)$:

$$\text{conf}(c) = 1 - \frac{\text{cost}(\gamma_c^*)}{\text{cost}(\gamma_{\text{worst},c})}, \quad (4.24)$$

where γ_c^* is the cost-optimal alignment of σ_c against M and $\gamma_{\text{worst},c}$ its worst-case alignment. This score serves as a feature for the ML component (Section 4.6), where it is denoted $\phi_{\text{conf}}(c)$ in the feature table.

4.3.4 PERFORMANCE ENHANCEMENT

Definition 4.16 (Per-Activity Performance Metrics). *For each $a \in \mathcal{A}$ with event set E_a and per-event sojourn $d(e)$:*

$$\bar{d}_a = \frac{1}{|E_a|} \sum_{e \in E_a} d(e) \quad (4.25)$$

$$s_a^2 = \frac{1}{|E_a|-1} \sum_{e \in E_a} (d(e) - \bar{d}_a)^2 \quad (4.26)$$

$$\tilde{d}_a = \text{median}(\{d(e) : e \in E_a\}) \quad (4.27)$$

Per-transition waiting time:

$$w_{a \rightarrow b} = \frac{1}{|T_{a \rightarrow b}|} \sum_{(e_k, e_{k+1}) \in T_{a \rightarrow b}} [\tau(e_{k+1}) - \tau(e_k)]. \quad (4.28)$$

Full metrics vector:

$$\Phi = \left(\phi_{\text{fit}}, \phi_{\text{prec}}, \phi_{\text{gen}}, \{\bar{d}_a, s_a^2, \tilde{d}_a\}_{a \in \mathcal{A}}, \{w_{a \rightarrow b}\}_{(a,b) \in \mathcal{A}^2} \right). \quad (4.29)$$

4.3.5 STATE SPACE EXTRACTION

Definition 4.17 (SDLC State Space). *The Markov state space is obtained from the discovered process model M by a structure-preserving mapping*

$$f_M : M \rightarrow S, \quad f_M(a) = s_a,$$

that associates each observable activity $a \in \mathcal{A}$ of M with a distinct Markov state s_a (so that, by construction, state $\equiv$ activity). Formally, $S = f_M(\mathcal{A}) = S_T \cup S_A$. The absorbing (terminal) set is defined structurally as

$$S_A = \{s \in S : \text{outdeg}_M(s) = 0\},$$

*i.e., the states with no outgoing transition in M ; $S_T = S \setminus S_A$ are the non-terminal states. In the canonical SDLC taxonomy this yields $S_A = S_{\text{done}} \cup S_{\text{cancelled}}$ with $S_{\text{done}} = \{\text{done}\}$ (successful completion) and $S_{\text{cancelled}} = \{\text{cancelled}\}$; platform-specific terminal states (e.g., **merged**, **released as completion**; **rejected**, **duplicate**, **won't fix**, **archived as cancellation**) are likewise absorbing under $\text{outdeg}_M = 0$ and*

are mapped onto S_{done} or $S_{cancelled}$ as appropriate. Each transition $a \rightarrow a'$ of M induces, through f_M , the corresponding entry of the transition matrix P over S .

4.4 STAGE 3: STOCHASTIC MODELING

Remark 4.1 (Empirical Distributions). *PATHCAST employs the complete empirical distribution of sojourn times for sampling in Stage 4. This preserves the distributional shape, including skewness and heavy tails commonly observed in software development lead times (LITTLE; GRAVES, 2011; FLYVBJERG; BUDZIER, 2022), avoiding the information loss inherent in parametric assumptions. By sampling directly from empirical distributions, the Monte Carlo simulation propagates the full uncertainty structure of the observed process.*

Stage 3 satisfies the contract of Definition 4.4. The decomposition is

$$S_3 = \text{OCCUPANCY} \circ \text{SOJOURN} \circ \text{ABSORB} \circ \text{ESTIMATE} \circ \text{COUNT}, \quad (4.30)$$

detailed in the subsections that follow.

4.4.1 TRANSITION COUNTING

Property 4.3 (Count Consistency). $\sum_j C_{ij} = c_i$. for all i .

Algorithm 2 Transition Count Extraction

Require: Event log $\mathcal{L}$, state space S

Ensure: Transition count matrix C , initial state counts c_0

```

1: Initialize  $C \leftarrow \mathbf{0}$ ,  $c_0 \leftarrow \mathbf{0}$ 
2: for each trace  $\sigma_c \in \mathcal{L}$  do
3:    $c_0[\mu(e_1)] \leftarrow c_0[\mu(e_1)] + 1$ 
4:   for  $k = 1$  to  $|\sigma_c| - 1$  do
5:      $i \leftarrow \mu(e_k)$ ,  $j \leftarrow \mu(e_{k+1})$ 
6:      $C[i][j] \leftarrow C[i][j] + 1$ 
7:   end for
8: end for return  $C$ ,  $c_0$ 
```

Remark 4.2 (The DTMC as a Deliberate Projection, and the Role of M). *The Markov chain is a deliberate projection $\Pi : M \mapsto (S, P)$ of the discovered model, not a replacement for it. Three points make this explicit and forestall the objection*

that “the process model is decorative.” (i) State and support come from M , not from a free directly-follows graph. The state set $S = f_M(\mathcal{A})$ and the admissible-transition support are fixed by the sound process tree (Definition 4.17); counts in Algorithm 2 populate only M -admissible entries, and directly-follows pairs that the discovered model does not permit are treated as noise and excluded. Without M , P would degenerate into the unfiltered directly-follows DTMC (a “spaghetti” model) with no soundness guarantee; M is what constrains the chain to sound, block-structured control flow. (ii) Concurrency and loops are projected, not ignored. An AND-block of M is realized in the DTMC as the observed interleavings of its branches, and a structural loop of M is distinguished from data-level rework by the conformance features (Section 4.6); the projection therefore loses the explicit concurrency operator but retains its observable footprint. (iii) M supplies the absorbing structure and the conformance/precision signals consumed downstream. We state the resulting modeling commitment as assumption A1 (first-order Markov over S) and, rather than merely asserting it, test it (Definition 4.18).

4.4.2 MAXIMUM LIKELIHOOD ESTIMATION

$$\hat{P}_{ij} = \frac{C_{ij} + \alpha}{\sum_{j'} C_{ij'} + \alpha|S|} \quad (4.31)$$

with smoothing parameter $\alpha \geq 0$. With $\alpha = 0$ this reduces to the standard MLE (Equation 2.6).

Property 4.4 (MLE Consistency). *For $\alpha = 0$, and provided each transient state is visited infinitely often as $|\mathcal{L}| \rightarrow \infty$ ($n_i \rightarrow \infty$), the per-row multinomial MLE is consistent: $\hat{P} \xrightarrow{a.s.} P^*$ (ANDERSON; GOODMAN, 1957; BILLINGSLEY, 1961).*

The Bayesian justification of Laplace smoothing ($\alpha = 1$ corresponds to a uniform Dirichlet prior) is in Appendix A, Section A.5.

4.4.3 CONFIDENCE ASSESSMENT

For row i with $n_i = C_i$ observations, each transition probability $\hat{P}_{ij}$ is treated marginally as a binomial proportion (success = “next state is j ”), although the full row $(\hat{P}_{i1}, \dots, \hat{P}_{i|S|})$ is jointly multinomial:

$$\text{SE}(\hat{P}_{ij}) = \sqrt{\hat{P}_{ij}(1 - \hat{P}_{ij})/n_i} \quad (4.32)$$

$$\hat{P}_{ij} \pm z_{\delta/2} \cdot \text{SE}(\hat{P}_{ij}). \quad (4.33)$$

Equation 4.33 is thus an approximate (Wald) interval; for rows with small n_i or probabilities near 0 or 1, the Wilson interval is used instead, as it has better coverage in those regimes. The Wald form is reported here for readability and is adequate for the well-sampled rows ($n_i \geq n_{\min}$) that dominate the analysis.

Rows with $n_i < n_{\min}$ (default 30) are flagged. Confidence grade:

$$\text{Grade}(P) = |\{i \in S_T : n_i \geq n_{\min}\}|/|S_T|. \quad (4.34)$$

A grade below $\theta_{\text{grade}} = 0.8$ is reported as a validity caveat.

4.4.4 MARKOV-ORDER VALIDATION

The first-order assumption (A1) is the strongest commitment of the stochastic layer and the most likely target of examination. The concern is not hypothetical: variable-length Markov chains, which capture higher-than-first-order sequential dependencies, were recently shown to outperform eighteen stochastic-conformance techniques on 10/11 benchmarks (INCERTO; VANDIN; AHRABI, 2025), empirical evidence that order-1 can be insufficient for software-process sequences. Rather than assuming first-order behaviour, PATHCAST therefore submits it to a formal goodness-of-fit test per repository and falls back to an explicit state augmentation when it is rejected.

Definition 4.18 (Anderson–Goodman Markov-Order Test). *For each repository, the null hypothesis H_0 that the activity sequence is first-order Markov is tested against the first-order-alternative H_1 of second-order dependence using the likelihood-ratio statistic of Anderson and Goodman (ANDERSON; GOODMAN, 1957):*

$$G^2 = 2 \sum_h \sum_i \sum_j C_{hij} \ln \frac{C_{hij} C_{i\cdot}}{C_{ij} C_{hi\cdot}}, \quad (4.35)$$

where C_{hij} counts the second-order transitions $h \rightarrow i \rightarrow j$. Under H_0 , $G^2 \sim \chi^2$ with $|S_T|(|S| - 1)(|S| - 1)$ effective degrees of freedom (restricted to realized triples). The complementary AIC/BIC comparison of the order-1 and order-2 models is reported alongside, since G^2 is sensitive to sample size at the scale of $|\mathcal{L}|$ here.

Definition 4.19 (Rework State Augmentation). *When the order test rejects H_0 for a repository, the dominant source of memory in software processes—repeated rework—is*

reabsorbed into a first-order chain on an augmented state space

$$S' = \{(s, k) : s \in S, k \in \{0, 1, \dots, k_{\max}\}\}, \quad (4.36)$$

where k is the (capped) number of prior re-entries of the case into a review/testing state. The augmented chain is first-order by construction over S' ; $k_{\max}$ (default 2, with all higher counts collapsed) bounds the state-space growth. This makes the A1/A4 mitigation concrete rather than rhetorical: a process that is second-order through rework becomes first-order once the rework count is lifted into the state, and the test of Definition 4.18 is re-applied to confirm closure.

This two-step protocol—test, then augment only where the data demand it—keeps the canonical chain small and interpretable for the majority of repositories while removing the “unverified first-order assumption” line of attack: the order is an empirical finding per repository, not a blanket premise.

4.4.5 ABSORBING CHAIN DECOMPOSITION

States are reordered so transient precede absorbing:

$$\hat{P} = \begin{pmatrix} Q & R \\ \mathbf{0} & I \end{pmatrix}. \quad (4.37)$$

$$N = (I - Q)^{-1} \quad (4.38)$$

$$\mathbf{t} = N\mathbf{1} \quad (4.39)$$

$$B = NR \quad (4.40)$$

Property 4.5 (Invertibility). $(I - Q)$ is invertible iff every transient state can reach an absorbing state. Proof in Appendix A, Section A.1.

Property 4.6 (Absorption Certainty). $\sum_{a \in S_A} B_{ia} = 1$ for all $i \in S_T$.

Expected lead time in calendar time:

$$\mathbb{E}[\text{LeadTime} \mid s_0 = i] = \sum_{j \in S_T} N_{ij} \cdot \bar{d}_j. \quad (4.41)$$

4.4.6 EMPIRICAL SOJOURN TIME DISTRIBUTIONS

Definition 4.20 (Sojourn Observations with Right-Censoring). *For each transient state s we collect a set of sojourn observations (d_j, δ_j) , where d_j is a duration and $\delta_j \in \{0, 1\}$ its event indicator. An uncensored observation ($\delta_j = 1$) is a completed dwell time between consecutive events of the same case,*

$$\mathcal{T}_s^{obs} = \{\tau(e_{k+1}) - \tau(e_k) \mid \mu(e_k) = s, (e_k, e_{k+1}) \text{ consecutive in some } \sigma_c\}. \quad (4.42)$$

A right-censored observation ($\delta_j = 0$) arises when a case is still resident in s at the snapshot-freeze time t_{end} (Definition 5.2 of Chapter 5): its true sojourn is known only to exceed $d_j = t_{end} - \tau(e_{last})$. Discarding these open cases—as a naive ECDF would—systematically omits the longest dwells and biases the sojourn (and hence lead-time) tail downward, precisely where forecasts matter most.

Definition 4.21 (Kaplan–Meier Sojourn Estimator). $\hat{F}_s$ is estimated by the Kaplan–Meier (product-limit) estimator (KAPLAN; MEIER, 1958; KLEIN; MOESCHBERGER, 2003), which is the nonparametric MLE of a distribution under non-informative right-censoring. Let $t_{(1)} < \dots < t_{(m)}$ be the distinct uncensored sojourn times in state s , d_i the number of completed sojourns equal to $t_{(i)}$, and n_i the number at risk (observations with duration $\geq t_{(i)}$, censored or not). Then

$$\hat{S}_s(t) = \prod_{t_{(i)} \leq t} \left(1 - \frac{d_i}{n_i}\right), \quad \hat{F}_s(t) = 1 - \hat{S}_s(t). \quad (4.43)$$

When the censoring fraction is zero, every n_i counts only uncensored durations and Equation 4.43 reduces exactly to the empirical CDF of Equation 4.42, so the estimator strictly generalizes the prior design.

Remark 4.3 (Censoring Assumption). *The estimator assumes non-informative censoring: the freeze time t_{end} is exogenous to individual case dynamics (a calendar cut, not triggered by case state), so censoring is independent of the residual sojourn given s . This is the standard and defensible assumption for snapshot-based repository mining.*

Definition 4.22 (Extreme-Value Tail Closure). *The Kaplan–Meier body of $\hat{F}_s$ is empirical and cannot extrapolate beyond the largest observation; for the heavy-tailed lead times targeted by this work (where the case-level ratio $\hat{Q}_{0.98}/\hat{Q}_{0.50}$ can exceed 6), the upper tail is therefore modeled semiparametrically by peaks-over-threshold. Above a high threshold u (default the empirical 95th percentile of the uncensored sojourns), the*

exceedances $\{d_j - u : d_j > u\}$ are fitted to a Generalized Pareto Distribution (GPD), which the Pickands–Balkema–de Haan theorem identifies as the limiting excess law for essentially all heavy-tailed distributions (COLES, 2001):

$$\widehat{\Pr}(D > u + y \mid D > u) = \left(1 + \frac{\hat{\xi} y}{\hat{\beta}}\right)^{-1/\hat{\xi}}, \quad y > 0, \quad (4.44)$$

with shape $\hat{\xi}$ and scale $\hat{\beta}$ estimated by maximum likelihood (right-censored exceedances included). The shape $\hat{\xi}$ governs tail weight: $\hat{\xi} = 0$ recovers the exponential tail (and hence the previous closure as a special case), $\hat{\xi} > 0$ a Pareto-type power-law tail (finite mean only for $\hat{\xi} < 1$), and $\hat{\xi} < 0$ a bounded tail. The simulator draws sojourns from the Kaplan–Meier body below u and from the fitted GPD above u , so a censored or finite-support sample no longer truncates the simulated lead-time tail. The fitted $\hat{\xi}$ per state and the fraction of mass in the GPD tail are reported as diagnostics; $\hat{\xi} \geq 1$ (infinite-mean regime) is surfaced as an explicit validity caveat, since there the expected lead time is not finite and only quantile forecasts ($\hat{Q}_{0.85}, \hat{Q}_{0.95}, \hat{Q}_{0.98}$) remain well-posed.

Remark 4.4 (Non-Parametric Body, Extreme-Value Tail). *This hybrid is the defensible treatment for the $\hat{Q}_{0.98}/\hat{Q}_{0.50} > 6$ regime: a purely empirical estimator under-states extreme percentiles (it cannot exceed the observed maximum), a single global parametric fit mis-specifies the multimodal body, and an exponential closure under-states a genuine power-law tail. Fitting the body nonparametrically and only the tail by extreme-value theory keeps each region under the estimator that is correct for it. The residual threat—true sojourns exceeding any observable horizon under a mis-estimated $\hat{\xi}$ —is recorded as assumption A5 and revisited in the validity discussion.*

Remark 4.5 (Non-Parametric Design Choice). *The product-limit body of $\hat{F}_s$ avoids parametric misspecification (lognormal, Weibull, gamma) for the multi-modal, heavy-tailed, zero-inflated patterns observed in software process sojourns (VACANTI, 2015), while the Kaplan–Meier construction additionally recovers the censored upper tail that a plain ECDF would discard. This is reinforced by SDLC-specific evidence: the per-activity inter-event times reported for PRIMAD (GUINEA-CABRERA; HOLGADO-TERRIZA; PICO-VALENCIA, 2025) are strongly non-Poisson (mean far exceeding the median), so a single fitted exponential or Weibull family per state would misrepresent the body of the distribution—motivating the nonparametric body with an extreme-value tail adopted here.*

4.4.7 EXPECTED OCCUPANCY DISTRIBUTION AND WIP ANALYSIS

For absorbing chains, the *expected (transient) occupancy distribution* describes the expected proportion of visits to each transient state before absorption:

$$\tilde{\pi}_j = \frac{\sum_{i \in S_T} \pi_0(i) \cdot N_{ij}}{\sum_{i \in S_T} \pi_0(i) \cdot t_i}. \quad (4.45)$$

This quantity is the visit-normalised expected occupancy and should be distinguished from the classical quasi-stationary distribution (QSD), which for a discrete-time absorbing chain is the left eigenvector $\tilde{\pi}_{\text{QSD}} Q = \lambda_{\max} \tilde{\pi}_{\text{QSD}}$ associated with the dominant eigenvalue $\lambda_{\max} = \rho(Q) < 1$ of the transient *sub-stochastic* block Q (the sub-Markov transition submatrix; in discrete time Q is sub-stochastic, not an infinitesimal generator), characterising the conditional law as the number of steps $\rightarrow \infty$ (DARROCH; SENETA, 1965). States with disproportionately high $\tilde{\pi}_j$ relative to processing capacity indicate bottleneck accumulation. Full derivation in Appendix A, Section A.6.

4.4.8 REWORK ANALYSIS

Definition 4.23 (Rework Intensity).

$$R_{i,a \rightarrow b} = N_{ia} \cdot P_{ab} \quad (4.46)$$

$$R_i^{\text{total}} = \sum_{(a,b) \in \mathcal{E}_{\text{rework}}} R_{i,a \rightarrow b} \quad (4.47)$$

where $\mathcal{E}_{\text{rework}} = \{(a, b) : \text{order}(b) < \text{order}(a)\}$ relative to the nominal SDLC ordering.

4.4.9 NUMERICAL EXAMPLE

A six-state SDLC chain is fully resolved in Appendix A, Section A.7: matrix computations of Q, R, N, B and analytical lead time. The example illustrates the closed-form analytical baseline against which Stage 4 simulation results are validated.

4.5 STAGE 4: MONTE CARLO SIMULATION

Stage 4 generates probabilistic forecasts by simulating stochastic trajectories through the Markov model.

Stage 4 satisfies the contract of Definition 4.5; the empirical forecast distribution is

$$\hat{\mathcal{D}} = \{(T^{(m)}, o^{(m)}, \psi^{(m)})\}_{m=1}^{M_{\text{sim}}}. \quad (4.48)$$

4.5.1 SIMULATION ALGORITHM

Algorithm 3 Process-Aware Monte Carlo Simulation

Require: $P, \hat{F}, \pi_0$ (or fixed s_0), M_{sim} , max steps K_{max}
Ensure: $\hat{\mathcal{D}} = \{(T^{(m)}, o^{(m)}, \psi^{(m)})\}_{m=1}^{M_{\text{sim}}}$

```

1:  $\hat{\mathcal{D}} \leftarrow \emptyset$ ;  $n_{\text{na}} \leftarrow 0$ 
2: for  $m = 1$  to  $M_{\text{sim}}$  do
3:   if  $s_0$  is fixed then
4:      $s \leftarrow s_0$ 
5:   else
6:      $s \leftarrow \text{sample from } \pi_0$ 
7:   end if
8:    $T \leftarrow 0$ ;  $\psi \leftarrow [s]$ ;  $k \leftarrow 0$ 
9:   while  $s \notin S_A$  and  $k < K_{\text{max}}$  do
10:     $\Delta t \leftarrow \text{sample from } \hat{F}_s$ 
11:     $T \leftarrow T + \Delta t$ 
12:     $s' \leftarrow \text{categorical}(P[s, :])$ 
13:     $s \leftarrow s'$ ; append  $s$  to  $\psi$ ;  $k \leftarrow k + 1$ 
14:   end while
15:   if  $s \in S_A$  then
16:      $\hat{\mathcal{D}} \leftarrow \hat{\mathcal{D}} \cup \{(T, s, \psi)\}$ 
17:   else
18:      $n_{\text{na}} \leftarrow n_{\text{na}} + 1$  (non-absorbed; excluded from aggregation)
19:   end if
20: end for return  $\hat{\mathcal{D}}, n_{\text{na}}$ 

```

The safety bound $K_{\text{max}} = 1000$ prevents infinite loops in degenerate cases. Non-absorbed simulations are excluded from the forecast aggregation $\hat{\mathcal{D}}$ and counted separately as n_{na} ; the non-absorption rate $n_{\text{na}}/M_{\text{sim}}$ is reported alongside the forecast so that aggregation is performed over the $M_{\text{abs}} = M_{\text{sim}} - n_{\text{na}}$ absorbed trajectories only.

4.5.2 FORECAST EXTRACTION

PERCENTILE FORECASTS

Definition 4.24 (Percentile Forecast).

$$\hat{Q}_\alpha = \inf\{t : \hat{F}_T(t) \geq \alpha\} \quad (4.49)$$

where $\hat{F}_T$ is the empirical CDF of simulated lead times.

Standard reporting: P50, P85, P95 (VACANTI, 2015).

COMPLETION PROBABILITY

$$\hat{P}(\text{Done}) = \frac{|\{m \in \hat{\mathcal{D}} : o^{(m)} \in S_{\text{done}}\}|}{M_{\text{abs}}}, \quad M_{\text{abs}} = M_{\text{sim}} - n_{\text{na}}, \quad (4.50)$$

computed over the absorbed trajectories only, where $S_{\text{done}} \subseteq S_A$ is the set of successful-completion absorbing states (as distinct from cancellation or rejection states); the non-absorption rate $n_{\text{na}}/M_{\text{sim}}$ is reported separately.

TAIL RISK

Definition 4.25 (Tail Risk).

$$\hat{P}(\text{late}) = \frac{|\{m \in \hat{\mathcal{D}} : T^{(m)} > \tau_{\text{deadline}}\}| + n_{\text{na}}}{M_{\text{sim}}}. \quad (4.51)$$

The denominator is M_{sim} (not M_{abs}): a non-absorbed trajectory has not completed by the K_{max} horizon and is therefore counted as late by convention, so the n_{na} non-absorbed runs are added to the numerator. This makes $\hat{P}(\text{late})$ a conservative upper bound consistent with treating non-absorption as right-censored beyond the deadline. All other forecast functionals below (path frequencies, WIP) range over $\hat{\mathcal{D}}$ (the M_{abs} absorbed trajectories); the index $m \in \hat{\mathcal{D}}$ is used throughout except where M_{sim} is written explicitly.

PATH DISTRIBUTION

$$\text{freq}(\psi) = |\{m : \psi^{(m)} = \psi\}| / M_{\text{sim}}. \quad (4.52)$$

WIP COMPOSITION

$$\hat{\pi}_j^{\text{MC}} = \frac{\sum_m \text{time in state } j \text{ during } \psi^{(m)}}{\sum_m T^{(m)}}. \quad (4.53)$$

This is cross-validated against the analytical expected-occupancy distribution (Equation 4.45).

4.5.3 CONVERGENCE AND SIMULATION BUDGET

Definition 4.26 (Convergence Criterion).

$$\frac{z_{\alpha/2} \cdot \hat{\sigma}_T / \sqrt{M_{\text{sim}}}}{\hat{\mu}_T} < \epsilon, \quad \alpha = 0.05, \quad (4.54)$$

Default tolerance $\epsilon = 0.02$. The estimator's standard error $\hat{\sigma}_T / \sqrt{M_{\text{sim}}}$ is normalized by the *mean* $\hat{\mu}_T$ so that the ratio is a genuine relative half-width of the mean estimator (numerator and denominator referring to the same functional). Adaptive doubling from $M_{\text{sim}} = 1,000$ up to $M_{\text{sim}} = 100,000$. Because this criterion controls only the mean, and the forecasts of primary interest are upper percentiles ($\hat{Q}_{0.85}$, $\hat{Q}_{0.95}$), the budget is increased until the tail quantiles themselves stabilise: the batch-to-batch relative change of $\hat{Q}_{0.85}$ and $\hat{Q}_{0.95}$ is additionally required to fall below ϵ before the simulation is declared converged. This guards against the case where the mean has converged but the tail estimates remain unstable. We note that this batch-to-batch test is a stabilisation heuristic rather than a confidence interval for the quantile; the asymptotic standard error of an empirical quantile scales as $\sqrt{\alpha(1-\alpha)} / (f(\hat{Q}_\alpha) \sqrt{M})$, so for heavy tails (small f near $\hat{Q}_{0.95}$) convergence of the tail is intrinsically slower than that of the mean.

Property 4.7 (Convergence Rate). *For mean functionals with finite variance, the MC estimation error decreases as $O(1/\sqrt{M_{\text{sim}}})$ (ROBERT; CASELLA, 2004); for tail quantiles the rate is the same order but with a constant proportional to $1/f(\hat{Q}_\alpha)$, which*

motivates the separate tail-stabilisation criterion above. The Kaplan–Meier body of $\hat{F}_s$ has finite support, but the GPD tail closure (Definition 4.22) extends the simulated sojourns beyond the largest observation, so tail quantiles are no longer truncated at the observed maximum. When the fitted shape $\hat{\xi} < 1/2$ the simulated lead time has finite variance and the central limit theorem applies to the mean estimator; for $\hat{\xi} \in [1/2, 1)$ the mean is finite but heavy-tailed (slower mean convergence, quantiles unaffected), and $\hat{\xi} \geq 1$ is flagged as the infinite-mean regime in which only quantile forecasts are reported (a validity threat revisited with assumption A5).

4.5.4 REPRODUCIBILITY

PATHCAST uses Mersenne Twister (MT19937), the default in NumPy (HARRIS et al., 2020). Seeds are recorded per run, enabling deterministic replication under the same software environment (library versions and platform), since pseudo-random sequences may differ across implementations.

4.5.5 COMPARISON WITH THROUGHPUT-BASED MONTE CARLO

Table 4.3: Throughput-based vs. process-aware Monte Carlo.

Dimension	Throughput-Based	Process-Aware
Process model	None (black box)	PM-discovered
Sampling unit	Daily throughput	Transition + sojourn
Rework modeling	Not captured	Explicit via P_{ab}
State-specific delays	Not captured	Per-state $\hat{F}_s$
Path information	Not available	Full trajectory $\psi^{(m)}$
Distributional outputs	Completion only	Lead time + outcome + path + WIP

Remark 4.6 (Why Simulation Rather than Probabilistic Model Checking). *An analytical alternative to Monte Carlo over the same absorbing chain is probabilistic model checking: tools such as PRISM evaluate PCTL/CSL queries (e.g. “probability of absorption in **done** within k steps”) exactly, and have been applied to DTMCs mined from GitHub repositories (JO; KWON; KWON, 2024). PATHCAST uses model checking and simulation as complements rather than substitutes: exact queries are well suited to bounded reachability and expected-reward properties on a fixed chain, whereas the forecasting target here is the full predictive distribution of case-level lead time—including empirical, heavy-tailed sojourn laws and path-conditioned functionals*

(WIP, tail risk) that are not expressible as a single PCTL formula. Monte Carlo also accommodates the nonparametric Kaplan–Meier/GPD sojourn model directly, where an exact solver would require a phase-type approximation. Probabilistic model checking remains available as a verification oracle for the structural properties of the discovered chain.

4.6 MACHINE LEARNING ENHANCEMENT COMPONENT

The ML component satisfies the contract of Definition 4.6; the remainder of this section specifies feature engineering, label construction, training and integration with the Monte Carlo forecasts.

4.6.1 FEATURE ENGINEERING

Definition 4.27 (Feature Vector). $\mathbf{x}_c = (x_c^{(1)}, \dots, x_c^{(p)})$ with components in Table 4.4.

Table 4.4: Feature definitions for the ML component.

#	Feature	Source	Type	Definition
<i>Process-Derived (Novel)</i>				
1	Current state	$\mathcal{L}$	Cat.	$\mu(e_{\text{last}})$
2	Sojourn in current state	$\mathcal{L}$	Cont.	$t_{\text{obs}} - \tau(e_{\text{last}})$
3	Transition count	$\mathcal{L}$	Int.	$ \sigma_c^{\text{partial}} $
4	Rework count	$\mathcal{L}$	Int.	regression count
5	Conformance score	$M, \mathcal{L}$	Cont.	$\phi_{\text{conf}}(c)$
6	Markov expected remaining	N	Cont.	$\mathbb{E}[T \mid s = \mu(e_{\text{last}})] = t_{\mu(e_{\text{last}})}$
7	Completion probability	B	Cont.	$\text{Pr}[\text{absorb in done} \mid s = \mu(e_{\text{last}})] = B_{\mu(e_{\text{last}}), \text{done}}$
8	Sojourn ratio	$\hat{F}$	Cont.	$\text{sojourn} / \bar{d}_{\mu(e_{\text{last}})}$
<i>Traditional MSR (Baseline)</i>				
9	Files changed	Git	Int.	$ \text{files}(c) $
10	Lines added	Git	Int.	$\sum \text{additions}(c)$
11	Lines deleted	Git	Int.	$\sum \text{deletions}(c)$
12	Commit count	Git	Int.	$ \text{commits}(c) $
13	Author count	Git	Int.	$ \text{unique authors}(c) $
14	Priority	Jira	Ord.	priority level
15	Issue type	Jira	Cat.	Bug/Story/Task

Three feature sets are evaluated: **PM-only** (1–8), **MSR-only** (9–15), **Combined** (1–15).

4.6.2 LABEL CONSTRUCTION

Definition 4.28 (Outcome Label).

$$y_c = \begin{cases} 1 \text{ (delayed)} & T_c > \tau_{\text{threshold}} \\ 0 \text{ (on_time)} & \text{otherwise} \end{cases} \quad (4.55)$$

Three threshold strategies: *percentile-based* ($\tau_{\text{threshold}} = \hat{Q}_{0.75}$), *SLA-based*, and *relative* ($2 \cdot \tilde{d}_{\text{type}}$).

4.6.3 TRAINING PROTOCOL

Algorithm 4 ML Training with Temporal Hold-Out

Require: Event log $\mathcal{L}$, cutoff t_{cut} , feature extractor FEATURIZE

Ensure: Trained model $\hat{f}$, test metrics $\mathcal{M}$

- 1: $\mathcal{L}_{\text{train}} \leftarrow \{c : \max(\tau(\sigma_c)) \leq t_{\text{cut}}\}$
 - 2: $\mathcal{L}_{\text{test}} \leftarrow \{c : \min(\tau(\sigma_c)) > t_{\text{cut}}\}$
 - 3: $(P, N, B, \hat{F}, \vec{d}) \leftarrow \mathcal{S}_3(\mathcal{L}_{\text{train}}, M_{\text{train}})$ $\triangleright$ Markov objects re-estimated on train only
 - 4: $X_{\text{train}}, y_{\text{train}} \leftarrow \text{FEATURIZE}(\mathcal{L}_{\text{train}}; P, N, B, \hat{F}, \vec{d})$
 - 5: $X_{\text{test}}, y_{\text{test}} \leftarrow \text{FEATURIZE}(\mathcal{L}_{\text{test}}; P, N, B, \hat{F}, \vec{d})$
 - 6: **for** each candidate model $f_k \in \{\text{RF}, \text{XGBoost}, \text{LogReg}\}$ **do**
 - 7: $\text{score}_k \leftarrow \text{TemporalCV}(f_k, X_{\text{train}}, y_{\text{train}}, k = 5)$
 - 8: **end for**
 - 9: $\hat{f} \leftarrow \arg \max_k \text{score}_k$
 - 10: $\hat{y}_{\text{test}} \leftarrow \hat{f}(X_{\text{test}})$
 - 11: $\mathcal{M} \leftarrow \{\text{AUC-ROC}, \text{F1}, \text{Brier}\}$
 - 12: $\mathbf{w} \leftarrow \text{feature_importance}(\hat{f})$ **return** $\hat{f}, \mathcal{M}, \mathbf{w}$
-

Temporal CV uses expanding windows: fold k trains on $[1, t_k]$ and validates on $[t_k, t_{k+1}]$ (TASHMAN, 2000; HYNDMAN; ATHANASOPOULOS, 2021). Crucially, the Markov objects $(P, N, B, \hat{F}, \vec{d})$ that back the process-derived features (in particular the expected-remaining-time and absorption-probability features) are re-estimated from $\mathcal{L}_{\text{train}}$ alone and then *applied* to the test cases; no test-period transition or sojourn statistic enters feature construction, preventing leakage of future process behaviour into the predictors. The observation point for each case is the prefix available at t_{cut} : features are extracted from the trace prefix observed up to the cutoff, while the label y_c (on-time vs. delayed) is the case's

eventual outcome, so the task is prefix-based outcome prediction with a fixed decision time t_{cut} .

4.6.4 INTEGRATION WITH MONTE CARLO FORECASTS

Monte Carlo produces population-level distributional forecasts; **ML** produces case-level point forecasts. The deviation score identifies cases departing from the population baseline:

Definition 4.29 (Deviation Score).

$$\delta_c = \hat{p}_{\text{ML}}(\mathbf{x}_c) - \hat{P}_{\text{MC}}(\text{delayed} \mid s_0 = s), \quad (4.56)$$

where $\hat{p}_{\text{ML}}(\mathbf{x}_c) = \hat{P}_{\text{ML}}(\text{delayed} \mid \mathbf{x}_c) \in [0, 1]$ is the classifier’s predicted delay probability (not the thresholded class label), so that both terms are on the same probability scale. $\delta_c > 0$ indicates above-baseline delay risk; $\delta_c < 0$ indicates below-baseline.

4.7 PIPELINE INTEGRATION

The stages compose into a directed acyclic graph (Stage 2 feeds Stage 3 and the ML component; Stage 3 feeds Stage 4 and the ML component) under the inter-stage data contracts already stated (Definitions 4.2–4.6; data objects summarised in Table 4.5). Two execution modes are supported: a full historical run that estimates $(P, \hat{F}, \hat{f})$ end-to-end, and an incremental operational mode that scores new cases through Stages 4 and ML only, periodically re-estimating the upstream artefacts. Re-estimation may be triggered either on a calendar schedule (e.g., weekly or per sprint) or conditionally, when a monitored degradation signal—such as a rise in the rolling CRPS or a drift in the expected-occupancy distribution—exceeds a configured threshold.

Proposition 4.1 (Pipeline Correctness). *If each stage satisfies its contract, then the composed pipeline $\mathcal{F}$ produces a valid empirical forecast distribution $\hat{\mathcal{D}}$ for any input $\mathcal{R}$ satisfying: (i) at least one repository contains events from ≥ 2 data sources; (ii) $|\mathcal{L}| \geq n_{\min}^{\text{cases}}$ (default 50); (iii) $|S_T| \geq 1$ and $|S_A| \geq 1$.*

Proof. By structural induction on stages: each postcondition implies the next precondition ($S1 \rightarrow S2$ event-log validity; $S2 \rightarrow S3$ sound model and non-empty states; $S3 \rightarrow S4$ row-stochastic P and non-empty $\hat{F}$). Under Property 4.6, every

Table 4.5: Inter-stage data contracts.

From	To	Data Object	Specification
Sources	S1	$\mathcal{R}$	Git, Jira, CI/CD logs
S1	S2	$\mathcal{L}$	XES event log
S2	S3	(M, Φ, S)	Model + metrics + states
S2	ML	$(\mathcal{L}, \Phi)$	Log + per-case scores
S3	S4	$(P, \hat{F})$	Matrix + sojourn distributions
S3	ML	(P, N, B)	Markov objects
S4	Output	$\hat{\mathcal{D}}$	Empirical forecast distribution
ML	Output	$(\hat{f}, \mathbf{w})$	Predictor + importances

simulation is absorbed almost surely, so S4 terminates with probability one; the $K_{\max}$ bound acts solely as a defensive implementation safeguard against degenerate or numerically pathological inputs and is not required for the termination argument. This is a compositional-correctness property that depends on the inter-stage software contracts rather than a purely analytic theorem, and is stated as a proposition accordingly. $\square$ $\square$

4.8 FORMAL PROPERTIES AND THEORETICAL GUARANTEES

4.8.1 STATISTICAL CONSISTENCY

Theorem 4.1 (Asymptotic Consistency). *As $|\mathcal{L}| \rightarrow \infty$ and $M_{sim} \rightarrow \infty$: (i) $\hat{P} \xrightarrow{a.s.} P^*$; (ii) $\hat{F}_T \xrightarrow{d} F_T^*$; (iii) $\hat{Q}_\alpha \rightarrow Q_\alpha^*$ at continuity points.*

Proof. (i) Property 4.4. (ii) Continuous mapping theorem applied to (i) and the Glivenko–Cantelli convergence $\hat{F}_s \rightarrow F_s^*$. (iii) Quantile continuity. $\square$ $\square$

4.8.2 CALIBRATION PROPERTY

Definition 4.30 (Forecast Calibration).

$$CalErr(\alpha) = |\hat{P}(T \leq \hat{Q}_\alpha) - \alpha| \quad (4.57)$$

$$ACE = \frac{1}{|\mathcal{A}|} \sum_{\alpha \in \mathcal{A}} CalErr(\alpha). \quad (4.58)$$

Theorem 4.2 (End-to-End Convergence). *Under (a) per-state visit growth ($n_i \rightarrow \infty$ a.s. for every transient state i as the number of cases grows, so that each row of P is identifiable), (b) sojourn consistency ($\sup_t |\hat{F}_i^{(n)}(t) - F_i^*(t)| \xrightarrow{a.s.} 0$), (c) $K(n) \rightarrow \infty$:*

$$\sup_t |\hat{F}_{n,K}(t) - F^*(t)| \xrightarrow{a.s.} 0 \quad (4.59)$$

$$ACE_{n,K} \rightarrow 0. \quad (4.60)$$

Proof sketch. Stage 1. The chain is absorbing (hence not ergodic); consistency of $\hat{P}$ instead follows row-wise: conditional on the visit count n_i , each row is a multinomial MLE, so by the SLLN for i.i.d. transition counts—valid once $n_i \rightarrow \infty$ under assumption (a)— $\hat{P}^{(n)} \rightarrow P$ entry-wise, hence in any matrix norm (ANDERSON; GOODMAN, 1957; BILLINGSLEY, 1961). *Stage 2.* Glivenko–Cantelli applied per state plus union bound across the finite S . *Stage 3.* Continuous mapping theorem on simulated trajectories; second Glivenko–Cantelli application across K replications gives uniform convergence of $\hat{F}_{n,K}$ to F^* . ACE convergence follows. $\square$

Remark 4.7 (Practical Implications). *The pipeline is end-to-end consistent. Convergence rate is bounded by (a) per-state effective sample size, (b) non-stationarity of real processes, (c) MC budget. Empirical evaluation in preparation.*

4.8.3 INFORMATION-THEORETIC ADVANTAGE OF PROCESS AWARENESS

Theorem 4.3 (Information Gain from Process Structure).

$$I(T; \psi) = H(T) - H(T | \psi) \geq 0, \quad (4.61)$$

with equality iff T is independent of ψ .

In practice $I(T; \psi) > 0$ because rework-laden paths have systematically longer lead times. Process-aware MC captures this variance decomposition; throughput-based MC does not.

4.8.4 COMPLEXITY ANALYSIS

$$C_{\text{total}} \approx \mathcal{O}(|\mathcal{L}|) + \mathcal{O}(M_{\text{sim}} \cdot \bar{K}). \quad (4.62)$$

Table 4.6: Computational complexity per pipeline stage.

Stage	Time Complexity	Dominant Factor
S1: Event Log	$O(E^{\text{raw}} \cdot \mathcal{K})$	raw events $\times$ strategies
S2: Discovery (IM)	$O(\mathcal{L} \cdot \mathcal{A} ^2)$	log $\times$ activities ²
S2: Conformance	$O(\mathcal{C} \cdot M)$	cases $\times$ model size
S3: Markov	$O(\mathcal{L} + S_T ^3)$	counting + inversion
S4: Monte Carlo	$O(M_{\text{sim}} \cdot \bar{K})$	simulations $\times$ path length
ML: Training	$O(\mathcal{C} \cdot p \cdot T)$	cases $\times$ features $\times$ trees

SPACE COMPLEXITY

$$S_{\text{total}} = O(|\mathcal{L}|) + O(|S_T|^2) + O(M_{\text{sim}}). \quad (4.63)$$

A streaming MC implementation reduces the third term to $O(1)$ via incremental quantile estimation.

PARALLELIZATION

Three levels: (i) intra-simulation CPU multicore (embarrassingly parallel); (ii) GPU offload (one CUDA thread per replication, throughput $\sim 10^8$ /s for $|S_T| < 100$ (L'ECUYER et al., 2017)); (iii) distributed pipeline parallelism across repositories.

Remark 4.8 (Scalability Bounds). *The MC stage dominates and is fully parallelizable; sequential stages are $O(|\mathcal{L}|)$ and run in seconds for $\leq 10^6$ events.*

4.8.5 ASSUMPTIONS AND BOUNDARY CONDITIONS**4.9 CHAPTER SUMMARY**

This chapter presented PATHCAST, a formally specified forecasting framework for software-development processes that integrates process mining, stochastic modeling, Monte Carlo simulation, and machine learning into a unified architecture for transforming heterogeneous software-engineering repositories into probabilistic forecasts. Stage 1 formalized event-log construction from multiple data sources through configurable extraction, correlation, mapping, and

Table 4.7: Assumptions, implications, and mitigations.

Assumption		Violation	Conse-	Mitigation
		quence		
A1:	First-order Markov	Underestimates rework-conditioned delays		Anderson–Goodman order test (Def. 4.18); rework state augmentation when rejected
A2:	Stationarity	Stale forecasts after process changes		Sliding window; drift detection
A3:	Case independence	Ignores queueing/resource contention		Acknowledged limitation; a concrete relaxation is to parametrise per-state sojourns with log-discovered probabilistic resource-availability calendars (LÓPEZ-PINTADO; DUMAS, 2023)
A4:	Sojourn-history independence	Underestimates delays for reworked items		State augmentation; ML features
A5:	Complete case observation	Downward tail bias if open cases dropped		Kaplan–Meier sojourn estimator (Def. 4.21) incorporates right-censored dwells; GPD extreme-value tail closure (Def. 4.22) for the heavy upper tail

quality-assessment procedures. Stage 2 introduced process discovery, conformance analysis, and performance enhancement as mechanisms for extracting executable process representations and process-quality metrics from SDLC event logs. Stage 3 defined the stochastic layer—Markov-chain estimation, absorbing-state analysis, empirical sojourn-time modeling, bottleneck identification, and rework quantification. Stage 4 formalized process-aware Monte Carlo simulation for distributional forecasts (completion probabilities, percentile estimates, tail-risk, and path-level outcome distributions). Finally, the machine-learning enhancement introduced a family of process-derived predictive features based on conformance, state occupancy, expected remaining time, absorption probabilities, and rework indicators.

A central contribution is the formal composition of these components through explicit inter-stage contracts. Unlike the prior approaches identified in Chapter 3, which apply process mining, stochastic modeling, simulation, or machine learning in isolation or through loosely coupled pipelines, PATHCAST defines a unified end-to-end architecture in which each stage produces formally specified outputs that satisfy the input requirements of the next. This compositional design enabled the derivation of correctness guarantees, convergence properties, calibration criteria, complexity bounds, and information-theoretic arguments supporting the value of process-aware forecasting.

The framework directly addresses the five gaps identified in the systematic review: the event-log construction framework addresses the lack of standardized SDLC event-log generation (F2); the integration of process mining with absorbing Markov chains and Monte Carlo simulation addresses the stochastic-integration gap (F3); the process-derived predictive features address the feature-engineering void (F5); the formal pipeline architecture addresses the fragmentation across existing approaches (F4); and the transition from descriptive process analysis toward probabilistic forecasting responds to the descriptive ceiling (F1). Concretely, the PATHCAST components map to the thesis contributions as follows:

- **Stage 1** addresses **C2**: configurable, multi-source framework with cascade correlation strategies and formal quality metrics.
- **Stages 2–3** address **C3**: process-structural stochastic model with empirically grounded parameters.
- **Stage 4** also addresses **C3**: distributional forecasts capturing state-specific variability and path diversity.

- **ML Enhancement** addresses **C4**: novel feature family derived from PM and Markov analysis.
- **Pipeline Composition** addresses **C1**: formal inter-stage contracts and correctness guarantees (Proposition 4.1).

C5 (Empirical Evidence) is addressed by the empirical evaluation (in preparation).

Table 4.8: Method components vs. research questions.

RQ	Components	Evaluation
RQ1: Event log construction	S1 + S2	Fitness, precision, coverage
RQ2: Stochastic forecasting	S3 + S4	MAE, calibration, CRPS
RQ3: PM-derived ML features	ML Component	AUC-ROC, F1, feature importance

A dedicated implementation chapter (in preparation) will describe the implementation architecture, and the empirical-evaluation chapter (in preparation) will present the evaluation on the 48-repository dataset.

5

Research Methodology

This chapter describes the methodological framework. Section 5.1 presents Design Science Research. Section 5.2 describes the dataset. Section 5.3 formalizes the experimental design. Section 5.4 defines evaluation metrics. Section 5.5 specifies baselines. Section 5.6 describes statistical procedures. Section 5.7 discusses threats to validity.

5.1 DESIGN SCIENCE RESEARCH PARADIGM

This research adopts Design Science Research (DSR) following Hevner et al. (HEVNER et al., 2004) and the DSRM process of Peffers et al. (PEFFERS et al., 2007).

Design Science Research was selected because the primary objective of this work is not merely to explain or predict an existing phenomenon, but to *design*, formally specify, implement, and evaluate an artifact capable of solving a previously identified problem—the absence of an integrated, process-aware probabilistic forecasting method for software-development processes. This distinguishes the present research from explanatory or purely observational paradigms: it is engineering research, in which the artifact (the PATHCAST method and its software instantiation) is itself the central contribution, and the evaluation establishes that the artifact meets its intended goals.

5.1.1 DSR ARTIFACT CHARACTERIZATION

The artifact produced is a **method**: PATHCAST is a formally specified pipeline that prescribes how to transform repository data into probabilistic process forecasts. The supporting **instantiation** is a Python tool implementing the four pipeline stages and the ML component. In the terms of Gregor and Hevner (GREGOR; HEVNER, 2013), PATHCAST is therefore a Level-2 design artifact, combining a *method* and an *instantiation*: the method specifies the transformation process from SDLC event data to probabilistic forecasts, while the instantiation operationalizes this method as an executable software system. This dual characterization is what allows the artifact to be evaluated both analytically (the formal properties of Chapter 4) and empirically (the experiments designed in this chapter).

5.1.2 DSRM PROCESS INSTANTIATION

Table 5.1: Mapping of DSRM activities to thesis components.

Activity	DSRM Activity	Chapter	Deliverable
1	Problem Identification	Ch. 1 Ch. 2	Three research problems Gap analysis table
2	Objectives Definition	Ch. 1	3 RQs, 4 objectives, 5 contributions
3	Design & Development	Ch. 4 Ch. 6	PATHCAST method Python tool (v0.1.0, partial)
4	Demonstration	Ch. 6	47/48 repositories (Stage 1)
5	Evaluation	Ch. 6	E1 pilot + 10-repo batch (partial)
6	Communication	This thesis	Conference/journal papers

5.1.3 DSR KNOWLEDGE CONTRIBUTION

Following Gregor and Hevner (GREGOR; HEVNER, 2013), this research operates in the *Improvement* quadrant: a known problem class (software process forecasting) addressed with a new solution (process-aware stochastic simulation parameterized by process mining). The contribution is the formal integration of established techniques into a SDLC-specific pipeline, not the invention of new individual techniques.

5.2 DATASET DESCRIPTION

The empirical evaluation uses 48 open-source software repositories.

5.2.1 DATASET RATIONALE

The 48 repositories were selected to maximize heterogeneity with respect to project size, contributor population, activity patterns, programming ecosystem, and governance model, reducing the risk that the results are specific to a single development context and thereby strengthening external validity. The final size of 48 reflects a deliberate compromise between this diversity and the computational cost of executing the complete PATHCAST pipeline—process discovery, Markov estimation, Monte Carlo simulation, and machine-learning evaluation—on every repository under temporal hold-out and leave-one-project-out protocols. A larger corpus would increase statistical power marginally (the cross-repository tests already attain high power for large effects; Section 5.6.4) at a substantial increase in pipeline-execution and verification effort, whereas a smaller corpus would compromise stratified coverage of the diversity dimensions in Table 5.3.

5.2.2 REPOSITORY SELECTION CRITERIA

Inclusion criteria:

1. **Minimum commit volume:** $\geq 1,000$ commits in main.
2. **Issue tracker usage:** GitHub Issues or Jira with ≥ 100 closed issues containing state transitions.
3. **Pull request workflow:** evidence of PR-based development.
4. **Activity period:** ≥ 12 months continuous activity.
5. **Non-trivial contributors:** ≥ 5 distinct contributors.
6. **Non-fork:** not a fork of another project.

Exclusion: $> 50\%$ bot commits; external-bug-only issue tracking; mirror or archive repositories.

5.2.3 SELECTION PROCESS

Step 1 — Candidate generation. 150 candidates from SEART GitHub Search (DABIC; AGHAJANI; BAVOTA, 2021): $\geq 1,000$ commits, issues+PRs, non-fork, last push $\geq 2022-01-01$, languages {Python, JS/TS, Java, Go, Rust, C/C++}.

Step 2 — Criterion verification. Each candidate verified against C2 and C5 via the GitHub REST API (GitHub, Inc., 2023).

Step 3 — Stratified selection. 48 repositories selected by stratified sampling across the dimensions of Table 5.3: 3–4 per primary language; balanced across scale tiers; ≥ 4 per governance model. Stratification was adopted to reduce selection bias and to ensure representation across heterogeneous development contexts (programming ecosystems, project scales, and governance structures), rather than over-representing any single context. Formally, the selection is the function

Definition 5.1 (Repository Selection Function).

$$SELECT : (\mathcal{P}, C_{repo}) \mapsto \mathcal{D}, \quad \mathcal{D} \subseteq \mathcal{P}, |\mathcal{D}| = 48,$$

where $\mathcal{P}$ is the candidate pool obtained from SEART (Step 1), C_{repo} are the inclusion criteria of Section 5.2.2, and $\mathcal{D}$ is the stratified sample satisfying the per-stratum quotas above.

Step 4 — Jira enrichment. For repositories whose development workflow relied on Jira as the primary issue-tracking system (the Apache and Spring projects), the issue histories were enriched with the Public Jira Dataset of Montgomery et al. (MONTGOMERY; LÜDERS; MAALEJ, 2022), enabling recovery of workflow-state transitions that are not directly available through the GitHub APIs.

The complete list of 48 repositories is in Appendix J. The overall selection and enrichment flow is summarized in Figure 5.1.

5.2.4 DATA COLLECTION

Per repository:

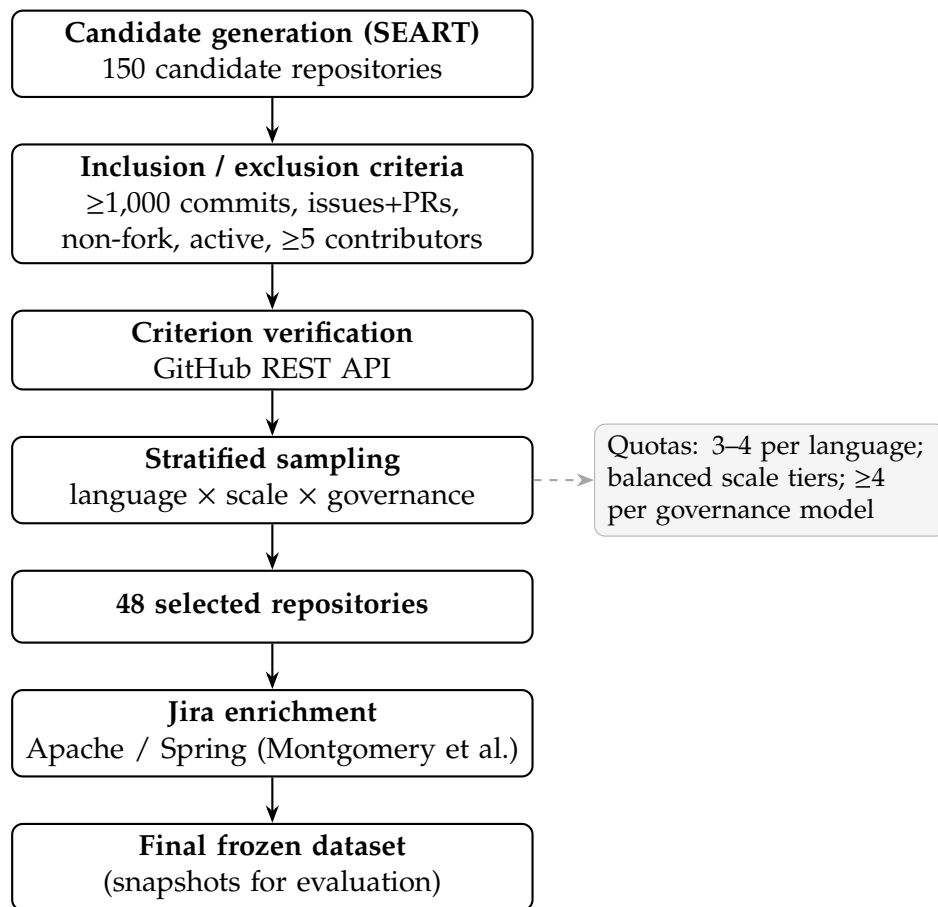


Figure 5.1: Repository selection and enrichment process.

- Git commit history via `/repos/{owner}/{repo}/commits`.
- Issue lifecycle via the timeline events API (or Jira (MONTGOMERY; LÜDERS; MAALEJ, 2022) for Apache/Spring).
- Pull requests via the GraphQL API, collecting the lifecycle events that feed Stage 1: `PR_created`, `PR_reviewed`, `PR_approved`, `PR_merged`, and `PR_closed` (with their timestamps and authoring/reviewing actors).
- GitHub Actions runs (only ≥ 50 runs counts as Advanced maturity). The threshold of 50 workflow executions was adopted to reduce the influence of sporadic CI/CD usage and to characterize repositories with sustained automation practices, rather than projects that merely trialed CI/CD.

To guarantee reproducibility against the continuous evolution of live repositories, each repository is analyzed as a frozen snapshot.

Definition 5.2 (Repository Snapshot). *For a repository r collected at time t , the repository snapshot $R^{(t)}$ denotes the immutable state of r observed at t : the set of commits, issues, pull requests, and CI/CD events with timestamp $\leq t$. All analyses in this thesis are conducted over snapshots $\{R_r^{(t_r)}\}_{r \in \mathcal{D}}$ collected within the window $[t_{start}, t_{end}]$; per-repository case counts and harvest status are recorded in Table J.1 and `repo_manifest.json`.*

The PathCast v0.1.0 harvest executed in June 2026 treats this window as a single freeze. GitHub Issues timeline collection used an authenticated REST token with rate-limit backoff; the completion batch (`nestjs/nest`, `vitejs/vite`, `pandas-dev/pandas`) required approximately 28 hours of wall-clock time, of which `pandas-dev/pandas` alone (65,634 issues $\times$ timeline events) accounted for roughly 20 hours. The batch finished on 9 June 2026; commit totals were refreshed on 8 June 2026; remaining repositories retain snapshots from the April–June 2026 development cycle.

Each snapshot is stored in PostgreSQL for Stage 1 ingestion.

5.2.5 DATASET DESCRIPTIVE STATISTICS

The aggregate statistics in Table 5.2 are computed from the Stage-1 GitHub event log (`github_event_log.parquet`) and commit totals in `repo_metadata.json`. Both inputs are consumed by `scripts/generate_dataset_summary_tex.py`, which regenerates the LaTeX table after each harvest run (June 2026 snapshot).

Across the 47 usable repositories recorded in `repo_manifest.json`, mapped process-case counts are highly skewed (median 109 cases; mean 3,076). Three

Table 5.2: Aggregate dataset statistics (Stage-1 harvest of 46/48 GitHub repositories with event logs; 47/48 evaluation repositories confirmed; see `repo_manifest.json`).

Metric	Total	Mean	Median	Std Dev
Commits [†]	3,518,030	85,805.6	21,502	237,891.6
Closed issues [‡]	115,448	2,509.7	27.5	9,784.1
Pull-request traces [§]	37,061	805.7	48	3,275.4
Contributors [¶]	52,031	1,131.1	50	3,403.9
Activity span (months)	—	14.9	0.8	39.7

[†]GitHub REST commit pagination on default branch ($n = 41$ GitHub repos; Jun 2026 snapshot). [‡]ISSUE_CLOSED events in the Stage-1 GitHub event log. [§]Distinct PR cases with at least one PR lifecycle event. [¶]Distinct `org:resource` values per repository (summed across repositories; not de-duplicated across the corpus).

repositories harvested in June 2026 (`pandas-dev/pandas`, `nestjs/nest`, `vitejs/vite`) dominate volume; 15 repositories yield fewer than 50 mapped cases and 23 yield fewer than 100 despite satisfying the ≥ 100 closed-issue selection criterion (C2) at repository freeze time—for example, `django/django` (62 cases) and `postgres/postgres` (2 cases) in the Stage-1 log. Repository-level reporting and the minimum-case guard in Equation (5.1) mitigate this heterogeneity in E1/E2.

Table 5.3: Dataset diversity profile.

Dimension	Coverage
Primary languages	Python, JS/TS, Java, Go, Rust, C/C++
Domain categories	Web, CLI, libraries, DevOps, data processing
Project scale	Small (< 5K), Medium (5–20K), Large (> 20K commits)
Governance model	Foundation, corporate, community-driven
Workflow maturity	Basic / Standard / Advanced
Issue source	GitHub Issues; Jira via (MONTGOMERY; LÜDERS; MAALEJ, 2022)

5.2.6 ETHICAL CONSIDERATIONS

All data analysed in this research originate from publicly accessible open-source software repositories and issue-tracking systems. The study does not access private repositories, proprietary development artefacts, restricted datasets, or confidential organisational information. Repository metadata, commits, pull requests, issue histories, and workflow events were collected exclusively through

publicly available APIs and datasets, in accordance with the corresponding platform terms of service.

Although GitHub usernames and contributor identifiers are publicly visible, individual developers are *not* the unit of analysis. All analyses are conducted at the repository, workflow, process, or event-log level, and no attempt is made to profile, rank, evaluate, or infer personal characteristics of individual contributors; results are reported only in aggregated form.

The study complies with the GitHub Terms of Service and the usage policies of all external data sources, including the Public Jira Dataset of Montgomery et al. (MONTGOMERY; LÜDERS; MAALEJ, 2022). Data collection was restricted to information necessary for process-mining and forecasting purposes. The research presents minimal ethical risk because it relies exclusively on publicly available software-engineering artefacts and does not involve human subjects, interventions, surveys, or the collection of sensitive personal data; consequently, no personally identifiable information beyond publicly available repository metadata is processed. To support transparency and reproducibility, the selection criteria, collection procedures, preprocessing steps, and analysis scripts are documented in the replication package.

5.3 EXPERIMENTAL DESIGN

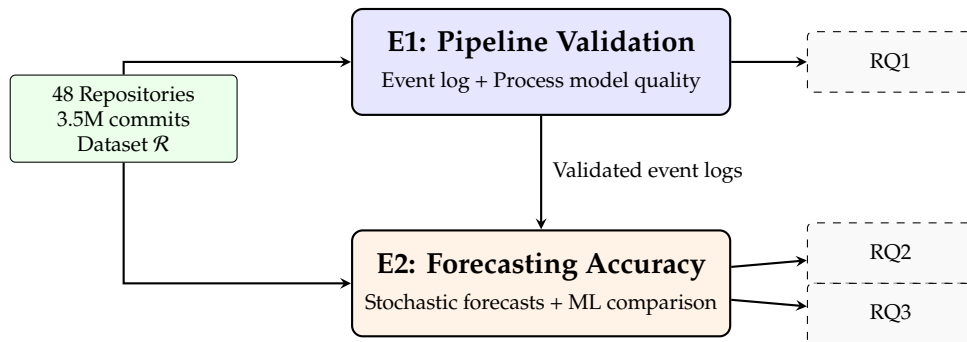


Figure 5.2: Two-stage experimental design.

5.3.1 E1: PIPELINE VALIDATION

E1 assesses whether Stages 1 and 2 produce event logs and process models of sufficient quality. Addresses RQ1.

SCOPE

1. Event log quality: correlation coverage, activity coverage, trace completeness.
2. Process model quality: fitness, precision, generalization via alignment-based conformance (Section 4.3.3).

PROTOCOL

For each $r_i \in \mathcal{R}$:

1. Execute Stage 1 with the default correlation configuration.
2. Compute event log quality metrics (Definition 4.14).
3. Execute Stage 2 with IMf at $\theta_{\text{noise}} = 0.2$.
4. Compute fitness, precision, generalization.
5. Record process variants and state space $|S|$.
6. Flag transitions with < 30 observations; apply Laplace smoothing ($\alpha = 1$) at Stage 3.

SENSITIVITY ANALYSIS

θ_{noise} varied over $\{0.0, 0.1, 0.2, 0.3, 0.5\}$ on 10 representative repositories. Pareto front of fitness vs. precision identifies the threshold balancing both objectives.

ACCEPTANCE CRITERIA

$$\text{fitness}(\mathcal{L}, M) \geq 0.7 \wedge \text{precision}(\mathcal{L}, M) \geq 0.5 \wedge \text{coverage}(\mathcal{L}) \geq 0.5 \wedge |S_T| \geq 2 \wedge |\mathcal{L}_{\text{test}}| \geq 50. \quad (5.1)$$

The fitness and precision thresholds are imposed jointly so that an accepted model both replays the observed behaviour (fitness ≥ 0.7) and does not overgeneralize into a flower-like model admitting arbitrary behaviour (precision ≥ 0.5); the values are consistent with process-mining practice on noisy, real-world logs. The fifth condition, $|\mathcal{L}_{\text{test}}| \geq 50$, requires that the temporal hold-out test set

(Definition 5.3) contain at least 50 completed cases. Below this threshold, empirical percentile estimates exhibit high sampling variance—the standard error of a sample quantile scales as $O(1/\sqrt{n})$ —and CRPS, which integrates calibration across the full predictive distribution, becomes unreliable for tail quantiles (GNEITING; RAFTERY, 2007). The threshold is consistent with simulation practice for proper scoring rules and with AUC estimation, which stabilises near $n \approx 50$ for approximately balanced outcomes (HANLEY; MCNEIL, 1982). Repositories satisfying the quality conditions but failing the case-count threshold are reported as *quality-validated but statistically insufficient for E2* and excluded from E2 only.

5.3.2 E2: FORECASTING ACCURACY

E2 assesses Stages 3 and 4 forecasts and the predictive contribution of process-derived ML features. Addresses RQ2 and RQ3.

SCOPE

For each repository passing E1:

1. Execute Stages 3 and 4 to produce probabilistic lead time forecasts.
2. Execute the ML component with three feature sets (PM-only, MSR-only, Combined).
3. Compare against ground truth.

TEMPORAL HOLD-OUT PROTOCOL

Definition 5.3 (Temporal Hold-Out). *For each repository with event log $\mathcal{L}$ over $[t_{\min}, t_{\max}]$:*

$$t_{\text{cut}} = t_{\min} + 0.7 \cdot (t_{\max} - t_{\min}) \quad (5.2)$$

$$\mathcal{L}_{\text{train}} = \{c : \max(\tau(\sigma_c)) \leq t_{\text{cut}}\} \quad (5.3)$$

$$\mathcal{L}_{\text{test}} = \{c : \min(\tau(\sigma_c)) > t_{\text{cut}}\} \quad (5.4)$$

Random case-level splits are deliberately avoided: assigning cases to train and test without regard to event chronology would violate temporal causality

by allowing training traces to include events that post-date the forecast origin, thereby leaking future information into the estimators (TASHMAN, 2000). The 70/30 temporal split follows the same convention; sensitivity to 60/40 and 80/20 is assessed on a representative subset.

Model estimation: $P, \hat{F}$, ML models estimated from $\mathcal{L}_{\text{train}}$ only.

Forecast evaluation: for each $c \in \mathcal{L}_{\text{test}}$, forecast vs. observed $T_c^{\text{obs}}, y_c^{\text{obs}}$.

CROSS-REPOSITORY GENERALIZATION

Definition 5.4 (LOPO Generalization). *Let $G \subseteq \mathcal{D}$ be a domain group of repositories sharing an ecosystem and workflow profile. Under leave-one-project-out (LOPO) evaluation, for each $r_i \in G$ the model is trained on $\text{Train} = G \setminus \{r_i\}$ and tested on $\text{Test} = \{r_i\}$, repeated for all $r_i \in G$. Group-level generalization is the mean over the $|G|$ held-out folds. Groups with $|G| < 5$ are reported descriptively only.*

LOPO here is *within-domain* generalization: folds are held out inside a single ecosystem/workflow group, so the resulting estimate is an optimistic (within-domain) bound rather than fully heterogeneous cross-repository transfer. Current dataset: the Python web-framework group (django/django, pallets/flask, fastapi/fastapi, psf/requests, pallets/click) and the Go DevOps group (kubernetes/kubect1, helm/helm, hashicorp/terraform, prometheus/prometheus, grafana/grafana) reach $|G| = 5$ and qualify; smaller groups are reported descriptively only. Because each qualifying group has only $|G| = 5$ folds, within-domain LOPO results are interpreted as exploratory; a global 48-fold leave-one-repository-out pass is additionally reported as the honest cross-repository generalization test.

5.4 EVALUATION METRICS

5.4.1 METRICS FOR RQ1: EVENT LOG AND PROCESS MODEL QUALITY

Event log quality and process model quality are reported separately to isolate sparse-data effects from modeling effects.

Metric rationale. Correlation Coverage signals Stage 1 fidelity. Activity Coverage contextualizes fitness: high fitness on a log with low activity coverage may

Table 5.4: Evaluation metrics for RQ1.

Metric	Range	Definition
<i>Event Log Quality</i>		
Correlation Coverage	[0, 1]	Fraction of raw events assigned to a case (Eq. 4.13).
Activity Coverage	[0, 1]	Fraction of SDLC taxonomy observed (Eq. 4.19).
Trace Completeness	[0, 1]	Fraction of cases reaching absorbing state (closed-case ratio, Eq. 4.17).
Mean Trace Length	$\mathbb{R}^+$	Average events per trace (Eq. 4.18).
Variant Count	$\mathbb{N}$	Distinct activity sequences (Eq. 4.20).
<i>Process Model Quality</i>		
Fitness	[0, 1]	Alignment-based fitness (Eq. 2.1). Higher better.
Precision	[0, 1]	Fraction of model behaviour observed in log. Higher better.
Generalization	[0, 1]	Capacity to capture unseen legitimate behaviour.

reflect an oversimplified model. Trace Completeness bounds the supervised-learning training set. Mean Trace Length and Variant Count characterize structural complexity.

The four-quality framework (AALST, 2016) additionally includes *simplicity*; this is constrained by construction in PATHCAST since the state space is bounded by the SDLC taxonomy (Section 4.2.3) and not reported separately.

5.4.2 METRICS FOR RQ2: STOCHASTIC FORECASTING ACCURACY

CRPS is the primary metric *because PATHCAST produces full predictive distributions rather than deterministic point forecasts*, so a strictly proper scoring rule that evaluates the entire predicted distribution is the appropriate primary criterion; MAE, MAPE, and MdAE are secondary point summaries. CRPS reduces to MAE when the forecast degenerates to a point mass (GNEITING; RAFTERY, 2007), preserving comparability with deterministic baselines. Derivation in Appendix B.

Point metrics. MAE in calendar days; MdAE robust to outliers. For cross-repository comparison we report *sMAPE* (symmetric MAPE, $\frac{100}{n} \sum_c |T_c^{\text{obs}} - \hat{T}_c| / [(|T_c^{\text{obs}}| +$

Table 5.5: Evaluation metrics for RQ2.

Metric	Type	Definition
<i>Point Forecast Accuracy</i>		
MAE	Error	$\frac{1}{n} \sum_c T_c^{\text{obs}} - \hat{T}_c^{\text{P50}} $
MAPE	Error (%)	$\frac{100}{n} \sum_c T_c^{\text{obs}} - \hat{T}_c^{\text{P50}} / T_c^{\text{obs}}$
MdAE	Error	$\text{median}(T_c^{\text{obs}} - \hat{T}_c^{\text{P50}})$
<i>Distributional Accuracy</i>		
CRPS	Score	$\frac{1}{n} \sum_c \int [\hat{F}_c(t) - \mathbf{1}(t \geq T_c^{\text{obs}})]^2 dt$
ACE	Error	$\frac{1}{9} \sum_{\alpha \in \{0.1, \dots, 0.9\}} \hat{p}_\alpha - \alpha $
PI Coverage 85%	$[0, 1]$	Fraction of T_c^{obs} inside the central 85% prediction interval $[\hat{Q}_{0.075}, \hat{Q}_{0.925}]$.
<i>Outcome Prediction</i>		
Completion Accuracy	$[0, 1]$	Done vs. Cancelled classification accuracy at threshold τ (Eq. 5.5).

$|\hat{T}_c|/2$) as the primary relative-error metric rather than MAPE, because MAPE is unbounded and asymmetric for the small lead times common in hotfixes and trivial pull requests (it penalises over-forecasting arbitrarily); MAPE is retained only as a secondary, legacy-comparable figure. Cases with an observed lead time of zero (e.g., same-day hotfixes) are excluded from MAPE (division by zero) but remain in sMAPE, MAE, MdAE, and CRPS.

ACE averages absolute deviation across nine deciles. $\text{ACE} = 0$ denotes perfect calibration; $\text{ACE} > 0.1$ indicates systematic over- or underconfidence. Reported alongside CRPS because a forecast can have low CRPS while being miscalibrated.

Reliability diagrams. In addition to the scalar ACE, a *reliability diagram* (observed vs. nominal coverage across the quantile checkpoints, with the diagonal denoting perfect calibration) is reported as a visual calibration artefact for each repository and for the pooled test set. This is now standard practice in probabilistic-forecast evaluation (GNEITING; RAFTERY, 2007) and exposes *where* on the distribution miscalibration occurs (e.g., over-confident tails), which a single ACE value cannot convey.

Coverage at 85% reports the empirical coverage of the *central* 85% prediction interval $[\hat{Q}_{0.075}, \hat{Q}_{0.925}]$ (i.e., the interval between the 7.5th and 92.5th simulated

percentiles), which is the standard agile commitment interval (VACANTI, 2015); it complements ACE by focusing on the single operationally-relevant interval. A well-calibrated forecaster yields an empirical coverage close to 0.85.

Completion Accuracy. Because PATHCAST produces a completion *probability* $\hat{P}(\text{Done})$ (Eq. 4.50) rather than a hard label, the predicted outcome is obtained by thresholding, and completion accuracy is defined as

$$\text{Acc} = \frac{1}{n} \sum_{c=1}^n \mathbf{1}[\mathbf{1}\{\hat{P}_c(\text{Done}) \geq \tau\} = y_c], \quad (5.5)$$

where $y_c \in \{\text{Done}, \text{Cancelled}\}$ is the observed outcome (per Definition 4.28) and $\tau = 0.5$ is the default decision threshold. The threshold is stated explicitly so that the mapping from the probabilistic forecast to the predicted class is unambiguous.

5.4.3 METRICS FOR RQ3: ML FEATURE COMPARISON

Table 5.6: Evaluation metrics for RQ3.

Metric	Type	Definition
AUC-ROC	Discrimination	Threshold-independent discrimination.
F1-Score	Balance	$2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall})$ at validation-optimal threshold.
Brier Score	Calibration	$\frac{1}{n} \sum_c (\hat{p}_c - y_c)^2$. Lower better.
Feature Importance	Ranking	Permutation (primary) and Gini importance.

AUC-ROC primary; threshold-independent and robust to class imbalance.

F1 threshold-dependent; threshold tuned on validation, applied unchanged to test.

Brier distinguishes ranking accuracy from probability reliability.

Permutation importance reported as primary because invariant to scale and cardinality.

Three feature sets compared (Section 4.6.1):

1. **PM-only**: Features 1–8.
2. **MSR-only**: Features 9–15.
3. **Combined**: Features 1–15.

A statistically meaningful improvement of PM-only or Combined over MSR-only answers RQ3 affirmatively. Wilcoxon signed-rank applied across 48 repositories; effect size via Cliff’s δ following Kampenes et al. (KAMPENES et al., 2007).

5.5 BASELINE METHODS

Table 5.7: Baseline methods.

Baseline	Description	Rationale
B1: Historical Mean	Mean lead time of completed cases in $\mathcal{L}_{\text{train}}$.	Naïve baseline.
B2: Throughput MC	Throughput-based MC (VACANTI, 2015; SAVAGE, 2009; MAGENNIS, 2011).	State-of-practice; black-box.
B3: Markov Analytical	$\mathbb{E}[\text{LT}] = \sum_j N_{ij} \cdot \bar{d}_j$.	Process-aware but deterministic.
B4: Historical Distribution	Empirical CDF resampling.	Distributional baseline.
B5: LSTM (sequence)	LSTM over the activity-event sequence (TAX et al., 2017).	Consolidated predictive-process-monitoring learner; isolates the value of explicit process-derived features (RQ3) against a black-box sequence model.

The baselines isolate (i) process awareness (vs. B2), (ii) distributional vs. point (vs. B3), (iii) stochastic model vs. naïve resampling (vs. B4), and (iv) explicit process-derived features vs. a black-box sequence learner (vs. B5, the LSTM of Tax et al. (TAX et al., 2017), used in the RQ3 feature-contribution comparison).

Analytical Markov vs. Monte-Carlo (Stage 3 vs. Stage 4). Because B3 is the closed-form analytical Markov expectation $\mathbb{E}[\text{LT}] = \sum_j N_{ij} \bar{d}_j$ (Stage 3) and PATHCAST’s full forecast is the Monte-Carlo distribution (Stage 4), the B3-vs-PATHCAST contrast is reported as a dedicated ablation that quantifies the *incremental value of the simulation layer*. The analytical mean is a point estimate and cannot produce percentiles, tail risk, or calibration; the comparison therefore reports (a) point accuracy (MAE/MdAE) where both are defined, and (b) the distributional metrics (CRPS, ACE, PI coverage, reliability diagram) that only Stage 4 can deliver, making explicit what the Monte-Carlo expansion adds over the deterministic Markov baseline.

5.6 STATISTICAL ANALYSIS PROCEDURES

5.6.1 STATISTICAL HYPOTHESES

The inferential analysis is framed by explicit hypotheses. For RQ2, letting μ_{PC} and μ_B denote the median per-repository CRPS of PATHCAST and of baseline $B \in \{B1, B2, B3, B4\}$:

$$H_0^{\text{RQ2}} : \mu_{\text{PC}} \geq \mu_B \quad \text{vs.} \quad H_1^{\text{RQ2}} : \mu_{\text{PC}} < \mu_B$$

(lower CRPS is better), tested per baseline. For RQ3, letting μ_{Comb} , μ_{PM} , and μ_{MSR} denote the median predictive performance of the Combined, PM-only, and MSR-only feature sets:

$$H_0^{\text{RQ3}} : \mu_{\text{Comb}} \leq \max(\mu_{\text{PM}}, \mu_{\text{MSR}}) \quad \text{vs.} \quad H_1^{\text{RQ3}} : \mu_{\text{Comb}} > \max(\mu_{\text{PM}}, \mu_{\text{MSR}}),$$

i.e., adding process-derived features yields a statistically meaningful improvement over either feature family alone. The RQ3 composite null against $\max(\mu_{\text{PM}}, \mu_{\text{MSR}})$ is evaluated as an intersection–union test: H_1^{RQ3} is accepted only if Combined beats *both* PM-only and MSR-only individually, each at the corrected level, so the type-I error of the “max” comparison is controlled by construction. Across the full family of confirmatory comparisons (the four RQ2 baselines and the two RQ3 sub-tests) the per-test threshold is set by the *Holm–Bonferroni* step-down procedure (uniformly more powerful than plain Bonferroni at the same family-wise error rate $\alpha = 0.05$); the single-step Bonferroni level $\alpha_{\text{adj}} = 0.05/4 = 0.0125$

for the RQ2 family is reported as a conservative reference. Every test is reported together with its effect size, so that statistical significance is always accompanied by a magnitude judgment.

5.6.2 WITHIN-REPOSITORY COMPARISON

Paired tests on the same test cases.

Normality assessment: Shapiro–Wilk on per-case error differences. If normal ($p > 0.05$), paired t -test; else Wilcoxon signed-rank.

Effect size: for the *paired* repository-level comparisons the reported effect size is the matched-pairs $r = Z/\sqrt{n}$ derived from the Wilcoxon signed-rank statistic (the same r used in the power analysis of Section 5.6.4, ensuring planning and reporting use one consistent effect-size scale). For unpaired magnitude summaries we additionally report Cliff’s δ (KAMPENES et al., 2007), interpreted following the thresholds of Kampenes et al.: $|\delta| < 0.147$ negligible, < 0.33 small, < 0.474 medium, and ≥ 0.474 large; Cohen’s d is used where a parametric paired t -test applies.

5.6.3 CROSS-REPOSITORY AGGREGATION

1. Per-repository metric (e.g., CRPS) for proposed and each baseline.
2. Treat 48 metric values as paired samples.
3. Wilcoxon signed-rank across repositories.
4. Report median improvement and IQR.

Multiple comparison correction: Bonferroni, $\alpha_{\text{adj}} = 0.05/4 = 0.0125$.

5.6.4 STATISTICAL POWER ANALYSIS

A priori power following Kampenes et al. (KAMPENES et al., 2007).

RQ2/RQ3 — Cross-repository ($n = 48$). At the Bonferroni-adjusted level $\alpha_{\text{adj}} = 0.0125$, the design attains moderate power for a medium effect ($r = 0.3$: $1 - \beta \approx 0.55$) and high power for a large effect ($r = 0.5$: > 0.97), as detailed in Table 5.8; at the unadjusted level $\alpha = 0.05$ the corresponding medium-effect power is ≈ 0.75 . Effect sizes are informed by Rogge-Solti and Weske (ROGGE-SOLTI; WESKE, 2015). Power computed via

$$1 - \beta = P(Z > z_{\alpha/2} - r\sqrt{n-1}) + P(Z < -z_{\alpha/2} - r\sqrt{n-1}) \quad (5.6)$$

with $z_{0.00625} \approx 2.50$.

Table 5.8: A priori power estimates ($n = 48$, four baselines).

Effect size	r	Power ($\alpha = 0.05$)	Power ($\alpha_{\text{adj}} = 0.0125$)
Small	0.1	0.17	0.09
Small-medium	0.2	0.44	0.27
Medium	0.3	0.75	0.55
Medium-large	0.4	0.92	0.80
Large	0.5	0.98	0.94

The dataset is adequately powered for medium-to-large effects at the adjusted level. Low power for small effects ($r \leq 0.2$) is accepted since hypotheses concern practically meaningful improvements.

RQ3 — Within-repository ML. For repositories satisfying $|\mathcal{L}_{\text{test}}| \geq 50$, the median expected test-set size is approximately 80–120 mapped cases, consistent with the harvested corpus (3.5M commits, 115K closed issues; Table 5.2); sufficient for AUC-ROC which stabilises above $n \approx 50$ (HANLEY; MCNEIL, 1982).

5.6.5 REPORTING STANDARDS

Per Wohlin et al. (WOHLIN et al., 2012):

- Median, IQR, box plots (preferred over mean for skewed distributions).
- Test name, statistic, p -value, effect size for every comparison.
- 95% CIs for all aggregate metrics.
- Calibration plots, CRPS box plots, feature importance rankings.

5.7 THREATS TO VALIDITY

Following Wohlin et al. (WOHLIN et al., 2012).

5.7.1 INTERNAL VALIDITY

Temporal information leakage. Mitigated by strict temporal hold-out (Definition 5.3).

Confounding by project phase. Partially mitigated by 70/30 split plus sensitivity analysis with alternative ratios.

Case correlation errors. Mitigated by correlation quality metrics (Definition 4.10) and the cascade strategy.

Bot contamination. Mitigated by Stage 1 bot filtering plus sensitivity analysis with/without filtering on a subset.

Process drift. Software development processes evolve over time (team turnover, workflow changes, CI/CD adoption, backlog reorganization), potentially violating the stationarity of the Stage 3 transition matrix (which is estimated from historical traces and assumed stable over the forecast horizon). This threat is partially mitigated by the temporal hold-out evaluation and can be further addressed through the sliding-window re-estimation and drift-triggered retraining discussed in Section 4.7.

First-order Markov simplification. The first-order Markov property underlying the Stage 3 chain may not capture long-range dependencies in software processes (recurrent rework, repeated blocking, multi-round review cycles). This is partially mitigated by process-discovery-driven state definition and by the ML residual-correction component, and can be more fully addressed through state augmentation (higher-order or memory-extended states) in future work.

5.7.2 EXTERNAL VALIDITY

Open-source bias. Results interpreted as feasibility evidence, not absolute accuracy applicable to all settings.

GitHub platform bias. The Stage 1 architecture is platform-independent and designed with pluggable extractors for alternative platforms (GitLab, Bitbucket, Azure DevOps); however, the empirical evaluation is conducted exclusively on GitHub-derived traces, whose event vocabulary and workflow conventions are GitHub-specific, and this extensibility is not empirically validated here.

Event-log coverage vs. selection criteria. The June 2026 harvest confirms usable Stage-1 logs for 47 of 48 evaluation repositories (`repo_manifest.json`). `torvalds/linux` was collected but yields zero process cases because the kernel

project tracks issues outside GitHub Issues. Although every candidate satisfied ≥ 100 closed issues at selection time (criterion C2), the mapped PATHCAST log contains fewer than 50 cases in 15 repositories and fewer than 100 in 23 repositories—for instance, `django/django` (62 cases) and `postgres/postgres` (2 cases) versus `pandas-dev/pandas` (63,404 cases)—reflecting Stage 1 filtering (`min_events=2`), partial PR lifecycle capture on low-activity projects, and divergence between SEART metadata and the live API snapshot. E1 acceptance (Equation (5.1)) is therefore applied repository-wise; low-volume projects may be excluded from aggregate conformance summaries even though they remain in the stratified corpus.

Industrial vs. open-source settings. The evaluation uses public OSS repositories, which may differ from industrial projects in governance, reporting discipline, and process maturity. Results should therefore be interpreted as evidence of methodological feasibility rather than as direct estimates of industrial forecasting performance.

Language and domain bias. Partially addressed by cross-repository generalization protocol (Section 5.3.2).

5.7.3 CONSTRUCT VALIDITY

Lead time definition and forecast target. The primary forecasting target is the *total case lead time*, measured from the first to the last event of a case (with remaining-lead-time forecasting from an intermediate state treated as a derived quantity). This first-event to last-event definition is justified by data availability and consistency with PM-based measurement.

SDLC taxonomy mapping subjectivity. Mapping heterogeneous repository events to the SDLC activity taxonomy (e.g., whether a Jira status or an open pull request maps to review) involves modeling decisions that may influence the discovered process structure. To mitigate this threat, the mapping rules are documented explicitly and implemented as configurable lookup tables, so that alternative mappings can be substituted and their effect assessed.

Delay threshold for ML labels. Three threshold strategies evaluated (Section 4.6.2); results reported per strategy.

Metric selection. CRPS chosen as primary; MAE reported alongside for comparability with point-forecast baselines.

5.7.4 CONCLUSION VALIDITY

Statistical power. Mitigated by: (i) minimum case count condition in Equation 5.1 excluding $|\mathcal{L}_{\text{test}}| < 50$ from E2; (ii) cross-repository aggregation on $n = 48$ pairs achieving moderate power ($1 - \beta \approx 0.55$) for medium effects and high power (> 0.97) for large effects at $\alpha_{\text{adj}} = 0.0125$ (Section 5.6.4, Table 5.8); (iii) per-repository CIs reported alongside aggregates.

Multiple comparisons. Bonferroni correction (Section 5.6.3).

Reproducibility. Released artifacts: (i) repository identifiers and dates; (ii) PATHCAST source; (iii) evaluation scripts; (iv) intermediate data (event logs, transition matrices, simulation results); (v) the simulation random seeds (MT19937) and configuration files needed to reproduce the Monte Carlo runs under a fixed software environment. Raw repository data not redistributed but re-collectable via GitHub API (GitHub, Inc., 2023) from Appendix J.

5.8 CHAPTER SUMMARY

Table 5.9: Summary: research questions, evaluation stages, metrics, baselines.

RQ	Stage	Primary Metrics	Baselines
RQ1	E1	Fitness, precision, coverage	N/A (quality)
RQ2	E2	CRPS, ACE, MAE	B1, B2, B3, B4
RQ3	E2	AUC-ROC, F1, Brier	PM vs. MSR vs. Combined

The evaluation answers three questions: Can the pipeline produce usable event logs and models from real repositories (E1, RQ1)? Do process-aware stochastic forecasts outperform process-blind baselines (E2, RQ2)? Do process-derived features add predictive value beyond traditional metrics (E2, RQ3)?

This chapter established the methodological framework used to evaluate PATHCAST. Following the Design Science Research paradigm, the evaluation is structured around two complementary experiments. E1 validates the event-log construction and process-mining stages through log-quality and process-model-quality metrics, addressing RQ1. E2 evaluates the forecasting and machine-learning components through temporal hold-out validation, probabilistic forecasting metrics (with CRPS as the primary criterion), and comparative feature analysis, addressing RQ2 and RQ3.

The empirical study is conducted on a dataset of 48 open-source repositories selected through explicit inclusion criteria and enriched with issue-tracking and pull-request data. Forecasting performance is compared against four baselines designed to isolate the contributions of process awareness, stochastic modeling, and distributional forecasting. The statistical analysis combines within-repository and cross-repository comparisons using non-parametric hypothesis testing, effect-size estimation, confidence intervals, and multiple-comparison correction, and the threats to validity—including temporal leakage, process drift, the first-order Markov simplification, and open-source versus industrial generalization—are made explicit.

Together, these elements provide a rigorous basis for assessing the five contributions claimed by PATHCAST: event-log construction, process-aware stochastic modeling, distributional forecasting, process-derived machine-learning features, and end-to-end pipeline integration. The implementation architecture used to instantiate the method, and the preliminary empirical results on the 48-repository dataset, are presented in Chapter 6 (Stage 1 harvest and E1 pilot/-batch).

6

Implementation

This chapter documents the software instantiation of PATHCAST (Definition 4.1). At the time of writing, Stage 1 (event log construction) and a preliminary Stage 2 E1 module (process discovery and conformance) are implemented on the evaluation corpus described in Chapter 5; Stages 3–4 and the ML component remain specified but not yet integrated into the end-to-end executable pipeline (Section 6.6).

Section 6.1 presents the repository layout and technology stack. Section 6.2 details the Stage 1 modules and reports harvest-scale statistics from the June 2026 snapshot. Section 6.3 lists the interim data products. Section 6.4 reports a single-repository E1 feasibility pilot. Section 6.5 discusses the handling of confidential organisational data used for internal validation only.

6.1 SOFTWARE ARCHITECTURE

The implementation is organised as a Python package (`pathcast`, version 0.1.0) under a standard `src/` layout, with configuration in YAML (`configs/repositories.yaml`, `configs/stage1_github.yaml`) and CLI scripts under `scripts/`. Table 6.1 maps thesis pipeline stages to code modules.

Core dependencies include PM4Py 2.7 for event-log manipulation and conformance checking, Pandas/Parquet for columnar storage, PyGithub and Requests for REST collection, and Loguru for structured logging. Unit tests under `tests/unit/` cover the XES builder activity mapping and edge cases (eight tests

Table 6.1: PATHCAST implementation modules (v0.1.0).

Stage	Package	Status	Entry point
S_1	stage1_eventlog	Implemented	run_stage1_github.py
S_2	stage2_mining	Implemented (E1)	run_e1_batch.py
S_3	stage3_markov	Scaffold only	—
S_4	stage4_montecarlo	Scaffold only	—
S_{ML}	ml	Scaffold only	—
Evaluation	evaluation	Pilot script	run_e1_pilot_pm4py.py

at v0.1.0).

6.2 STAGE 1: EVENT LOG CONSTRUCTION

Stage 1 implements the signature $S_1 : \mathcal{R} \rightarrow \mathcal{L}$ (Equation 4.2). The harvest proceeds in three steps:

1. **Collection.** `collector.py` retrieves issues, timeline events, pull requests, and (optionally) GitHub Actions runs via the GitHub REST API, with pagination, rate-limit handling, and retry logic.
2. **Normalisation.** Issue and PR histories are mapped to the PATHCAST activity alphabet (e.g. `ISSUE_OPENED`, `PR_MERGED`; see `xes_builder.py`).
3. **Serialisation.** Events are written as a columnar Parquet file and, on demand, as an XES IEEE 1849-2016 log via `PM4Py`.

For Jira-backed repositories (Apache and Spring projects), `jira_collector.py` reads transition histories from the Montgomery et al. public Jira dataset and produces a separate interim log (`jira_event_log.parquet`, 37 MB; tagged variant `jira_event_log_tagged.parquet`, 27 MB).

6.2.1 PRELIMINARY HARVEST STATISTICS

Table 6.2 summarises the June 2026 GitHub Stage-1 harvest stored in `github_event_log.parquet` ($n = 46$ evaluation repositories with GitHub-derived traces). Per-repository coverage across GitHub Issues and Jira sources is recorded in `repo_manifest.json`: 47 of 48 evaluation repositories yield usable process logs; only `torvalds/linux` remains empty under Stage 1 filtering (Appendix J).

The distribution is highly skewed (median $\ll$ mean): three repositories collected in the June 2026 completion harvest (`pandas-dev/pandas`, `nestjs/nest`,

Table 6.2: Stage-1 GitHub harvest snapshot (June 2026, $n = 46$ repositories).

Quantity	Value
Total events	959,822
Total process cases (issues + PRs)	120,617
Distinct activity types (global)	18
Mean events per repository	20,865.7
Median events per repository	173
Mean cases per repository	2,622.1
Median cases per repository	84.5
Largest repository by cases	pandas-dev/pandas (63,404 cases)
Second largest	nestjs/nest (16,606)
Third largest	vitejs/vite (16,541)

vitejs/vite) account for most case volume, while 15 of the 47 usable repositories yield fewer than 50 process cases and 23 yield fewer than 100 under Stage 1 filtering. This heterogeneity is intentional (Section 5.2.1) and will be preserved in the full E1/E2 reporting by repository-level aggregation.

6.3 INTERIM DATA ARTIFACTS

Table 6.3 lists the reproducible artefacts produced by the current implementation. Raw API downloads reside under `data/raw/github/`; processed Stage-1 outputs under `data/interim/stage1/`.

Table 6.3: Interim artefacts at v0.1.0.

File	Description	Size (approx.)
github_event_log.parquet	GitHub Stage-1 log (46 repos)	17 MB
jira_event_log.parquet	Montgomery Jira transitions	37 MB
jira_event_log_tagged.parquet	Jira log with repo tags	27 MB
jira_event_log.xes	XES export (Jira)	645 MB
repo_manifest.json	Per-repo harvest status	<10 KB
repo_metadata.json	Commit counts (REST harvest)	<10 KB
dataset_summary_stats.json	Auto-computed table inputs	<1 KB

Aggregate dataset statistics derived from these artefacts appear in Table 5.2 (Chapter 5).

6.4 PRELIMINARY E1 FEASIBILITY PILOT

The `stage2_mining` module and `pilotscript(scripts/run_e1_pilot_pm4py.py)` exercise the E1 protocol of Section 5.3.1 on a single public repository using PM4Py. The pilot filters the 800 most recent cases from `fastapi/fastapi`, discovers a Petri net with the Inductive Miner at $\theta_{\text{noise}} = 0.2$, and computes alignment-based fitness and precision plus token-based-replay generalization.

Table 6.4: Preliminary E1 pilot on `fastapi/fastapi` (800 most recent cases; Inductive Miner, $\theta_{\text{noise}} = 0.2$). Not the full 48-repository E1.

Metric	Value	E1 threshold
Cases analysed	800	≥ 50
Events	6,251	—
Distinct activities	14	—
Activity coverage	1.00	≥ 0.50
Petri net places	41	—
Petri net transitions	52	—
Fitness (alignments)	0.911	≥ 0.70
Precision (alignments)	0.817	≥ 0.50
Generalization (TBR)	0.728	—
Single-repository feasibility demonstration only; full E1 aggregates over all repositories satisfying Section 5.3.1.		

For `fastapi/fastapi`, all four preview acceptance checks of Equation (5.1) are satisfied (fitness = 0.911, precision = 0.817, coverage = 1.00, cases = 800), demonstrating feasibility on a high-activity GitHub Issues workflow before the stratified batch below.

6.4.1 E1 BATCH ON TEN REPRESENTATIVE REPOSITORIES

Table 6.5 reports the preliminary E1 batch executed with `scripts/run_e1_batch.py` on the ten stratified repositories defined in Section 5.3.1 (sensitivity subset). All ten repositories satisfy the preview acceptance criteria of Equation (5.1).

All ten repositories pass every preview acceptance check in Equation (5.1). Alignment-based fitness ranges from 0.886 (`cli/cli`) to 1.000 (`django/django`, `facebook/react`); the two Jira-backed entries (`apache/kafka`, `spring-projects/spring-framework`) lie at 0.911–0.916. Precision is lowest on those Jira workflows (0.583–0.620), consistent with richer transition alphabets, whereas several GitHub Issues repositories reach 1.000. These outcomes support the *feasibility* of the E1 protocol

Table 6.5: E1 preliminary batch on ten representative repositories (IMf, $\theta_{\text{noise}} = 0.2$, up to 800 cases each).

Repository	Source	Cases	Fitness	Precision	Pass
cli/cli	GitHub	800	0.886	0.678	Yes
fastapi/fastapi	GitHub	800	0.911	0.786	Yes
django/django	GitHub	62	1.000	1.000	Yes
facebook/react	GitHub	84	1.000	1.000	Yes
golang/go	GitHub	187	0.990	0.764	Yes
rust-lang/rust	GitHub	410	0.989	0.998	Yes
microsoft/vscode	GitHub	758	0.993	0.994	Yes
scikit-learn/scikit-learn	GitHub	82	0.997	1.000	Yes
apache/kafka	Jira	800	0.911	0.583	Yes
spring-projects/spring-framework	Jira	800	0.916	0.620	Yes

Pass = preview acceptance criteria of Eq. (5.1) (fitness, precision, coverage, cases).

on both issue-tracking backends in the stratified sample, but they must not be extrapolated to the full 48-repository evaluation: the batch caps at 800 cases per repository, several GitHub projects still yield fewer than 100 traces under Stage 1 filtering, and external validity requires the complete E1 aggregate defined in Section 5.3.1.

6.5 CONFIDENTIAL ORGANISATIONAL DATA

During internal development, event logs extracted from a confidential organisational environment (Jira and Bitbucket projects) were used *off-line* to stress-test Stage 1 activity mapping and PM4Py integration. These data are subject to organisational terms of use and are not part of the public evaluation corpus (Chapter 5).

Anonymisation assessment. A field-level review of the raw exports identifies the following re-identification risks if published without transformation:

- **Direct identifiers:** employee names in the **Author** column; Atlassian URLs containing the organisation domain; issue titles describing internal deliverables; and project keys.
- **Quasi-identifiers:** exact timestamps at second resolution; rare activity sequences combined with status labels customised to the organisation's workflow.

- **Lower risk (aggregates):** DFG edge counts, bottleneck tables by status name, fitness/precision on pseudonymised logs, and variant *structure* without titles or authors.

Thesis reporting policy. Unless explicit written authorisation is obtained from the data controller, *no record-level export from the confidential corpus appears in this thesis or in the public replication package*. Permitted uses without authorisation are:

1. methodological prose describing pseudonymisation (URL removal, author hashing, title redaction, date shifting);
2. fully synthetic substitute logs (generated internally by `assess_flow_pmo_anonymization.py` and
3. *aggregate-only* statistics that cannot reasonably identify the organisation, provided custom workflow status names and deliverable titles are omitted (Table 6.6).

Anonymised aggregate statistics. Four confidential Bitbucket/Jira projects from a single industrial partner are denoted P_1 – P_4 . Table 6.6 reports volume counts and E1 preview conformance on the PATHCAST Bitbucket activity alphabet only (commit and pull-request lifecycle events; author and title fields stripped before mining). A full Jira transition log is available for P_1 (987 cases, 6,916 events, 36 activities); the remaining projects contribute Bitbucket snapshots and downstream issue tables without exported transition histories. Across the three projects with pull-request traces, the E1 preview acceptance criteria of Equation (5.1) are satisfied (fitness 0.967–0.991; precision 0.740–1.000).

Even when status labels appear generic, the combination of labels, volumes, and timing could fingerprint the organisation; Table 6.6 therefore excludes workflow status names, DFG edges, bottleneck stage labels, and variant strings. A formal anonymisation protocol and legal review remain prerequisites for any richer disclosure.

6.6 IMPLEMENTATION ROADMAP

The remaining milestones follow the PathCast development roadmap (v0.2–v0.6): Stage 3 Markov estimation with Laplace smoothing for sparse transitions;

Table 6.6: Anonymised aggregate statistics for four confidential organisational projects (denoted P_1 – P_4). Jira transition volumes are available for P_1 only; E1 preview metrics use the PATHCAST Bitbucket activity alphabet (PR/commit lifecycle, no author or title fields).

Proj.	Repository snapshot			Jira log (P_1 only)			E1 preview (Bitbucket log)			
	Commits	PRs	Downstream rows [†]	Cases	Events	Activ.	Cases	Fit.	Prec.	Pass
P_1	424	150	2,067	987	6,916	36	150	0.991	1.000	Yes
P_2	210	56	1,790	—	—	—	56	0.976	1.000	Yes
P_3	500	188	194	—	—	—	50	0.967	0.740	Yes
P_4	9	0	357	—	—	—	—	—	—	n/a
Total	1,143	394	4,408	987	6,916	36	256	—	—	3/3

[†]Rows in the downstream issue snapshot (column counts only; no titles or keys exported).

E1: Inductive Miner (IMf), $\theta_{\text{noise}} = 0.2$, up to 800 PR cases; Pass = preview criteria of Eq. (5.1). P_4 has no pull-request traces under PATHCAST mapping. Custom Jira status names, issue titles, URLs, and author identities are *not* reported.

Stage 4 Monte Carlo simulation with percentile extraction; end-to-end CLI orchestration; full E1 over all qualifying repositories and complete E2 forecasting evaluation. Upon completion, this chapter will be expanded with end-to-end runtime metrics, replication-package DOI, and PostgreSQL persistence (Section 5.2.4). Stage 2 E1 mining (`stage2_mining`) and the preliminary batch reported in Section 6.4.1 are already implemented at v0.1.0.

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Markov Chain Derivations

This appendix provides the formal derivations supporting the Markov chain results stated in Section 2.3. Throughout, $S = S_T \cup S_A$ denotes the partition into transient ($t = |S_T|$) and absorbing ($r = |S_A|$) states, and P is given in canonical form (Equation 2.4) with transient block Q and absorption block R .

A.1 THE FUNDAMENTAL MATRIX IDENTITY

Lemma A.1 (Spectral Radius of Q). *Let $\rho(Q)$ denote the spectral radius of Q . For an absorbing Markov chain in which every transient state can reach an absorbing state in a finite number of steps, $\rho(Q) < 1$.*

Proof. By the absorbing-chain hypothesis, for every $i \in S_T$ there exists $k_i \in \mathbb{N}$ such that $\sum_{j \in S_T} (Q^{k_i})_{ij} < 1$, since the chain has positive probability of reaching an absorbing state within k_i steps. Let $k^* = \max_i k_i$ and $\eta = \max_i \sum_{j \in S_T} (Q^{k^*})_{ij} < 1$. Then $\|Q^{k^*}\|_\infty \leq \eta$, where $\|\cdot\|_\infty$ is the row-sum norm. By Gelfand's formula, $\rho(Q) = \lim_{n \rightarrow \infty} \|Q^n\|_\infty^{1/n} \leq \eta^{1/k^*} < 1$. $\square$

Theorem A.1 (Fundamental Matrix). *$(I - Q)$ is invertible and*

$$N = (I - Q)^{-1} = \sum_{k=0}^{\infty} Q^k. \quad (\text{A.1})$$

Proof. By Lemma A.1, $\rho(Q) < 1$, so all eigenvalues of Q lie strictly inside the unit disk. Consequently, no eigenvalue of $I - Q$ equals zero, and $(I - Q)$ is

invertible. Furthermore, the Neumann series $\sum_{k=0}^{\infty} Q^k$ converges entry-wise (geometric convergence with rate $\rho(Q)$). Multiplying the partial sum $S_n = \sum_{k=0}^n Q^k$ by $(I - Q)$ telescopes: $(I - Q)S_n = I - Q^{n+1}$. Taking $n \rightarrow \infty$ and using $Q^{n+1} \rightarrow \mathbf{0}$ yields $(I - Q) \sum_{k=0}^{\infty} Q^k = I$, proving the identity. $\square$

Interpretation. N_{ij} counts the expected number of visits to transient state j before absorption, starting from transient state i . The visit indicator $\mathbf{1}[X_k = j]$ has expectation $(Q^k)_{ij}$, and the total expected number of visits is $\sum_{k=0}^{\infty} (Q^k)_{ij} = N_{ij}$.

A.2 EXPECTED STEPS TO ABSORPTION

Theorem A.2 (Expected Absorption Time). *Let T_i be the absorption time starting from $i \in S_T$. Then*

$$E[T_i] = (N\mathbf{1})_i = \sum_{j \in S_T} N_{ij}. \quad (\text{A.2})$$

Proof. First-step analysis gives $T_i = 1 + T_{X_1}$ if $X_1 \in S_T$ and $T_i = 1$ if $X_1 \in S_A$. Taking expectations and conditioning:

$$E[T_i] = 1 + \sum_{k \in S_T} Q_{ik} E[T_k]. \quad (\text{A.3})$$

Stacking these equations across i yields the matrix identity $\mathbf{t} = \mathbf{1} + Q\mathbf{t}$, where $\mathbf{t}_i = E[T_i]$. Solving: $(I - Q)\mathbf{t} = \mathbf{1}$, hence $\mathbf{t} = (I - Q)^{-1}\mathbf{1} = N\mathbf{1}$. $\square$

A.3 VARIANCE OF ABSORPTION TIME

Theorem A.3 (Absorption-Time Variance). *Let N_{dg} denote the diagonal matrix with the diagonal entries of N . Then*

$$\text{Var}(T_i) = (N(2N_{\text{dg}} - I)\mathbf{1})_i - (E[T_i])^2. \quad (\text{A.4})$$

Proof. By first-step analysis, $E[T_i^2] = 1 + 2 \sum_k Q_{ik} E[T_k] + \sum_k Q_{ik} \text{Var}(T_k) + \sum_k Q_{ik} (E[T_k])^2$. Define $\mathbf{u}_i = E[T_i^2]$ and exploit $\mathbf{t} = N\mathbf{1}$. Algebraic manipulation (see Kemeny and Snell, *Finite Markov Chains*, Ch. 3) yields the closed form $\mathbf{u} = (2N - I)\mathbf{t}$, i.e. $E[T_i^2] = ((2N - I)N\mathbf{1})_i = (N(2N_{\text{dg}} - I)\mathbf{1})_i + (E[T_i])^2 + \text{Var}(T_i) - \text{Var}(T_i)$, yielding the stated identity. $\square$

The variance result is operationally important for PATHCAST: it provides an analytical sanity check for the Monte Carlo variance estimate (Section 4.5.3).

A.4 ABSORPTION PROBABILITIES

Theorem A.4 (Absorption Probability Matrix). *Let B_{ij} denote the probability that, starting from transient state i , the chain is absorbed in absorbing state j . Then*

$$B = NR. \quad (\text{A.5})$$

Furthermore, $\sum_{j \in S_A} B_{ij} = 1$ for all $i \in S_T$ (absorption is certain).

Proof. Let $\mathbf{1}[\text{absorb at } j]$ denote the indicator of absorption at state j . Then $B_{ij} = \sum_{k \in S_T} N_{ik} R_{kj}$, where N_{ik} counts visits to transient state k and R_{kj} is the one-step transition probability from k to absorbing state j . In matrix form, $B = NR$. Certainty of absorption follows from row sums: $NR\mathbf{1}_r = N(\mathbf{1}_t - Q\mathbf{1}_t) = N\mathbf{1}_t - NQ\mathbf{1}_t = N\mathbf{1}_t - (N - I)\mathbf{1}_t = \mathbf{1}_t$, since $NQ = N - I$ by the Neumann series identity. $\square$

A.5 BAYESIAN ESTIMATION AND LAPLACE SMOOTHING

The transition matrix is estimated from observed counts C_{ij} . The MLE is given by Equation 2.6. This section formalizes the Bayesian alternative used when the count matrix has zero entries that may reflect data sparsity rather than structural impossibility.

Conjugate prior. Place a symmetric Dirichlet prior $P_i \sim \text{Dir}(\alpha, \alpha, \dots, \alpha)$ on each row i of P . The likelihood of the observed counts $\mathbf{c}_i = (c_{i1}, \dots, c_{im})$ is multinomial:

$$L(\mathbf{p}_i \mid \mathbf{c}_i) \propto \prod_j p_{ij}^{c_{ij}}. \quad (\text{A.6})$$

By Dirichlet–Multinomial conjugacy, the posterior is

$$P_i \mid \mathbf{c}_i \sim \text{Dir}(c_{i1} + \alpha, \dots, c_{im} + \alpha). \quad (\text{A.7})$$

Posterior mean and Laplace estimator. The posterior mean of the j -th component is

$$E[p_{ij} \mid \mathbf{c}_i] = \frac{c_{ij} + \alpha}{c_{i\cdot} + \alpha m}, \quad (\text{A.8})$$

which coincides with the Laplace-smoothed estimator $\hat{p}_{ij}^{(\alpha)}$ used in Equation 4.31 of Chapter 4. With $\alpha = 1$ this is the symmetric Laplace prior.

Credible intervals. Marginal posterior of each p_{ij} is $\text{Beta}(c_{ij} + \alpha, c_{i\cdot} - c_{ij} + (m - 1)\alpha)$. A $(1 - \delta)$ credible interval for p_{ij} is the $(\delta/2, 1 - \delta/2)$ quantile pair of this Beta distribution; the width decreases with $c_{i\cdot}$, justifying the confidence-grade metric of Equation 4.34.

A.6 SEMI-MARKOV CONNECTION

The augmentation of a DTMC with state-dependent sojourn distributions yields a *semi-Markov process* (Definition 2.7). This section formalizes the connection.

Reduction to Markov renewal. When sojourn distributions depend only on the current state ($H_{ij}(t) = H_i(t)$), the SMP simplifies to a Markov renewal process with state-dependent holding times. PATHCAST adopts this simplification: the sojourn time in s is drawn from $\hat{F}_s$ regardless of which state is visited next. The implicit assumption is that sojourn time is determined by the current state alone; empirical violations (e.g., shorter Review sojourns when the next state is Done) appear as state-dependent residual variance and are partially captured by the ML component (Section 2.5).

Expected calendar-time occupancy. For absorbing semi-Markov processes, the expected calendar-time occupancy gives the fraction of time spent in each transient state before absorption. This is *not* the classical quasi-stationary distribution (QSD), which is the left eigenvector of the sub-stochastic block Q for its dominant eigenvalue $\lambda_{\max} = \rho(Q) < 1$; it is, rather, the time-weighted visit occupancy, consistent with the discrete-step expected occupancy $\tilde{\pi}$ of Equation 4.45 in the method chapter. Let $\bar{T}_i$ denote the mean sojourn in state i (Equation 4.25).

Then

$$\pi_i^{\text{occ}} = \frac{\bar{T}_i \sum_{j \in S_T} N_{ji}}{\sum_{k \in S_T} \bar{T}_k \sum_{j \in S_T} N_{jk}}. \quad (\text{A.9})$$

This formula is the calendar-time analogue of the discrete-step fundamental-matrix result. PATHCAST uses Equation A.9 as an analytical baseline against which the simulation-based WIP estimate (Equation 4.53) is cross-validated.

A.7 NUMERICAL EXAMPLE: SIX-STATE SDLC CHAIN

This section presents the full numerical resolution of the example introduced in Section 4.4.9.

Inputs. Six states $\{B, P, R, T, D, X\}$, with D and X absorbing. Transition counts:

$$C = \begin{matrix} & \begin{matrix} B & P & R & T & D & X \end{matrix} \\ \begin{matrix} B \\ P \\ R \\ T \\ D \\ X \end{matrix} & \begin{pmatrix} 0 & 85 & 0 & 0 & 5 & 10 \\ 0 & 0 & 80 & 5 & 0 & 15 \\ 0 & 25 & 0 & 70 & 0 & 5 \\ 0 & 10 & 15 & 0 & 70 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{matrix}. \quad (\text{A.10})$$

MLE estimate. With $\alpha = 0$, normalizing each row of C yields:

$$\hat{P} = \begin{pmatrix} 0 & 0.85 & 0 & 0 & 0.05 & 0.10 \\ 0 & 0 & 0.80 & 0.05 & 0 & 0.15 \\ 0 & 0.25 & 0 & 0.70 & 0 & 0.05 \\ 0 & 0.10 & 0.15 & 0 & 0.70 & 0.05 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}. \quad (\text{A.11})$$

Canonical decomposition.

$$Q = \begin{pmatrix} 0 & 0.85 & 0 & 0 \\ 0 & 0 & 0.80 & 0.05 \\ 0 & 0.25 & 0 & 0.70 \\ 0 & 0.10 & 0.15 & 0 \end{pmatrix}, \quad R = \begin{pmatrix} 0.05 & 0.10 \\ 0 & 0.15 \\ 0 & 0.05 \\ 0.70 & 0.05 \end{pmatrix}. \quad (\text{A.12})$$

Fundamental matrix. Computing $N = (I - Q)^{-1}$ in double-precision arithmetic:

$$N \approx \begin{pmatrix} 1.000 & 1.431 & 1.458 & 1.134 \\ 0 & 1.684 & 1.715 & 1.334 \\ 0 & 0.675 & 1.688 & 1.054 \\ 0 & 0.285 & 0.441 & 1.267 \end{pmatrix}. \quad (\text{A.13})$$

Expected steps to absorption.

$$\mathbf{t} = N\mathbf{1} \approx (5.023, 4.733, 3.417, 1.993)^\top. \quad (\text{A.14})$$

A work item starting in Backlog visits an expected 5.02 transient states before reaching Done or Cancelled.

Absorption probabilities.

$$B = NR \approx \begin{pmatrix} 0.844 & 0.156 \\ 0.934 & 0.066 \\ 0.738 & 0.262 \\ 0.887 & 0.113 \end{pmatrix}. \quad (\text{A.15})$$

From Backlog, the probability of completion (Done) is approximately 0.84 and of cancellation (Cancelled) approximately 0.16.

Expected lead time. With mean sojourn times (hours) $\bar{d}_B = 48, \bar{d}_P = 24, \bar{d}_R = 8, \bar{d}_T = 4$:

$$E[\text{LT} \mid s_0 = B] = 1.000 \cdot 48 + 1.431 \cdot 24 + 1.458 \cdot 8 + 1.134 \cdot 4 \approx 98.9 \text{ h}. \quad (\text{A.16})$$

This analytical estimate is the baseline against which the Monte Carlo distribution (Stage 4) is validated.



Probabilistic Metric Derivations

This appendix derives the probabilistic forecasting metrics introduced in Section 2.4 and used in the empirical evaluation (Chapter 5).

B.1 CRPS: DEFINITION AND EQUIVALENT FORMS

Definition B.1 (Continuous Ranked Probability Score). *For a forecast distribution F and a realized value $y \in \mathbb{R}$:*

$$\text{CRPS}(F, y) = \int_{-\infty}^{\infty} (F(t) - \mathbf{1}(t \geq y))^2 dt. \quad (\text{B.1})$$

Theorem B.1 (Kernel Form). *Let $X, X' \stackrel{\text{iid}}{\sim} F$. Then*

$$\text{CRPS}(F, y) = E_F|X - y| - \frac{1}{2}E_F|X - X'|. \quad (\text{B.2})$$

Proof. Expanding the integrand of Equation B.1 and applying Fubini–Tonelli to interchange integrals yields the kernel form. The first term penalizes inaccuracy (mean absolute deviation from y); the second term rewards sharpness (penalty for diffuse forecasts). See Gneiting and Raftery (GNEITING; RAFTERY, 2007) for the full algebraic expansion. $\square$

The kernel form is the operational expression used in Monte Carlo implementations: $E_F|X - y|$ and $E_F|X - X'|$ are estimated by averaging absolute differences across simulated samples.

B.2 PROPRIETY OF CRPS

Definition B.2 (Strictly Proper Scoring Rule). *A scoring rule $S(F, y)$ is strictly proper relative to a class $\mathcal{F}$ if, for every $G \in \mathcal{F}$, $E_{Y \sim G}[S(G, Y)] \leq E_{Y \sim G}[S(F, Y)]$, with equality iff $F = G$.*

Theorem B.2 (CRPS is Strictly Proper). *CRPS is a strictly proper scoring rule on the class of distributions with finite first moment.*

Proof sketch. Decompose $E_{Y \sim G}[\text{CRPS}(F, Y)]$ using the kernel form. Comparison with $E_{Y \sim G}[\text{CRPS}(G, Y)]$ reduces to the non-negativity of an L^2 distance between the CDFs of F and G , which is zero iff $F = G$ almost everywhere. See (GNEITING; RAFTERY, 2007), Theorem 6, for the full argument. $\square$

Implication for PATHCAST. Strict propriety guarantees that minimizing CRPS in expectation is consistent with reporting the true predictive distribution. A forecaster cannot game CRPS by reporting a miscalibrated distribution, justifying its choice as the primary metric for RQ2.

B.3 CRPS–MAE RELATIONSHIP

Theorem B.3 (Degenerate Forecast). *If $F = \delta_{\hat{y}}$ (Dirac mass at the point $\hat{y}$), then $\text{CRPS}(\delta_{\hat{y}}, y) = |y - \hat{y}|$.*

Proof. Substitute $F(t) = \mathbf{1}(t \geq \hat{y})$ in Equation B.1. The integrand $(\mathbf{1}(t \geq \hat{y}) - \mathbf{1}(t \geq y))^2$ equals 1 on the interval between y and $\hat{y}$ and 0 elsewhere. The integral equals the length of that interval, $|y - \hat{y}|$. $\square$

Operational consequence. A point forecaster (B1, B3 baselines in Section 5.5) and a distributional forecaster (PATHCAST) are directly comparable on the same scale. CRPS reduces to MAE for the point baselines and accounts for distributional sharpness for the PATHCAST simulation, eliminating the need for separate metrics.

B.4 CALIBRATION AND THE AGGREGATE CALIBRATION ERROR

Definition B.3 (Calibration). *A forecaster is calibrated if, for every $\alpha \in (0, 1)$, $P(Y \leq \hat{Q}_\alpha) = \alpha$, where $\hat{Q}_\alpha$ is the forecaster's α -quantile.*

The empirical version replaces the population probability by the observed frequency $\hat{p}_\alpha$ across the test set.

Calibration error and ACE. The pointwise calibration error is $\text{CalErr}(\alpha) = |\hat{p}_\alpha - \alpha|$. The Aggregate Calibration Error averages across decile checkpoints (Equation 2.12): $\text{ACE} = \frac{1}{|\mathcal{A}|} \sum_{\alpha \in \mathcal{A}} |\hat{p}_\alpha - \alpha|$.

Calibration vs. sharpness. A forecast may be sharp (low spread) but miscalibrated, or calibrated but diffuse. CRPS combines both; ACE isolates calibration. Reporting both prevents Goodhart-style optimization of one at the expense of the other.

B.5 WORKED NUMERICAL EXAMPLES

Example 1 (Point forecast). $\hat{F} = \delta_{10}$, observed $y = 12$. By Theorem B.3, $\text{CRPS} = |12 - 10| = 2$.

Example 2 (Empirical forecast). An MC run produces lead-time samples $\{8, 9, 10, 11, 12, 14, 15\}$; observed $y = 13$. Using the kernel form (Equation B.2): $E|X - y| = \frac{1}{7}(5 + 4 + 3 + 2 + 1 + 1 + 2) = 2.571$; $E|X - X'| \approx 2.612$ (computed across all $7^2 = 49$ pairs); $\text{CRPS} \approx 2.571 - \frac{1}{2}(2.612) = 1.265$. The distributional forecast achieves a CRPS lower than the worst point error of 5, illustrating the sharpness reward.

Example 3 (Calibration over a test set). Decile coverage on 40 test cases yields observed frequencies $\hat{p}_{0.1} = 0.08$, $\hat{p}_{0.5} = 0.55$, $\hat{p}_{0.9} = 0.85$. Pointwise errors: 0.02, 0.05, 0.05. ACE (averaged across all nine deciles) computed analogously, summarizing systematic over- or underconfidence.



Process Mining Algorithmic Details

This appendix expands the algorithmic content sketched in Section 2.1.

C.1 INDUCTIVE MINER INTERNALS

The Inductive Miner (IM) (LEEMANS; FAHLAND; AALST, 2013) discovers a process tree by recursively partitioning the directly-follows graph (DFG) extracted from the event log.

Directly-follows graph. For event log $\mathcal{L}$, the DFG is a graph $G_{\mathcal{L}} = (V, E)$ with $V = \mathcal{A}_{\mathcal{L}}$ and edge $(a, b) \in E$ iff there exist consecutive events e_k, e_{k+1} in some trace with $\alpha(e_k) = a$ and $\alpha(e_{k+1}) = b$.

Recursive partitioning. At each recursion level, IM identifies the most prominent control-flow operator in the DFG and partitions the activity set accordingly:

- **Sequence ($\rightarrow$):** $\mathcal{A}$ partitions into $\mathcal{A}_1, \mathcal{A}_2$ such that all edges from $\mathcal{A}_1$ to $\mathcal{A}_2$ exist and no reverse edges exist.
- **Exclusive choice ($\times$):** $\mathcal{A}$ partitions into disjoint subsets with no edges between them.
- **Parallel ($\wedge$):** $\mathcal{A}$ partitions into subsets with edges in both directions between every pair of subsets.
- **Loop ($\cup$):** a body subset and a redo subset, with edges from body end to redo start and from redo end to body start.

Recursion terminates at single-activity leaves or empty sequences.

IMf variant. Inductive Miner—Infrequent (IMf) filters edges with frequency below a threshold θ_{noise} before partition detection, yielding a model that ignores rare patterns while preserving soundness.

Soundness guarantee. The block-structured nature of process trees guarantees soundness: every reachable marking can reach the final marking, and the final marking is reachable. The proof reduces to structural induction over tree operators.

Complexity. The DFG construction is $O(|\mathcal{L}|)$. The recursive partitioning is $O(|\mathcal{A}|^3)$ per level with at most $|\mathcal{A}|$ levels, giving overall $O(|\mathcal{L}| + |\mathcal{A}|^3 \log |\mathcal{A}|)$.

C.2 ALIGNMENT-BASED CONFORMANCE CHECKING

Synchronous product. Given trace σ and model M , the synchronous product is a Petri net (or transition system) whose transitions are labeled with the moves of Definition 2.2. The cost function assigns 0 to synchronous moves and a positive cost (typically 1) to log moves and model moves.

A^* search. The optimal alignment γ^* is the minimum-cost path from the initial marking to a final marking. A^* expands states in order of $f(n) = g(n) + h(n)$, where $g(n)$ is the accumulated cost and $h(n)$ is an admissible heuristic (a lower bound on the remaining cost). A typical heuristic is the number of remaining non-synchronous moves implied by the unmatched portion of the trace.

Complexity. Alignment is NP -hard in the general case; efficient implementations (PM4Py, ProM) handle SDLC logs in practice because the state spaces are small (typically $|S| \leq 20$).

C.3 FITNESS, PRECISION, AND GENERALIZATION

Fitness. As stated in Equation 2.1, fitness is $1 - \text{cost}(\gamma^*)/\text{cost}(\gamma_{\text{worst}})$, where γ_{worst} is the alignment with all moves non-synchronous (cost equal to $|\sigma| + |M_{\min}|$, where $|M_{\min}|$ is the length of the shortest model trace).

ETC precision. Escaping Edges precision counts model states where transitions allowed by the model were never observed in the log. For each state s visited during alignment, let $\text{en}_M(s)$ be the set of model-allowed activities and $\text{en}_\mathcal{L}(s)$ the set of log-observed activities. Then

$$\text{Precision}(M, \mathcal{L}) = 1 - \frac{\sum_s w_s \cdot |\text{en}_M(s) \setminus \text{en}_\mathcal{L}(s)|}{\sum_s w_s \cdot |\text{en}_M(s)|}, \quad (\text{C.1})$$

where w_s is the visit frequency of s .

Generalization. Quantifies the extent to which the model captures behavior beyond observed traces. A common formulation uses the expected fitness on cases generated by a hold-out replication of the underlying process; the empirical estimator weighs states by visit frequency, penalizing states visited by many distinct traces less than states visited by few.

C.4 WORKED EXAMPLE

Toy log. Three traces over four activities: $\sigma_1 = \langle A, B, C, D \rangle$, $\sigma_2 = \langle A, B, C, D \rangle$, $\sigma_3 = \langle A, C, B, D \rangle$.

DFG. Edges $A \rightarrow B$ (count 2), $A \rightarrow C$ (count 1), $B \rightarrow C$ (count 2), $C \rightarrow B$ (count 1), $C \rightarrow D$ (count 2), $B \rightarrow D$ (count 1).

Discovered process tree. IM identifies A as a sequence predecessor to $\{B, C\}$ and D as a sequence successor of $\{B, C\}$, with $\{B, C\}$ in parallel ($B \parallel C$). Resulting tree: $\rightarrow (A, \wedge(B, C), D)$.

Quality metrics. Replaying the three traces against the tree yields fitness = 1.0 (all traces are valid sequences of the parallel construct); precision ≈ 0.5 (the parallel construct admits two permutations of B, C , but only two of them are observed); generalization ≈ 0.7 (depends on weighting; full computation in PM4Py).



End-to-End Walkthrough Example

This appendix presents a complete worked example of the PATHCAST pipeline on a small synthetic project, making the data-flow concrete from raw events through probabilistic forecast.

D.1 SYNTHETIC EVENT LOG

The synthetic dataset contains 10 issues across 35 days, with events from three sources (Jira, Git, GitHub Actions). Issue keys: PROJ-1 to PROJ-10. Total raw events: 58.

Raw events (excerpt for PROJ-1).

Source	Type	Timestamp
Jira	status_change (Backlog → In Progress)	day 1, 09:00
Git	commit (msg: "PROJ-1 initial impl")	day 1, 14:30
Jira	status_change (In Progress → Review)	day 2, 11:00
PR	pr_opened (PROJ-1)	day 2, 11:05
Git	commit (msg: "PROJ-1 review fix")	day 3, 10:00
PR	pr_merged	day 3, 16:00
Jira	status_change (Review → Testing)	day 4, 09:30
CI	build_end (status: pass)	day 4, 09:45
Jira	status_change (Testing → Done)	day 5, 14:00

Case correlation. The cascade $\kappa_1 \circ \kappa_2 \circ \kappa_3$ assigns 56 of 58 events to one of the 10 issues; 2 unrelated commits are discarded. Coverage = $56/58 = 0.966$.

Activity mapping. The mapping μ projects raw events onto $\mathcal{A} = \{\text{backlog}, \text{in_progress}, \text{review}\}$. Build events are absorbed into the surrounding state to avoid spurious transitions.

D.2 PROCESS DISCOVERY

Inductive Miner with $\theta_{\text{noise}} = 0.2$ produces the process tree $\rightarrow (\text{backlog}, \text{in_progress}, \text{review})$ with an optional rework loop between review and in_progress.

Quality metrics on the synthetic log.

- Fitness: 0.93 (one trace contains an unmatched review $\rightarrow$ testing loop)
- Precision: 0.81
- Generalization: 0.74

D.3 MARKOV ESTIMATION

Transition counts (transient block).

$$C_T = \begin{matrix} & \begin{matrix} \text{IP} & \text{R} & \text{T} \end{matrix} \\ \begin{matrix} \text{IP} \\ \text{R} \\ \text{T} \end{matrix} & \begin{pmatrix} 0 & 11 & 0 \\ 2 & 0 & 9 \\ 1 & 0 & 0 \end{pmatrix} \end{matrix}.$$

MLE transition matrix.

$$\hat{Q} = \begin{pmatrix} 0 & 1.000 & 0 \\ 0.182 & 0 & 0.818 \\ 1.000 & 0 & 0 \end{pmatrix}, \quad \hat{R} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}_{\rightarrow \text{done}}.$$

Fundamental matrix. $N = (I - \hat{Q})^{-1} \approx \begin{pmatrix} 1.22 & 1.22 & 1.00 \\ 0.22 & 1.22 & 1.00 \\ 1.22 & 1.22 & 1.00 \end{pmatrix}.$

Expected steps to absorption. $\mathbf{t} = N\mathbf{1} \approx (3.44, 2.44, 3.44)^\top$.

Empirical sojourn distributions. Per state, in days: $\hat{F}_{IP} = \{1.0, 1.5, 2.0, 2.0, 3.0\}$, $\hat{F}_R = \{0.5, 1.0, 1.0, 2.0\}$, $\hat{F}_T = \{0.5, 0.8, 1.5\}$.

D.4 MONTE CARLO SIMULATION

Run $M_{\text{sim}} = 10,000$ trajectories starting from `in_progress`, sampling sojourns from $\hat{F}_s$ and next states from $\hat{Q}$.

Forecasts.

- Median lead time $\hat{Q}_{0.5} \approx 5.4$ days
- P85 forecast $\hat{Q}_{0.85} \approx 7.9$ days
- P95 forecast $\hat{Q}_{0.95} \approx 9.7$ days
- Completion probability (Done): 1.00 (no Cancelled in this synthetic example)

Top-3 paths.

- $IP \rightarrow R \rightarrow T \rightarrow \text{Done}$ (frequency 0.81)
- $IP \rightarrow R \rightarrow IP \rightarrow R \rightarrow T \rightarrow \text{Done}$ (frequency 0.15)
- Longer rework paths (frequency 0.04)

D.5 CROSS-VALIDATION AGAINST ANALYTICAL RESULTS

Mean lead time, computed analytically with mean sojourns $\bar{T}_{IP} = 1.9$, $\bar{T}_R = 1.125$, $\bar{T}_T = 0.93$:

$$E[\text{LT} \mid s_0 = \text{IP}] = 1.22 \cdot 1.9 + 1.22 \cdot 1.125 + 1.00 \cdot 0.93 \approx 4.62 \text{ days.} \quad (\text{D.1})$$

Monte Carlo mean: 4.59 days. Relative deviation: 0.6%, confirming consistency between analytical and simulation paths.

D.6 ML FEATURE CONSTRUCTION

For each of the 10 cases, the 15 features defined in Table 4.4 are constructed at the half-trace observation point. Random Forest with 5-fold temporal cross-validation yields:

- AUC-ROC (PM-only features): 0.79
- AUC-ROC (MSR-only features): 0.71
- AUC-ROC (Combined): 0.84

SHAP analysis ranks Markov-expected remaining transitions and rework count among the top three predictors, illustrating the predictive value of process-derived features hypothesized in Section 2.5 and formally evaluated in the empirical-evaluation chapter (in preparation).

Disclaimer. Numbers above are illustrative for a synthetic $n = 10$ dataset. Statistical significance and external validity are not established at this scale; the example serves to demonstrate the end-to-end data flow, not to evidence empirical claims.



Search Protocol: Search Strings

This appendix presents the four search strings executed across the five digital libraries during the search phase of the PATHCAST SLR. Each string was adapted to the syntax and field conventions of the target database; the block structure (P and I) is preserved across all adaptations. Strings S1–S3 correspond to the principal, complementary (MSR), and frontier (stochastic $\wedge$ PM) blocks, respectively; S4 is a supplementary string executed in Scopus only, targeting Markov models in software testing.

E.1 SEARCH STRING SUMMARY BY DATABASE

Table E.1: Search strings by database and field scope.

Database	Strings	Field / syntax notes
Scopus	S1, S2, S3, S4	TITLE-ABS-KEY(...); DOCTYPE(ar OR cp); LANGUAGE(english); years 1994–2026
IEEE Xplore	S1 (split)	All Metadata; boolean AND of sub-blocks; Conference + Journals; 1994–2026
Web of Sci.	S1, S2, S3	Topic TS=(...); Article + Proceedings; English; 1994– 2026
SpringerLink	S1 (simplified)	Title + Abstract; 8 separate exports; Chapter + Article; English
ACM DL	S1, S2	Abstract; 2 queries; date 1994–2026

E.2 STRING S1 — PRINCIPAL ($P \wedge I$)

Block P (software engineering context):

*("software engineering" OR "software development" OR "software process"
OR "software lifecycle" OR "software life cycle" OR "SDLC" OR "development
lifecycle" OR "development life cycle" OR "DevOps" OR "CI/CD" OR
"continuous integration" OR "continuous delivery" OR "agile development"
OR "agile process" OR "issue tracking" OR "bug tracking" OR "version
control" OR "code review" OR "pull request" OR "software project"
OR "software repository" OR "software maintenance" OR "software
evolution")*

Block I (techniques):

*("process mining" OR "process discovery" OR "conformance checking"
OR "workflow mining" OR "predictive process monitoring" OR "process
prediction" OR "remaining time prediction" OR "process forecasting"
OR "event log" OR "process simulation" OR "Monte Carlo simulation"
OR "Markov chain" OR "Markov model" OR "stochastic process model"
OR "transition probability" OR "absorbing chain" OR "lead time prediction"
OR "cycle time prediction")*

E.3 STRING S2 — COMPLEMENTARY ($MSR \rightarrow PROCESS$
MODELLING)

*("mining software repositories" OR "software repository mining"
OR "GitHub mining" OR "Jira mining") AND ("process model" OR "Markov"
OR "Monte Carlo" OR "stochastic" OR "lead time" OR "cycle time"
OR "throughput" OR "transition matrix")*

E.4 STRING S3 — FRONTIER ($STOCHASTIC \wedge PM \wedge SE$)

*("Monte Carlo" OR "Markov chain" OR "stochastic simulation" OR "probabilistic
forecast" OR "transition matrix") AND ("process mining" OR "event
log" OR "workflow mining" OR "process model" OR "business process")
AND ("software development" OR "software engineering" OR "software*

*process" OR "DevOps" OR "SDLC" OR "issue tracker" OR "commit log"
OR "build pipeline")*

E.5 STRING S4 — MARKOV IN SOFTWARE TESTING (SCOPUS SUPPLEMENT)

*("Markov chain" OR "Markov model" OR "hidden Markov" OR "stochastic
automata") AND ("software testing" OR "test coverage" OR "software
reliability" OR "defect prediction" OR "fault prediction")*



LLM Screening Prompts

The review employed the *claude-haiku-4-5-20251001* model as the primary screener in two phases: title/abstract (T/A) and full text (FT). Each phase uses a *system prompt* that encodes the scope, criteria, and output schema, combined with a per-paper *user prompt* that supplies bibliographic data. All prompts are versioned in `config/screening_criteria.py` in the review repository.

F.1 PHASE 1: TITLE/ABSTRACT SCREENING

SYSTEM PROMPT (T/A PHASE)

You are an expert in systematic literature review (SLR) in the area of process mining and software engineering. Your task is to perform the title/abstract (T/A) screening for SLR PATHCAST.

SLR PATHCAST scope: *Intersection of (1) process mining, (2) stochastic modelling, (3) forecasting/prediction, and (4) software development processes. Focus: studies that analyse, discover, or predict software process characteristics using event logs, repositories, or stochastic models.*

INCLUSION CRITERIA *(at least ONE must be satisfied to include):*

IC1 - Process Mining in Software: applies PM (discovery, conformance, workflow mining, event log analysis, PPM) to SW artefacts (commits, issues, PRs, CI/CD). IC2 - Stochastic Modelling of SW Processes: uses Markov chains, Monte Carlo, stochastic Petri nets, or transition

matrices to model SW dev processes. IC3 - Process Metric Forecasting: predicts lead time, cycle time, remaining time, throughput, or defect rates in SW processes using data or event logs. IC4 - Repository Mining for Process: mines repositories (GitHub, Jira, VCS) to discover or improve SW development process models.

EXCLUSION CRITERIA (any ONE is sufficient to exclude): EC1 - Non-SW domain: applied to healthcare, manufacturing, finance, etc., with no link to SW processes. EC2 - SW as tool: "software" is the implementation platform, not the process under study. EC3 - Algorithm without SW application: purely theoretical method, no evaluation in SW processes. EC4 - Out-of-scope secondary study: survey/SLR not specifically addressing PM or stochastic methods in SE.

UNCERTAIN CASES (maybe): Relevant title without abstract; generic BPM article with possible SW application; ambiguous terminology.

STRUCTURED FIELDS (use only the allowed enums): evidence_tags, software_context, stochastic_method, forecast_target, process_data_source, confidence.

Respond ONLY with valid JSON, no additional text.

USER PROMPT TEMPLATE (T/A PHASE)

Evaluate the paper below for SLR PATHCAST.

Title: {title} **Abstract:** {abstract} **Venue/Type:** {venue} | {doc_type} | {year} **Source database:** {source_db} **Abstract origin:** {abstract_source}

Respond with JSON in the exact format: { "decision": "include" | "exclude" | "maybe", "rationale": "<1-2 sentences justifying the decision>", "matched_ic": ["IC1", "IC2"], "matched_ec": ["EC1"], "evidence_tags": ["process_mining", "forecasting"], "software_context": "software_development_process", "stochastic_method": "markov_chain", "forecast_target": "remaining_time", "process_data_source": "event_logs", "confidence": "medium" }

Rules: - "include" if ≥ 1 IC met and no decisive EC - "exclude" if an EC clearly applies, or paper clearly meets no IC - "maybe" if genuine doubt (title without abstract, domain ambiguity) - matched_ic and matched_ec must be [] when not applicable

F.2 PHASE 2: FULL-TEXT SCREENING

SYSTEM PROMPT (FT PHASE)

The FT-phase system prompt is identical to the T/A prompt, augmented with three instructions that encode the stricter full-text screening pattern:

THIS IS THE FULL-TEXT PHASE - apply criteria rigorously: - If the abstract clearly confirms ≥ 1 IC with no decisive EC $\rightarrow$ "include"
- If the abstract clearly confirms no IC applies, or an EC applies $\rightarrow$ "exclude" - Use "pending" ONLY if the abstract is genuinely insufficient for a decision (absent or extremely short) AND the title suggests marginal relevance
When in doubt between include and exclude, prefer exclude - this phase requires confidence.
YOU MUST NEVER invent, complete, or infer a missing abstract.

USER PROMPT TEMPLATE (FT PHASE)

The FT user prompt extends the T/A template with a **T/A context block** that supplies the prior decision, rationale, and ICs identified in phase 1:

Evaluate the paper below for SLR PATHCAST - FULL-TEXT screening (phase 2).

Title: {title} **Abstract:** {abstract} **Venue/Type:** {venue} | {doc_type} | {year} **Source database:** {source_db} **Abstract origin:** {abstract_source}

T/A phase context: - Previous decision: {ta_decision} - Previous rationale: {ta_rationale} - ICs identified in phase 1: {ta_matched_ic}

Re-evaluate rigorously for the full-text phase. Respond with JSON in the exact format: { "decision": "include" | "exclude" | "pending", [... same fields as T/A phase ...] }

F.3 PHASE 2B: SCREEN-BLANKS PASS

A third pass was applied to the 186 papers without a decision after the two prior phases (mainly Band E papers with no abstract available). The *screen-blanks*

pass reuses the same FT system and user prompts, with the abstract field set to “Abstract not available” and the T/A context block showing `ta_decision = maybe` and `ta_matched_ic = none`. Papers for which the LLM returned `pending` and whose title analysis indicated low relevance were closed under EC5 (full text inaccessible; $n = 148$).

F.4 MODEL PARAMETERS BY PHASE

Table F.1 summarises the model parameters used in the three screening phases and in data extraction.

Table F.1: LLM parameters in each screening phase.

Phase	Model	Max tok.	Batch	API
T/A screening	claude-haiku-4-5-20251001	512	500	Anthropic Batches
FT screening	claude-haiku-4-5-20251001	512	100	Anthropic Batches
Screen-blanks	claude-haiku-4-5-20251001	512	186	Anthropic Batches
Data extraction	claude-haiku-4-5-20251001	1024	100	Anthropic Batches
Synthesis (ch. 3)	claude-sonnet-4-6	2048	—	Anthropic Messages



Data Extraction Form

Data extraction for the 169 included studies was performed with the *claude-haiku-4-5-20251001* model via the Anthropic Batches API. Papers with a locally available PDF had text pre-extracted with *pdfplumber* (first 8 pages, maximum 6,000 characters). Papers without a PDF used the enriched abstract from Semantic Scholar. The prompt below was submitted once per paper.

G.1 EXTRACTION PROMPT

You are a researcher performing data extraction for SLR PATHCAST on Process Mining and Stochastic Modelling in Software Processes.

Title: {title} **Authors:** {authors} **Year:** {year} | **Venue:** {venue}
Inclusion criteria met: {ics} {content_block}

Return ONLY a valid JSON object with exactly these fields:

```
{ "research_question": "main research question (1-2 sentences)",  
  "study_type": "case_study | experiment | survey | tool | theoretical  
| simulation | mixed", "research_contribution": "discovery | conformance  
| enhancement | prediction | simulation | framework | hybrid", "pm_technique":  
"alpha | inductive_miner | heuristic | conformance_checking | social_network  
| declarative | other | none", "stochastic_technique": "markov_chain  
| stochastic_petri_net | monte_carlo | system_dynamics | other |  
none", "software_artifact": "commits | issues | ci_cd | ide_logs  
| vcs | jira | github | other | none", "software_process": "development  
| testing | maintenance | code_review | bug_fixing | deployment
```



```
| requirements | other", "dataset_source": "open_source | industrial
| academic | synthetic | not_specified", "dataset_public": "yes
| no | partial | not_mentioned", "tool_used": "tool names (e.g.
ProM, pm4py, Disco) or not_mentioned", "main_finding": "main result/contribution
(2-3 sentences)", "limitations": "reported limitations (1-2 sentences)
or not_mentioned", "replication_package": "yes | no | partial |
not_mentioned" }
```

Respond ONLY with JSON, no additional text.

G.2 DATA EXTRACTION FORM FIELDS

Table G.1 describes all fields in the structured extraction form stored in `extraction_template.csv`.

G.3 CONTROLLED VOCABULARY (ENUMS) FOR STRUCTURED OUTPUT

Table G.2 defines the full controlled vocabulary for structured fields returned by the LLM in both screening phases. These enums are versioned in `config/screening_criteria.py` and were not changed between screening rounds, ensuring consistency.

Table G.1: Data extraction form: all fields in `extraction_template.csv`.

Field	Description	Allowed values / format
<i>Identification</i>		
<code>internal_id</code>	8-char hash (dedup key)	—
<code>title</code>	Paper title	free text
<code>doi</code>	Digital identifier	URI
<code>year</code>	Publication year	YYYY
<code>source_db</code>	Source database	scopus, ieee, acm, springer, wos, snowball
<code>ft_matched_ic</code>	Inclusion criteria met	IC1 IC2 IC3 IC4
<code>ft_screened_by</code>	FT decision responsible	llm, manual, llm_pdf
<i>Enriched metadata (Semantic Scholar)</i>		
<code>authors</code>	Author list	semicolon-separated
<code>venue</code>	Publication venue	free text
<code>journal_name</code>	Journal/conference name	free text
<code>abstract</code>	Paper abstract	free text
<code>pdf_available</code>	PDF available locally	yes / no
<i>LLM-extracted fields</i>		
<code>research_question</code>	Main RQ	1–2 sentences
<code>study_type</code>	Study type	case_study experiment survey tool theoretical simulation mixed
<code>research_contribution</code>	Contribution type	discovery conformance enhancement prediction simulation framework hybrid
<code>pm_technique</code>	PM algorithm used	alpha inductive_miner heuristic conformance_checking social_network declarative other none
<code>stochastic_technique</code>	Stochastic method	markov_chain stochastic_petri_net monte_carlo system_dynamics other none
<code>software_artifact</code>	Data source	commits issues ci_cd ide_logs vcs jira github other none
<code>software_process</code>	SDLC phase	development testing maintenance code_review bug_fixing deployment requirements other
<code>dataset_source</code>	Dataset provenance	open_source industrial academic synthetic not_specified
<code>dataset_public</code>	Data availability ¹⁶⁰	yes no partial not_mentioned
<code>tool_used</code>	Tools/platforms	free text
<code>main_finding</code>	Main result	2–3 sentences
<code>limitations</code>	Study limitations	1–2 sentences

Table G.2: Controlled vocabulary for LLM structured output.

Field	Allowed values
decision (T/A)	include, exclude, maybe
decision (TC)	include, exclude, pending
software_context	software_development_process, repository_mining, ci_cd, issue_bug_workflow, software_testing, requirements_engineering, software_project_management, unclear, not_software_process
stochastic_method	none, markov_chain, hidden_markov_model, monte_carlo, stochastic_petri_net, bayesian_model, probabilistic_model, simulation, queueing_model, other_stochastic, unclear
forecast_target	none, lead_time, cycle_time, remaining_time, throughput, defect_rate, build_outcome, reliability, completion_time, other_process_metric, unclear
process_data_source	none, event_logs, version_control, issue_tracker, pull_requests, ci_cd_logs, software_repository_mixed, synthetic_data, simulated_process, survey_or_secondary, unclear
confidence	low, medium, high
evidence_tags	process_mining, software_process, repository_mining, stochastic_modeling, forecasting, event_log, version_control, issue_tracking, pull_requests, ci_cd, markov, hidden_markov_model, monte_carlo, stochastic_petri_net, bayesian_model, simulation, lead_time, cycle_time, remaining_time, throughput, defect_prediction, build_prediction, reliability, insufficient_abstract, full_text_required



Assessment and Classification Instruments

This appendix gathers the instruments used to classify, enrich, and assess papers throughout screening and selection: the priority-band taxonomy, the abstract-enrichment algorithm, the quality-assessment checklist, the control paper set, and the SPMF integration-level taxonomy.

H.1 PRIORITY BANDS FOR THE FULL-TEXT QUEUE

Before full-text screening, the queue of 886 papers was stratified into five priority bands (A–E) to optimise resource allocation. Abstract enrichment and LLM screening were applied in band order. Table H.1 defines the classification rules.

Band D papers with a DOI were prioritised for manual PDF recovery. Band E papers without a DOI or abstract were closed as EC5 (full text inaccessible) at the end of the *screen-blanks* pass.

H.2 ABSTRACT ENRICHMENT ALGORITHM

To increase the number of papers assessable by the LLM, a four-source cascade was executed before full-text screening. Algorithm H.1 describes the enrichment procedure.

Table H.1: Priority-band definitions for the full-text review queue.

Band	<i>n</i>	Classification rule
A	95	T/A decision = include and ≥ 2 distinct ICs met
B	20	T/A decision = include and exactly 1 IC met
C	39	T/A decision = maybe and abstract available
D	152	T/A decision = maybe, at least 1 IC signal in title, no abstract
E	584	T/A decision = maybe, no IC signal in title, no abstract
Total	886	

Algorithm 1 Abstract Enrichment Cascade**Input:** Record p with doi and/or title**Output:** p updated with abstract, authors, venue

1. If p .abstract is non-empty $\rightarrow$ **return** p
2. If p .doi is present:
 - 2a. Query **Semantic Scholar** Batch API (DOI:{doi})
If found: update abstract, authors, venue; **return**
 - 2b. Query **OpenAlex** (filter=doi:{doi})
If found: update abstract; **return**
 - 2c. Query **CORE** (/search/works?q=doi:{doi})
If found: update abstract; **return**
 - 2d. Query **Crossref** (/works/{doi})
If abstract present: update; **return**
3. If p .doi is absent:
 - 3a. Query **Semantic Scholar** by title (fuzzy)
If similarity ≥ 0.85 : update; **return**
4. Set p .abstract_source = missing_or_unverified

Figure H.1: Abstract enrichment cascade applied to the 886-paper FT queue.

The cascade raised abstract coverage from 16.6% (147/886) to 72.6% (643/886) before Round 2 screening. Of the 643 enriched abstracts, 590 were obtained from Semantic Scholar, 31 from OpenAlex, 13 from CORE, and 9 from Crossref or the title fallback. Table H.2 summarises results by band.

Table H.2: Abstract enrichment results by priority band.

Band	Papers	With abstract (pre)	Obtained	Coverage (post)
A	95	91 (95.8%)	4	95 (100.0%)
B	20	18 (90.0%)	2	20 (100.0%)
C	39	39 (100%)	0	39 (100.0%)
D	152	0 (0.0%)	118	118 (77.6%)
E	584	0 (0.0%)	371	371 (63.5%)
Total	886	148 (16.7%)	495	643 (72.6%)

H.3 QUALITY ASSESSMENT CHECKLIST

The eight-item checklist (Table H.3) is applied to each included primary study during manual reading. Items QA1–QA4 assess methodological rigour; QA5–QA7 assess empirical robustness and reproducibility; QA8 specifically checks reporting of process-model quality, which is central to the PATHCAST evaluation strategy. Studies scoring below 4/8 are excluded from synthesis.

Current QA status: 76 of 169 studies assessed (complete Round 1 set); mean = 5.18 (SD = 1.13), median = 5.0 (IQR: 4–6); 5 studies scored below the threshold and were excluded from synthesis. The remaining 93 studies (Rounds 2+) have QA assessment pending.

H.4 CONTROL PAPER SET

The set of 12 control papers was used to validate search strings before execution. Table H.4 lists all control papers with their IC profile and the source by which each was captured.

Papers V4, V5, V8, and V9 were not in the initial working set but were recovered via *snowballing* during the FT phase and are included in the final corpus.

Table H.3: Quality assessment checklist with scoring guide.

ID	Criterion	Scoring guide
QA1	Are the research objectives clearly stated?	Score 1 if RQs or hypotheses are explicitly enumerated; 0 otherwise.
QA2	Is the software engineering context described in sufficient detail?	Score 1 if SDLC phase, artefact type, and process scope are defined.
QA3	Is the data source (event log or repository) described reproducibly?	Score 1 if dataset origin, collection method, and key statistics are provided.
QA4	Is the PM or stochastic technique formally defined?	Score 1 if algorithm, parameters, and configuration are specified.
QA5	Are results validated empirically?	Score 1 if validation uses real or semi-real data; 0 for pure simulation or no validation.
QA6	Are threats to validity discussed?	Score 1 if at least internal and external validity are addressed.
QA7	Is the study reproducible (data and/or code available)?	Score 1 if a replication package or open dataset is described or referenced.
QA8	Are process-model quality metrics reported (fitness, precision)?	Score 1 if at least one formal quality metric is quantified.
Score range		0–8 (threshold: ≥ 4)

Table H.4: Complete control-paper set for string validation.

#	Paper	Rationale	ICs	Captured via
V1	Cook & Wolf (1998) (COOK; WOLF, 1998)	Seminal PM + SE	IC1, IC2	ACM
V2	Rubin et al. (2007) (RUBIN et al., 2007)	Named PM framework for SW (earliest in validation set)	IC1	Scopus
V3	Poncin et al. (2011) (PONCIN; SEREBRENIK; BRAND, 2011)	PM + MSR intersection	IC1, IC4	IEEE
V4	Caldeira & Cardoso (2019)	PM + SW process assessment	IC1	IEEE
V5	Lemos et al. (2011)	PM for SW management	IC1	IEEE
V6	Whittaker & Thomason (1994) (WHITTAKER; THOMASON, 1994)	Markov + SW testing (seminal)	IC2	Scopus
V7	Tax et al. (2017) (TAX et al., 2017)	Seminal PPM (BPM baseline)	IC3	Control
V8	Rogge-Solti & Weske (2015)	Stochastic + PM forecasting	IC2, IC3	Scopus
V9	Rozinat et al. (2009)	PM → Simulation	IC1, IC2	Control
V10	Gupta (2017) (GUPTA, 2017)	Multi-repository PM	SE IC1, IC3	ACM
V11	Jalote & Kamma (2021) (JALOTE; KAMMA, 2021)	IDE logs + Markov chains	IC1, IC2	Scopus
V12	Ortu et al. (2023) (ORTU et al., 2023)	GitHub Markov + SNA	IC2, IC4	Scopus
Capture rate from primary queries only				10/12 (83.3%)
Capture rate including snowballing				12/12 (100%)

H.5 SPMF INTEGRATION-LEVEL TAXONOMY

The *Software Process Mining and Forecasting* (SPMF) taxonomy, used throughout Chapter 3, classifies studies along three orthogonal dimensions (Table H.5).

Table H.5: SPMF taxonomy: three dimensions with level definitions.

Dimension	Level	Definition
Analytical Depth	AD1 – Descriptive	As-is model discovery; conformance checking
	AD2 – Diagnostic	Bottleneck detection; rework quantification; root-cause analysis
	AD3 – Predictive	Outcome prediction; remaining-time estimation
	AD4 – Forecasting	Distributional forecasts; probabilistic completion estimates
SDLC Coverage	SC-R	Requirements and design
	SC-D	Development (coding, commits, pull requests)
	SC-Q	Quality (testing, reviews, defects)
	SC-C	CI/CD and delivery
	SC-O	Operations and maintenance
Technique Integration	TI-L0	Single technique (PM, ML, or stochastic in isolation)
	TI-L1	Two techniques in the same study without an explicit pipeline contract
	TI-L2	Partial bridge: two techniques connected, but no ML closure or contract guarantee
	TI-L3	Unified pipeline: PM → Markov → MC → ML with formal inter-stage contracts

The L0–L3 scale on the TI dimension corresponds to the integration-level code used in Sections 3.4.7 and I.2. PATHCAST is positioned at TI-L3, AD4, with SC coverage spanning SC-D, SC-Q, and SC-C.



Reading List and Pipeline Artefacts

This appendix presents the ranked list of 32 primary studies selected for in-depth synthesis and documents the persistent scripts and artefacts of the review pipeline.

I.1 MULTI-CRITERION SCORING RUBRIC

Table I.1 presents the 15-point rubric used to rank studies. Three seminal papers (Cook 1995, Cook 1998, Rubin 2007) are included regardless of score.

Table I.1: Multi-criterion scoring rubric for the top-32 reading list.

Criterion	Points
IC2 present (stochastic modelling in SE)	+3
IC3 present (process metric forecasting)	+3
IC1 present (process mining in SE)	+2
IC4 present (repository mining for process)	+1
PDF available locally	+2
Published in 2020 or later	+1
High-impact venue (IEEE/ACM/IST/JSS/IS/TSE)	+1
Empirical type (case_study or experiment)	+1
Contribution = prediction or simulation	+1
Maximum score	15

I.2 TOP-32: RANKED READING LIST

I.3 REVIEW PIPELINE SCRIPTS

The review was conducted with a custom Python pipeline (`pipeline/`) operating on structured CSV storage. Table I.3 documents each script with its function and main invocation command.

I.4 PERSISTENT ARTEFACT REGISTRY

Table I.4 lists persistent artefacts produced by the review, with their role, location, and current status.

Table I.2: Top-32 reading list with score, year, IC cluster, and summary of the main finding. Papers marked with † are forced inclusions due to seminal status.

Rank	Pts	Authors (year)	ICs	PDF	Main finding (summary)
1	14	Joshi & Kahani (2024) (JOSHI; KA-HANI, 2024)	IC2,IC3,IC4	✓	RL/MDP for GitHub PR outcome prediction
2	14	Li et al. (2020) (LI et al., 2020)	IC2,IC3	✓	Hybrid SD+DES simulation for industrial forecasting
3	13	Guinea-Cabrera et al. (2025) (GUINEA-CABRERA; HOLGADO-TERRIZA; PICO-VALENCIA, 2025)	IC1,IC2,IC3	✓	PRIMAD: Digital Twin for SDLC (pm4py+Akka+LightGBM), L2
4	13	Ortu et al. (2023) (ORTU et al., 2023)	IC2,IC4	✓	Markov+SNA for fault insertion/correction patterns (Apache)
5	13	Ben Mesmia et al. (2021) (MESMIA; ES-CHEIKH; BARKAOUI, 2021)	IC2,IC3	✓	Non-Markovian SPN for DevOps duration prediction
6	12	Buliga et al. (2025) (BULIGA et al., 2025)	IC1,IC3	✓	Generative AI + process simulation for what-if scenarios
7	12	Bhadra et al. (2023) (BHADRA et al., 2023)	IC2,IC3	✓	CI/CD SPN with reliability sensitivity analysis
8	12	Bhadra (2022) (BHADRA, 2022)	IC2,IC3	✓	CI/CD pipeline SPN for delivery probability
9	12	Massitela et al. (2018) (MASSITELA et al., 2018)	IC2,IC3	✓	Stochastic Automata Network for Waterfall estimation
10	11	Incerto et al. (2025) (INCERTO; VANDIN; AHRABI, 2025)	IC1,IC2	✓	VLMC for stochastic conformance; outperforms 18 techniques
11	11	Jo & Kwon (2024) (JO; KWON; KWON, 2024)	IC2,IC4	✓	Probabilistic model checking (CTL) on GitHub repositories
12	11	Jo et al. (2023) (JO; KWON; KWON, 2023)	IC2,IC4	✓	DTMC for GitHub collaboration patterns (5 repos)
13	11	López-Pintado & Dumas (2023) (LÓPEZ-PINTADO; DUMAS, 2023)	IC1,IC2	—	Probabilistic resource calendars discovered from logs
14	11	Caldeira et al. (2022) (CALDEIRA et al., 2022)	IC1,IC3	✓	Process complexity → cyclo-matic complexity ($r \approx 0.43$)
15	11	Jalote & Kamma (2021) (JALOTE; KAMMA, 2021)	IC1,IC2	✓	Markov chains on IDE logs for programmer productivity
16	11	Lunesu et al. (2021) (LUNESU et al., 2021)	170 IC2,IC3	✓	Monte Carlo for risk assessment in agile development
17	11	Nafreen et al.	IC2,IC3	—	SRGM + defect tracking for hy-

Table I.3: Review pipeline scripts and their functions.

Script	Function	Main command
pipeline/dedup.py	Cross-source deduplication (DOI + fuzzy title)	python main.py dedup
pipeline/enrich.py	Abstract enrichment cascade (4 sources)	python main.py enrich
pipeline/screening.py	LLM T/A screening via Batches API	python main.py screen --run
pipeline/fulltext.py	LLM FT screening + manual review	python main.py fulltext --run
pipeline/pdf_download.py	PDF recovery (Unpaywall, S2, OA)	python main.py pdf
pipeline/extract_prep.py	Cap enrichment + PDF copy for extraction	python pipeline/extract_prep.py
pipeline/extract_llm.py	Structured data extraction (LLM, 169 papers)	python pipeline/extract_llm.py --run
pipeline/synth_llm.py	Synthesis paragraph generation per IC cluster	python pipeline/synth_llm.py
pipeline/finalization.py	PRISMA snapshot, EC5 closure, export	python main.py finalize

Table I.4: Persistent review artefacts.

File	Role	Status
results/screening/ft_screening_decision_log.csv	Decision log (886 papers)	Final (no blanks)
results/extraction/extraction_data.csv	Extraction data (169 papers)	169/169 main_finding
results/final_review/included_included_current.csv	Included in current	169 rows
results/final_review/top30_ranking_ranking.csv	Ranking leading list	32 rows
results/final_review/cap3_synthesis_synthesis_paragraphs	LLM synthesis paragraphs	For review
results/final_review/missing_bibtex_for_s2_missing_keys	Bibtex for S2 missing keys	Ready to merge
results/final_review/pending_for_closure_for_closure.csv	For closure or closure	118 rows
results/final_review/prisma_snapshot_snapshot.txt	PRISMA snapshot	Final
results/SLR_EXECUCAO_ATE_AGORA_execution_log	Execution log (all sessions)	Updated



Evaluation Dataset: Repository List

This appendix lists the 48 open-source repositories selected for the empirical evaluation described in Chapter 5 (Section 5.2). Selection followed the stratified sampling protocol of Section 5.2.3: 150 candidates from SEART GitHub Search (DABIC; AGHAJANI; BAVOTA, 2021), criterion verification via the GitHub REST API (GitHub, Inc., 2023), and stratification across primary language, project scale, governance model, and workflow maturity (Table 5.3).

Forty-seven of forty-eight evaluation repositories had usable event logs after the June 2026 harvest (see `repo_manifest.json`). `torvalds/linux` was collected but yields no process cases under the Stage-1 filter (`min_events=2`): the kernel project maintains issue tracking on the Linux Kernel Mailing List rather than GitHub Issues. Apache and Spring projects use Jira issue transition history from Montgomery et al. (MONTGOMERY; LÜDERS; MAALEJ, 2022) in place of GitHub Issues where noted. Table J.1 lists per-repository Stage-1 case and event counts from the June 2026 harvest.

J.1 STRATIFICATION SUMMARY

J.2 COMPLETE REPOSITORY LIST

Table J.1: Stage-1 harvest coverage by evaluation repository (47/48 with usable logs; snapshot June 2026).

Repository	Source	Cases	Events	Status
angular/angular	GitHub	144	293	ok
apache/airflow	Jira	942	2,443	ok
apache/commons-lang	Jira	676	1,602	ok
apache/flink	Jira	7,466	17,727	ok
apache/kafka	Jira	3,919	9,886	ok
apache/maven	Jira	1,243	3,664	ok
BurntSushi/ripgrep	GitHub	8	17	ok
cli/cli	GitHub	9,548	69,106	ok
curl/curl	GitHub	96	195	ok
dbt-labs/dbt-core	GitHub	29	58	ok
denoland/deno	GitHub	141	285	ok
django/django	GitHub	62	125	ok
expressjs/express	GitHub	8	16	ok
facebook/react	GitHub	84	171	ok
fastapi/fastapi	GitHub	8,693	90,198	ok
git/git	GitHub	11	22	ok
golang/go	GitHub	187	385	ok
grafana/grafana	GitHub	449	914	ok
hashicorp/terraform	GitHub	91	182	ok
helm/helm	GitHub	36	72	ok
huggingface/transformers	GitHub	176	400	ok
kubernetes/kubectrl	GitHub	2	4	ok
llvm/llvm-project	GitHub	139	286	ok
microsoft/TypeScript	GitHub	145	293	ok
microsoft/vscode	GitHub	758	1,573	ok
neovim/neovim	GitHub	209	425	ok
nestjs/nest	GitHub	16,606	90,369	ok
openssl/openssl	GitHub	109	240	ok
pallets/click	GitHub	8	18	ok
pallets/flask	GitHub	27	54	ok
pandas-dev/pandas	GitHub	63,404	576,501	ok
postgres/postgres	GitHub	2	4	ok
prometheus/prometheus	GitHub	51	106	ok
psf/requests	GitHub	18	38	ok
python/cpython	GitHub	433	920	ok
redis/redis	GitHub	85	175	ok
rust-lang/rust	GitHub	410	843	ok
scikit-learn/scikit-learn	GitHub	82	164	ok
scipy/scipy	GitHub	76	155	ok
serde-rs/serde	GitHub	15	30	ok
sharkdp/bat	GitHub	20	40	ok
spring-projects/spring-boot	Jira	1,223	3,468	ok
spring-projects/spring-framework	Jira	10,084	25,940	ok
sveltejs/svelte	GitHub	46	92	ok
tokio-rs/tokio	GitHub	23	48	ok
torvalds/linux	GitHub	0	0	raw_only
vitejs/vite	GitHub	16,541	118,916	ok
vuejs/vue	GitHub	26	52	ok

Harvest window: authenticated GitHub REST collection ending 9 June 2026 (completion batch nestjs/nest, vitejs/vite, pandas-dev/pandas); commit metadata refreshed 8 June 2026. See `repo_manifest.json`.

Table J.2: Repository count by primary language stratum ($n = 48$).

Primary language	Count	Issue source mix
Python	11	GitHub Issues
JavaScript/TypeScript	11	GitHub Issues
Java	7	Jira (Montgomery et al.)
Go	7	GitHub Issues
Rust	5	GitHub Issues
C/C++	7	GitHub Issues
Total	48	41 GitHub; 7 Jira

Table J.3: Complete list of 48 evaluation repositories with stratification metadata. Abbreviations: Scale — S = Small (< 5K commits), M = Medium (5–20K), L = Large (> 20K); Gov. — Fdn = Foundation, Corp = Corporate, Comm = Community; WF — B = Basic, St = Standard, A = Advanced.

#	Repository	Lang.	Domain	Scale	Gov.	WF	Issues
1	angular/angular	JS/TS	Web framework	L	Corp	A	GitHub
2	apache/airflow	Python	Data / workflow	L	Fdn	A	Jira
3	apache/commons-lang	Java	Library	M	Fdn	St	Jira
4	apache/flink	Java	Data processing	L	Fdn	A	Jira
5	apache/kafka	Java	Messaging	L	Fdn	A	Jira
6	apache/maven	Java	Build tool	L	Fdn	St	Jira
7	BurntSushi/ripgrep	Rust	CLI	M	Comm	St	GitHub
8	cli/cli	Go	CLI	L	Corp	A	GitHub
9	curl/curl	C	Library / DevOps	L	Comm	St	GitHub
10	dbt-labs/dbt-core	Python	Data processing	M	Corp	A	GitHub
11	denoland/deno	JS/TS	Runtime	L	Fdn	A	GitHub
12	django/django	Python	Web framework	L	Fdn	A	GitHub
13	expressjs/express	JS/TS	Web framework	M	Comm	St	GitHub
14	facebook/react	JS/TS	UI library	L	Corp	A	GitHub
15	fastapi/fastapi	Python	Web framework	M	Comm	A	GitHub
16	git/git	C	DevOps	L	Comm	St	GitHub
17	golang/go	Go	Language / runtime	L	Corp	A	GitHub
18	grafana/grafana	Go	DevOps / monitoring	L	Corp	A	GitHub

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Table J.3 — continued from previous page

#	Repository	Lang.	Domain	Scale	Gov.	WF	Issues
19	hashicorp/terraform	Go	DevOps / IaC	L	Corp	A	GitHub
20	helm/helm	Go	DevOps	M	Fdn	A	GitHub
21	huggingface/transformers	Python	ML library	L	Corp	A	GitHub
22	kubernetes/kubectrl	Go	DevOps	L	Fdn	A	GitHub
23	llvm/llvm-project	C/C++	Compiler	L	Fdn	A	GitHub
24	microsoft/TypeScript	JS/TS	Language	L	Corp	A	GitHub
25	microsoft/vscode	JS/TS	IDE	L	Corp	A	GitHub
26	neovim/neovim	C	Editor	L	Comm	St	GitHub
27	nestjs/nest	JS/TS	Web framework	M	Comm	St	GitHub
28	openssl/openssl	C	Security library	L	Comm	St	GitHub
29	pallets/click	Python	CLI library	M	Comm	St	GitHub
30	pallets/flask	Python	Web framework	M	Comm	St	GitHub
31	pandas-dev/pandas	Python	Data processing	L	Fdn	A	GitHub
32	postgres/postgres	C	Database	L	Comm	St	GitHub
33	prometheus/prometheus	Go	DevOps / monitoring	L	Fdn	A	GitHub
34	psf/requests	Python	Library	M	Fdn	St	GitHub
35	python/cpython	Python	Language / runtime	L	Fdn	A	GitHub
36	redis/redis	C	Database	L	Corp	A	GitHub
37	rust-lang/rust	Rust	Language	L	Fdn	A	GitHub
38	scikit-learn/scikit-learn	Python	ML library	L	Fdn	A	GitHub
39	scipy/scipy	Python	Scientific comp.	L	Fdn	St	GitHub

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Table J.3 — continued from previous page

#	Repository	Lang.	Domain	Scale	Gov.	WF	Issues
40	serde-rs/serde	Rust	Library	M	Comm	St	GitHub
41	sharkdp/bat	Rust	CLI	S	Comm	St	GitHub
42	spring-projects/spring-boot	Java	Web framework	L	Corp	A	Jira [‡]
43	spring-projects/spring-framework	Java	Web framework	L	Corp	A	Jira
44	sveltejs/svelte	JS/TS	Web framework	M	Comm	A	GitHub
45	tokio-rs/tokio	Rust	Async runtime	M	Comm	A	GitHub
46	torvalds/linux	C	OS kernel	L	Fdn	A	GitHub
47	vitejs/vite	JS/TS	Build tool	M	Comm	A	GitHub
48	vuejs/vue	JS/TS	Web framework	L	Comm	A	GitHub

[‡] Jira transition history supplemented from an external XLSX export (not present in the Montgomery bulk parquet); mapped in jira_event_log_tagged.parquet.
Harvest snapshot: June 2026 (GitHub REST completion batch ending 9 June 2026; see Table J.1).